

GPW WATS 2.02 FIX Order Gateway Specification (FIX 5.0).

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1. DISCLAIMER

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2. PREFACE

This document has been prepared by Warsaw Stock Exchange in order to help in the implementation process of GPW WATS trading platform.

2.1. TARGET AUDIENCE

This document has been prepared to development staff, Independent Software Vendors who produce software integrated with GPW WATS, analysts, market participants and all clients who want to deepen their knowledge about GPW WATS.

2.2. DOCUMENT PURPOSE

This document provides a technical description of FIX Order Gateway available in GPW WATS.

Bearing in mind this document has been prepared by Warsaw Stock Exchange in order to assist customers in their change deployment process which results from the trading platform replacement program and all efforts have been taken to ensure the accuracy of the document, some of the proposals mentioned in this document may still be subject to change or may require further supplements. If the final approach has not yet been taken it is clearly stated.

2.3. ASSOCIATED DOCUMENTS

GPW WATS FIX Order Gateway Specification is a part of GPW WATS documentation set.

Please check the following documents to learn about the construction of Trading System

- GPW WATS 1.01 Trading System

Please check the documentation of the trading protocols supported by GPW WATS.

- GPW WATS 2.01 Native Order Gateway Specification
- GPW WATS 2.02 FIX Order Gateway Specification (this document) .

Please check the description of the communication with Data Distribution Service.

- GPW WATS 3.01 Market Data Protocol

Please check the additional documentation which explains other services provided within GPW WATS.

- GPW WATS 4.01 Drop Copy Gateway
- GPW WATS 4.02 Post Trade Gateway
- GPW WATS 5.01 Risk Management Gateway.

Please check the additional documentation describing the following:

- GPW WATS 2.03 Rejection Codes
- GPW WATS 2.04 BenDec Message Definition Format
- GPW WATS 6.01 Connectivity.

It is recommended that you read the GPW WATS 1.01 Trading System document first.

3. DOCUMENT HISTORY

Version	Date	Description
0.51	29.06.2023	The initial publication of the documentation.
0.52	26.07.2023	Publication of v0.52.
0.53	18.08.2023	Message change: ExecutionReport field Price (44) has been change to required="N".
0.54	20.09.2023	Messages change: Logon (A): Fixed order of messages. Logout (5): SessionStatus field added. Header: ApplVerID field added.
0.56	10.11.2023	TestEvent (UT) message has been added.
0.57	30.11.2023	The following changes has been added: 4.2. Party Identifiers - updates(Clearing Firm and Contra Firm added, additional info added for shortcodes parties) 7. Messages - TestEvent message deleted. - PartyIDSource extended for messages(New order Single, AE, AR) to cover: - B = BIC - N = Legal Entity Identifier (LEI) - BusinessRejectReason enhancements, - SelfMatchPreventionID enabled as already implemented. 4.6. New chapter describing quotes handling. 4.4.11 Self-Trade prevention chapter added, 4.4.6 – Adnotation to new field 2431(ExecTypeReason).
0.58	18.12.2023	Publication of v0.58.
0.59	25.01.2024	Execution Report (8) message: ExecRestatementReason (378) has been replaced with ExecTypeReason (2431) field. 4.4.11 STP mechanism description has been extended and new ExecTypeReason=107 added. 5.3. Message throttling description has been extended. 7.16 SecondaryTradeReportID has been added to Trade Capture Report message. 7.17 SecondaryTradeReportID has been added to Trade Capture Report Ack message. 4.5.1.2 SecondaryTradeReportID has been added.

4. SERVICE DESCRIPTION

4.1. FIX GATEWAY SERVICE DESCRIPTION

FIX Order Gateway service provides a trading interface based on a FIX standard protocol.

FIX Order Gateway is a point-to-point service which relies on the TCP-IP protocol.

The main service functionalities are:

- order submission,
- order modification,
- order cancellation,
- transaction submission (block and cross trades)
- mass quotes

FIX Order Gateway service consists of two main layers:

Session layer- designed to maintain reliable and repeatable messaging between peers.

There are number of session layer messages to establish and maintain FIX connection between Trading Client Application and Order Gateway:

- Logon (A) and Logout (5) messages to start and terminate FIX connection,
- Heartbeat (0) and TestRequest (1) messages to maintain a FIX connection in periods of inactivity,
- ResendRequest (2) and SequenceReset (4) to recover lost messages,
- Reject (3) message to inform that received message cannot be further processed because it does not comply with the session layer rules.

All messages (including business messages) are examined on a session layer level, and only when verification is successful, messages are further processed.

The document follows a convention where messages used for exchanging information about orders and transactions are referred to as *business messages*. In the FIX specification, these messages are known as *application messages*.

Business layer – designed to achieve a business purpose, such as submitting and managing orders or transactions.

After FIX connection is established Trading Client Application and Order Gateway can exchange business layer messages. Each business message consists of a number of fields (standard header and trailer) as follows.

Trading Client Application sends business layer messages such as:

- New Order Single (D) to submit a new order,
- Order Cancel Replace Request (G) to change order data,
- Order Cancel Request (F) to cancel a previously submitted order,
- Trade Capture Report (AE) to report a block/cross trade,
- Mass Quote (i) to submit mass quote.

4.2. PARTY IDENTIFIERS

Members use short codes to safely exchange data about parties required by MiFID2/MiFIR for reporting purposes. Participants must additionally provide real data referred by the short codes, in a separate document.

A short code ranges from 4 to 4.294.967.295 ($2^{32} - 1$). The values 0, 1, 2, and 3 are reserved for special purposes. In FIX messages PartyID (448) field contains short codes.

Field PartyRole (452) specifies the role represented by the short code provided in the PartyID (448) field:

- Client member (Party Role = 3)

Client party role provides information about short codes which identify a natural person or firm or legal entity. Client party role is mandatory if the order is submitted with Order Capacity (528)=A or R. In case of aggregated orders or pending allocations the PartyID (448) field is set to 1 or 2 and PartyRoleQualifier should be set to 0 (None). PartyIDSource (447) is set to P – Short Code Identifier.

- Investment decision maker (Party Role = 122).

An Investment decision maker party role provides information about short codes which identify a natural person or algorithm responsible for the investment decision. An Investment decision maker identifier is mandatory if an order is submitted with Order Capacity (528) = P.

The PartyID for Investment decision maker value can range from 4 to 4.294.967.295. An Investment decision maker short code is linked to PartyRoleQualifier equal to 24 or 22. PartyRoleQualifier (24) is used when a natural person within the member is responsible for the investment decision. PartyRoleQualifier (22) is used when an algorithm is responsible for the investment decision. PartyIDSource (447) is set to P – Short Code Identifier.

- Executing trader (Party Role = 12).

An Executing trader party role provides information on short codes which identify a natural person or algorithm responsible for an order execution. The party role Executing trader is mandatory if Order Capacity is equal to both Agency and Principal(528)=A, R.or P). PartyID for Executing trader value ranges from 4 to 4.294.967.295. Executing trader short code is linked to PartyRoleQualifier equal to 24 or 22. PartyRoleQualifier (24) is used when a natural person within the member is responsible for order execution. PartyRoleQualifier (22) is used when an algorithm is responsible for order execution. When the time and location of order execution are determined by the client of the participant then the Executing trader's Party ID has the reserved code =3 (NONE) and PartyRoleQualifier should be set to 0 (None). PartyIDSource (447) is set to P – Short Code Identifier.

Parties section can also be used to specify additional information, PartyRole (452) can be used to specify role:

- Clearing Firm (Party Role = 4)

Clearing Firm party role can be optionally used for both CLOB orders and BLOCK transactions on FIX trading Gateway, PartyID (448) represents code identifying Clearing Firm and PartyIDSource (447) specify type of code provided in PartyID field. PartyID (448) is set to one of subsequent values:

- B = BIC
- D = Proprietary / Custom code
- N = Legal Entity Identifier (LEI)

PartyRoleQualifier (2376) is set to 0 as this value is not applicable to Clearing Firm Party Role.

- Contra Firm (Party Role = 17)

Contra Firm Party Role is mandatory for Single Sided TradeCaptureReports to identify second side of reported transaction. PartyID(448) represents code identifying Contra Firm Participant(second side of the transaction) and PartyIDSource (447) specify type of code provided in PartyID field and is set to D(Proprietary / Custom code).

Field PartyRoleQualifier (2376) provides further qualification of PartyRole (452):

- 0 = None (only valid for reserved short codes)
- 22 = Algorithm (applicable to PartyRole values 12 or 122)
- 23 = Firm or legal entity (LEI) (applicable to PartyRole value 3)
- 24 = Natural person (applicable to PartyRole values 3, 12, 122)

The table below provides the summary of the combination of fields in FIX Parties repeating group used to specify short codes.

Party Identifier	FIX Tags
Client – Natural person (National ID)	PartyID (448) = short code Party Role (452) = 3 PartyRoleQualifier (2376) = 24
Client – Legal entity (LEI)	PartyID (448) = short code Party Role (452) = 3 PartyRoleQualifier (2376) = 23
Client's aggregated orders	PartyID (448) = 1 (AGGR) Party Role (452) = 3 PartyRoleQualifier (2376) = 0
Client's pending allocation	PartyID (448) = 2 (PNAL) Party Role (452) = 3 PartyRoleQualifier (2376) = 0
Investment Decision Maker - Natural person (National ID)	PartyID (448) = short code Party Role (452) = 122 PartyRoleQualifier (2376) = 24
Investment Decision Maker - Algorithm	PartyID (448) = short code Party Role (452) = 122 PartyRoleQualifier (2376) = 22
Executing Trader - Natural person (National ID)	PartyID (448) = short code Party Role (452) = 12 PartyRoleQualifier (2376) = 24
Executing Trader - Algorithm	PartyID (448) = short code Party Role (452) = 12 PartyRoleQualifier (2376) = 22
Executing Trader - Order execution is determined by the client	PartyID (448) = 3 (Client) Party Role (452) = 12 PartyRoleQualifier (2376) = 0 (NONE)

4.3. SECURITY IDENTIFICATION

Instruments are identified in FIX protocol messages using numerical GPW WATS identifier. This identifier corresponds to InstrumentID in Market Data messages. The list of available Instrument IDs along with their ISINs and other basic characteristics will be provided on a daily basis via Market Data services (Instrument message).

GPW WATS instruments identifiers will not change over the entire life of an instrument and will not be reused after instrument termination for newly created instruments.

Relevant FIX tags:

SecurityIDSource (22) = 8 (Exchange Symbol)

SecurityID (48) = to InstrumentID in Market Data messages

4.4. ORDER MANAGEMENT

After FIX connection is established (see chapter 7. Establishing Connection) Participant is allowed by GPW WATS to submit new orders and manage existing orders.

New Order Single (35=D) message is used to submit New Order, Order Cancel Replace Request (35=G) is used to modify existing orders (which belongs to Participant), Order Cancel Request (35=F) concludes order lifecycle. After order is canceled, it is removed from GPW WATS. Participant cannot revert its status.

For each of the above messages, if it is received, successfully analyzed GPW WATS responds with Execution Report message (35=8) which reflect the current status and last action done on order.

Each of the above messages can be rejected on either session (35=3), business validation (35=j) or application (Execution Report (35=8) or OrderCancel Reject (35=9)) levels due to different reasons.

Figure 1. – Order lifecycle



Order lifecycle of a new order in GPW WATS starts when a new order is submitted and ends when it is cancelled (by user or System), expired by System based on its time validity parameter or when it is fully filled (after one or many trades).

4.4.1. ORDER IDENTIFIERS

The following Tags are utilized across FIX messages to identify Orders in GPW WATS:

- COrderId (11)
- OrderId (37)
- MDEntryID (278)
- OrigCOrderId (41)

4.4.1.1. ClOrdID (11)

ClOrdID is a mandatory field for each of the messages dedicated to create and manage orders sent by Participants:

- New Order Single (D)
- Order Cancel/Replace Request (G)
- Order Cancel Request (F)

or sent by GPW WATS:

- Execution Report (8).

ClOrdID is alphanumeric ASCII free-format string, which can include any character except a control character.

ClOrdID is assigned by the exchange member, GPW WATS populates previously assigned ClOrdID when it sends an Execution Report back to the exchange member.

ClOrdID is a transitional unique identifier of Client order. Thus ClOrdID reference to Order can be updated by each accepted modification ClOrdID message which refers to a particular order.

Client Order can be accessed by ClOrdID of the last message which successfully submits or modifies Participant order.

ClOrdID must be unique amongst all Participant's orders.

4.4.1.2. OrderID (37)

OrderID is a numeric format unique identifier of an order assigned by Trading System.

OrderID is unique across all trading days, all order books and all exchange members and it is persistent for the entire life of an order (i.e. it does not change after order modification).

GPW WATS provides newly generated OrderID in Execution Report returned to Participant each time a new order is submitted, subsequently the same OrderID is returned in Execution Reports when the order is modified or cancelled.

OrderID can be used by an exchange member (interchangeably with OrigClOrdID) when requesting an order modification (Order Cancel/Replace Request (G) message) or order cancellation (Order Cancel Request (F) message) to identify the relevant order which rests in Trading System. If an exchange member provides both OrderID and OrigClOrdID in the request to modify or cancel, Trading System will process the OrderID only and ignore OrigClOrdID.

4.4.1.3. MDEntryID (278)

The order identifier assigned by System and published in Market Data (public information). Corresponds to the publicOrderID field in Market Data OrderAdd (9), OrderModify (10), OrderDelete (11), OrderExecute (12) messages.

MDEntryID is a numeric format unique identifier of an order assigned by Trading System.

MDEntryID is unique across all order books and all exchange members, but only within a single trading day. On the next trading day, the numbering is reset and starts afresh (i.e. MDEntryID is not persistent for the entire life of an order). Moreover it also changes with each refill of an Iceberg order or each restatement of transfer orders (e.g. orders with time in force set to Good Till Cancel or Good Till Date).

Gateway provides MDEntryID for an order in FIX Execution Report (8) message only. PublicOrderID can be used to relate submitted orders to Market Data, however, orders are modified or cancelled only using the OrderID (37) or ClOrdID (11) fields. MDEntryID is a conditionally required field, when an order is not published in Market Data it is not assigned to an order. STOP orders before activation, orders with TimeInForce (59)= 3 (Immediate Or Cancel) or 4 (Fill Or Kill) which were fully cancelled are not published in Market Data and MDEntryID is not present in Execution Report.

4.4.1.4. OrigClOrdID (41)

OrigClOrdID is used to point order by referring to present ClOrdID of an order.

OrigClOrdID format is identical to ClOrdID - it is alphanumeric ASCII free-format string, which can include any character except a control character.

OrigClOrdID can be used by an exchange member (interchangeably with OrderID) when requesting an order modification (Order Cancel/Replace Request (G) message) or order cancellation (Order Cancel Request (F) message) to identify the relevant order which rests in Trading System. If an exchange member provides both OrderID and OrigClOrdID to modify or cancel, Trading System will process the OrderID only and ignore OrigClOrdID.

Figure 2. – Order identifiers usage an example



4.4.2. ORDER TYPES

Exchange members may submit the following Order Types in the New Order Single (D) message via FIX Order Gateway:

- Limit
- Market
- Market To Limit
- Iceberg (as Limit order type with DisplayQty)
- Stop Limit
- Stop Loss

4.4.2.1. Limit Order

Limit order is an order to buy or sell at a stipulated limit or better price, thus for buy order at its limit or lower price and for sell order at its limit or higher price.

Limit order can be executed with one or more counterpart resting orders until it is fully filled.

An unexecuted remainder of a limit order is added to the order book or is cancelled depending on TimeInForce (59).

Relevant FIX tags:

OrdType (40) = 2 (Limit)

Price (44) = Limit Price

4.4.2.2. Market Order

Market order is an order to buy or sell without a specified price. Market order is executed at the best opposing prices with one or more counterpart resting orders.

Unexecuted remainder of a market order is immediately cancelled.

During Continuous Trading market phase, market orders can only be submitted with IOC, FOK, VFA, VFC validity attributes, i.e. TimeInForce (59) = 2, 3, 4, 7.

During Auction market phase, market orders can only be submitted with VFA and VFC validity attributes, i.e. TimeInForce (59) = 2, 7.

Price (44) is not present for Market order type.

Relevant FIX tags:

OrdType (40) = 1 (Market)

TimeInForce (59) = 2 (At The Opening), or 3 (Immediate Or Cancel), or 4 (Fill Or Kill), or 7 (At The Closing)

4.4.2.3. Market To Limit

Market to limit order is an order to buy or sell without a specified price. Market to limit order is executed at the best opposing price level with one or more counterpart resting orders. Market to limit order cannot be executed on many price levels.

Unexecuted remainder of a market to limit order is immediately cancelled.

During Continuous Trading market phase, market orders can only be submitted with IOC, FOK, VFA, VFC validity attributes, i.e. TimeInForce (59) = 2, 3, 4, 7.

During Auction market phase, market orders can only be submitted with VFA and VFC validity attributes, i.e. TimeInForce (59) = 2, 7.

Price (44) is not present for Market to limit order type.

Relevant FIX tags:

OrdType (40) = K (Market to limit)

TimeInForce (59) = 2 (At The Opening), or 3 (Immediate Or Cancel), or 4 (Fill Or Kill), or 7 (At The Closing)

4.4.2.4. Iceberg

Iceberg order is a limit order, where only part of the total quantity is disclosed to the market via Market Data. The rest of the order is hidden (invisible) to the general market. Exchange member submitting an iceberg order must additionally specify the quantity that will be displayed initially, i.e. DisplayQty (1138), which:

- must be lower than the total order quantity,
- must be greater or equal to the minimum display quantity set in Trading System configuration,
- may be randomized by Trading System within a specified range.

Once the displayed quantity is completely filled, a new portion of the order is disclosed to the market according to the displayQty (1138) requested at the order entry (with possible randomization).

The value of each Iceberg Order (i.e. Price x OrderQty) at the point of entry or during any modification must be equal or greater than the minimum iceberg size set in the system configuration.

Iceberg orders can only be submitted with DAY, GTC, GTD validity attributes, i.e. TimeInForce (59) = 0, 1, 6.

Relevant FIX tags:

OrdType (40) = 2 (Limit)

Price (44) = Limit Price

DisplayQty (1138) = Initially Displayed Quantity

TimeInForce (59) = 0 (Day), or 1 (Good Till Cancel), or 6 (Good Till Date)

4.4.2.5. Stop Limit

Stop limit is an order to buy or sell at a stipulated limit or better price, which is activated and inserted into the order book upon reaching or exceeding (up for buy and down for sell) a predefined price level (called trigger price) by last traded price on the market. Until activation, the stop limit is hidden and cannot interact with the order book. Once activated, the stop limit is processed in the same manner as a limit order with the same validity attribute that the original stop order had prior to activation (nonetheless maintains its original order type = 4).

When submitting or modifying stop limit order, the following conditions must be met:

- for buy order: Price (44) $\geq$ Trigger Price (1102) $>$ Last Traded Price (LTP)
- for sell order: Price (44) $\leq$ Trigger Price (1102) $<$ Last Traded Price (LTP)

Stop limit can only be submitted with DAY, GTC, GTD validity attributes, i.e. TimeInForce (59) = 0, 1, 6.

Trigger price (1102) may be modified for an inactive stop order, but once the order is activated, trigger price can no longer be modified.

Relevant FIX tags:

- OrdType (40) = 4 (Stop Limit)
- Price (44) = Limit Price
- TriggerPrice (1102) = Activation Price
- TimeInForce (59) = 0 (Day), or 1 (Good Till Cancel), or 6 (Good Till Date)
- Triggered (1823) = 0 (Not triggered) or 2 (Stop order activated) - returned in Execution Report (8)
- ExecType (150) = L (Triggered) - returned one time after order is activated

4.4.2.6. Stop Loss

Stop loss is an order to buy or sell without a specified price, which is activated and inserted into the order book upon reaching or exceeding (up for buy and down for sell) a predefined price level (called trigger price) by the last traded price on the market. Until activation, stop loss is hidden and cannot interact with the order book. Once activated, stop loss is processed in the same manner as a market order with immediate or cancel (IOC) validity attribute (nonetheless maintains its original order type = 3).

When submitting or modifying stop loss order, the following conditions must be met:

- for buy order: trigger price (1102) > last traded price (LTP),
- for sell order: trigger price (1102) < last traded price (LTP).

Stop loss can only be submitted with DAY, GTC, GTD validity attributes, i.e. `TimelnForce (59) = 0, 1, 6`.

Trigger price (1102) may be modified for inactive stop order, but once the order is activated, trigger price can no longer be modified.

Relevant FIX tags:

- `OrdType (40) = 3` (Stop Loss)
- `TriggerPrice (1102) = Activation Price`
- `TimelnForce (59) = 0` (Day), or `1` (Good Till Cancel), or `6` (Good Till Date)
- `Triggered (1823) = 0` (Not triggered) or `2` (Stop order activated)-returned in Execution Report (8)
- `ExecType(150) = L` (Triggered) - returned one time after order is activated

4.4.3. ORDER VALIDITIES

Order validity determines an event or period after which an order is expired and is dropped from GPW WATS order books.

An order can be valid in the point in time when it is received by the system (IOC, FOK), until the precise timestamp current day (GTT), until the end of the current day (Day), until the end of specified day in the future (GTD), until it is cancelled (GTC), or only during the Trading Phase (VFA, VFC).

Exchange members may submit orders with the following validity attributes in the New Order Single (D) message via FIX Order Gateway:

- Day (DAY)
- Good Till Cancel (GTC)
- Good Till Date (GTD)
- Good Till Time (GTT)
- Immediate Or Cancel (IOC)
- Fill Or Kill (FOK)
- Valid For Auction (VFA)
- Valid For Closing (VFC)

Order validity is determined by `TimelnForce (59)` tag in conjunction with `ExpireDate (432)` or `ExpireTime (126)` tags.

Not all validity attributes may be used with each Order Type (40) throughout the various market phases.

4.4.3.1. Day (DAY)

Day order is valid only until the end of the current trading day. Unexecuted day orders are expired by Trading System at the end of the trading day during post-session processing (in Trading Closed market phase).

Day validity attribute may be used with each order type, except for market order and market to limit.

Relevant FIX tags:

TimeInForce (59) = 0 (Day)

4.4.3.2. Good Till Cancel (GTC)

Good till cancel order is valid until it is fully executed or canceled by the submitter or market operations (whatever comes first).

Good till cancel validity attribute may be used with each order type, except for market order and market to limit.

Relevant FIX tags:

TimeInForce (59) = 1 (Good Till Cancel)

4.4.3.3. Good Till Date (GTD)

Good till date order is valid until the end of the specified day.

If the expiry date specified for an order is not a trading day, the order becomes expired at the end of the previous trading day.

Good till date validity attribute may be used with each order type, except for Market Order and Market To Limit.

ExpireDate attribute is in a date format (without time component) e. g. "20220102" and can only be used in conjunction with (59) = 6

Relevant FIX tags:

TimeInForce (59) = 6 (Good Till Date)

ExpireDate (432) = Expiration Date

4.4.3.4. Good Till Time (GTT)

Good till time order is valid until the specified time of the current trading day.

If the expiry time specified for an order is later than the end of the last market phase, the order will become expired at the end of the current trading day.

Good till time validity attribute cannot be used with the future Expire Date (432).

Good till time validity attributes may be used with each order type, except for market order and market to limit.

Relevant FIX tags:

TimeInForce (59) = 6 (Good Till Date)

ExpireTime (126) = Expiration Time (UTC timestamp with current date)

4.4.3.5. Immediate Or Cancel (IOC)

Immediate or cancel order must be executed immediately, in full or partially, upon order entry. IOC orders may be matched against one or more opposing orders with the price limits within trade price collars. Unexecuted part of IOC order is automatically canceled by Trading System.

Immediate or cancel orders may be submitted during Continuous Trading market phase only.

Immediate or cancel validity attribute may be used with each order type, except for stop limit, stop loss and iceberg.

Relevant FIX tags:

TimeInForce (59) = 3 (immediate or cancel)

4.4.3.6. Fill Or Kill (FOK)

Fill or kill order has to be executed in full, immediately upon order entry. FOK orders may be matched against one or more opposing orders with the price limits within trade price collars. Unexecuted FOK order is automatically canceled by Trading System.

Fill or kill orders may be submitted during Continuous Trading market phase only.

Fill or kill validity attribute may be used with each order type, except for stop limit, stop loss and iceberg.

Relevant FIX tags:

TimeInForce (59) = 4 (Fill Or Kill)

4.4.3.7. Valid For Auction (VFA)

Valid for auction order is activated and inserted into the order book at the beginning of the next Auction market phase (whether scheduled or unscheduled). VFA order submitted in Continuous Trading market phase, is hidden until activation (i.e. it is not published via Market Data) and cannot interact with the order book. Once activated, VFA order is valid until the end of the Auction and becomes expired by Trading System immediately after the Uncrossing (if left unexecuted). VFA orders are valid for the current trading day only. Activated VFA orders retain their original priority timestamp from the point of submission.

Valid For Auction validity attribute may be used with each Order Type, except for Stop Limit, Stop Loss and Iceberg.

Relevant FIX tags:

TimeInForce (59) = 2 (At The Opening)

4.4.3.8. Valid For Closing (VFC)

Valid For Closing order is activated and only inserted into the order book at the beginning of the Closing Auction market phase, when Call sub-phase begins. VFC order submitted in Continuous Trading market phase or Auction other than Closing, is hidden until activation (i.e. it is not published via Market Data) and cannot interact with the order book. Once activated, VFC order is valid until the end of the Closing Auction and becomes expired by Trading System immediately after the Uncrossing (if left unexecuted). VFC orders are valid for the current trading day only. Activated VFC orders retain their original priority timestamp from the point of submission.

Valid For Closing validity attribute may be used with each Order Type, except for Stop Limit, Stop Loss and Iceberg.

Relevant FIX tags:

TimeInForce (59) = 7 (At The Closing)

4.4.4. ORDER CAPACITY (528)

OrderCapacity (528) defines the capacity of the firm that is submitting an order. These are the possible values:

- P - Principal mapped as DEAL (Dealing on own account)
- A - Agency mapped as AOTC (Any other capacity)
- R - Riskless Principal mapped as MTCH (Matched principal)

It is required to set the correct capacity field value by the exchange member.

4.4.5. MARKET MAKING

The liquidity provision activity is provided by the regular market makers under the terms of market making scheme agreement concluded with the Market and by the Issuer Market Maker agreements concluded with the Issuer of the instrument which is being traded. Regular Market Makers who submit their orders must set optional field OrderRestrictions (529) = 5 (Acting as Market Maker or Specialist).

According to MiFID2/MiFIR Regulations all orders submitted to the Market as part of the liquidity provision activity must be assigned with the "Liquidity Provision Activity" flag.

The liquidity provision activity can be performed by members of the Market and non-member firms which is differentiated by setting the OrderCapacity (528) value to "P" (Principal) and "A" (Agency) respectively. In addition

The table below provides a summary of the different combinations for the liquidity provision activity with FIX fields:

FIX field name(tag)	Regular market maker		Issuer market maker	
	Member	Non-member	Member	Non-member
OrderCapacity(528)	P = Principal	A = Agency	P = Principal	A = Agency
OrderAttributeType(2594)	2 = Liquidity provision activity order	2 = Liquidity provision activity order	2 = Liquidity provision activity order	2 = Liquidity provision activity order
OrderAttributeValue(2595)	Y	Y	Y	Y
PartyID - client short code (448)	NOT SET	SET	NOT SET	SET
OrderRestrictions(529)	5 = Acting as Market Maker or Specialist in the security	5 = Acting as Market Maker or Specialist in the security	NOT SET	NOT SET

4.4.6. ORDER STATUSES

During the order lifecycle FIX Order Gateway sends Execution Report (8) each time something happens to an order. There are two attributes, one OrdStatus (39) which reflect current order status and another ExecType (150) which refers to an event which has triggered Execution Report generation.

Order status can be unchanged across many Execution Reports, once ExecType reflects the last action performed on an order.

Information about the events on an order and its status is conveyed in FIX Execution Report (8) messages in the following tags:

4.4.6.1. OrdStatus (39)

Exchange member may receive from the FIX Order Gateway the following statuses for orders which have been submitted:

- New (0) - when an order is accepted into the order book until there is any execution on it (trade) or until an order is cancelled or expired. In case aggressive order was partially or fully filled there is no New status assigned to order. To help exchange member identify such a situation, new field ExecTypeReason(2431) = 110 (First trade on aggressive order) had been added to Execution Report.
- Partially Filled (1) - when there was at least one execution on an order until the order is fully filled (there is no remaining quantity on it) or until the order is cancelled or expired
- Filled (2) - when an order is fully executed (there is no remaining quantity on an order)
- Canceled (4) - when an order is canceled by an exchange member, Trading System itself (including situation when an incoming IOC/FOK order cannot be fully executed upon arrival) or by the Market Operations. Additional information about the reason for unsolicited cancellation is provided in the ExecTypeReason (2431) tag
- Rejected (8) - when an order is rejected by Trading System upon entry due to various business validations
- Expired (C) - when an order is automatically removed from the order book due to its validity condition specified in TimeInForce (59) tag (including removal of unexecuted part of VFA/VFC order after the Auction)

4.4.6.2. ExecType (150)

Exchange member may receive the following types of Execution Types from the FIX Order Gateway:

- New (0) - returned as a response to New Order Single when an order was submitted into book and not executed
- Canceled (4) - returned as a response to Order Cancel(35=F) when an order was successfully cancelled by Client, Trading System itself (including situation when an incoming IOC/FOK order cannot be fully executed upon arrival) or by market operations. Additional information about the reason for unsolicited cancellation is provided in the ExecTypeReason (2431) tag
- Replaced (5) - returned as a response to successful Order Cancel Replace(35=G) request
- Rejected (8) - returned as a response to New Order Single (35=D) or Order Cancel Replace(35=G) or Order Cancel (35=F) when there was an issue with message processing.
- Expired (C) - when an order is automatically removed from the order book due to its validity condition specified in TimeInForce (59) tag (including removal of unexecuted part of VFA/VFC order after the Auction)
- Restated (D) - returned when a new portion of an Iceberg order is disclosed to the market or at the beginning of a new trading day for orders submitted in the past but still valid (not cancelled or fully filled orders with validities such as GTC, GTD)

- Trade (F) - returned as a response to Partial or Full trade on order
- Trade Cancel (H)
- Triggered (L) - returned when order with additional activation condition is activated such as valid for auction / valid for closing orders (TimeInForce = 2 or 7) and stop loss / stop limit orders (OrdType = 3 or 4)

4.4.6.3. ExecTypeReason (2431)

ExecTypeReason (2431) provides additional information in the following situations:

- Order Restatement (100 - GT order restatement),
- Unsolicited Order Cancellation (ExecType = 4),
- Partial or full trade (ExecType = F).

ExecTypeReason (2431) can take following values:

- 100 - GT order restatement
- 101 - Iceberg order refill
- 102 - Cancel IOC/FOK order
- 103 - Cancel by STP
- 104 - Cancel by Market Operations
- 105 - Cancel on Trading Halt
- 106 - Cancel on Disconnect
- 107 - Cancel by Corporate Action
- 110 - First trade on aggressive order

4.4.6.4. Triggered (1823)

Triggered is an optional tag which is provided in Execution Report (8) messages, only relating to valid for auction / valid for closing orders (TimeInForce = 2 or 7) and stop loss / stop limit orders (OrdType = 3 or 4), and informs whether the order is already activated or not (at the point of Execution Report dispatch).

Triggered tag can take the following values:

- not triggered (0)
- triggered (1)
- stop order triggered (2)

4.4.7. CANCELLATION

Each submitted by New Order Add request order can be terminated by OrderCancel (35=F) request.

Only active orders (with Status (39) such as New, Partially Filled) can be cancelled. If an order was already Filled, Cancelled, Expired or Rejected it cannot be cancelled since GPW WATS treats such order as non-existent. An order can be referenced by one of two attributes:

OrderID (37) which is system unique ID which remains unchanged during the order lifecycle

origClOrdID which is Client unique order reference. Since it can be changed during the order lifecycle when an order is modified by Participant, it should always be last clOrdID.

If an exchange member provides both orderID and origClOrdID in the OrderCancel(35=F) request, Trading System will process the OrderID only and ignore origClOrdID.

4.4.8. MODIFICATION

Each submitted New Order Add request order can be terminated by OrderCancelReplace (35=G) request.

Only active orders (with Status (39) such as New, Partially Filled) can be modified. If an order was already Filled, Cancelled, Expired or Rejected it cannot be cancelled since GPW WATS treats such order as non-existent. An order can be referenced by one of two attributes:

- OrderID (37) which is a system unique ID remains unchanged during the order lifecycle
- origClOrdID which is Client unique order reference. Since it can be changed during the order lifecycle when an order is modified by Participant, it should always be last clOrdID.

If an exchange member provides both orderID and origClOrdID in the OrderCancelReplace (35=G), Trading System will process the OrderID only and ignore origClOrdID.

4.4.9. ACCOUNT STRUCTURE

There are two FIX conditional fields that can be used by Trading Member for clearing purposes: AccountType (581) and Account(1).

There are two possible types of accounts (i.e. AccountType (581) field):

1. Customer - account is carried on customer side of the books
2. House - house trader

4.4.10. ORDER BOOK RESTATEMENT

At the beginning of each trading day, exchange members receive Execution Report messages from Gateway informing them about orders that were submitted in the past but still valid (not cancelled, not fully filled orders with validities such as GTC, GTD, not expired).

More information can be found in the document D01.1 GPW WATS Trading System, chapter Trading Schedules.

Restated orders remain almost all attribute values from previous session such as OrderID (11), ClOrdID (37), OrdStatus(39), LeavesQty (151), CumQty (14). One exception is publicOrderID which is newly assigned returned in MDEntryID (278) in Execution Report.

Restated orders are marked in ExecutionReport with ExecType=D.

Relevant FIX tags:

ExecType (150) = D (Restated)

4.4.11. SELF-TRADE PREVENTION

The Self-Trade Prevention (STP) functionality enables the exchange members to avoid an unintentional execution of orders at the submitted by the same exchange member. The exchange member shall provide

the STP ID value at the order entry level to enable the trading platform to activate the STP functionality. There is no need for any additional configuration for the exchange members in the trading system. In FIX protocol in the New Order Single (D) and in the Execution Report (8) messages, an optional field SelfMatchPreventionID carrying the STP ID key, will be used. Self-trade The exchange will not dictate or validate the value of the STP ID as long as it is compliant with the format of the field. Allocation of the STP ID values is at the sole discretion of the exchange member.

Self-Trade Prevention mechanism works in scope of assigned SelfMatchPreventionID value.

In case incoming aggressive order could be executed with already present (passive) order System cancels passive order and exchange member receives the Execution Report with ExecTypeReason (2431) set to 107 - Cancel by Corporate Action and OrdStatus(39)=4.

Relevant FIX tag

SelfMatchPreventionID (2362)- numeric values

ExecTypeReason (2431) = 107 - Cancel by Corporate Action.

4.4.12. UNSOLICITED CANCELLATIONS

Orders can be cancelled by the Client or unsolicited by Market Operations or by System.

When unsolicited cancellation takes place, an exchange member receives the Execution Report with ExecTypeReason (2431) set to one of following values:

- Cancel on Trading Halt (105)
- Cancel on Connection Loss (106)
- Cancel by Market Operations (104)
- Cancel by STP (103)

4.5. TRADE CAPTURE REPORTS

The Trade Capture Reports (AE) are messages sent through FIX Order Gateway and through Post Trade Gateway. The Trade Capture Reports sent through FIX Order Entry Gateway are used to report Block Trades and Cross Trades to Market Operator for regulatory and post-trade transparency reasons. The Trade Capture Reports sent through FIX Post-trading Gateway are used for downstream processing to relay confirmed trades to parties involved in clearing and settling trades. The Trade Capture Reports sent through FIX Post-trading Gateway are not within the scope of this document.

4.5.1. TRADE CAPTURE REPORT IDENTIFIERS

4.5.1.1. Trade Report ID

The TradeReportID (tag 571) is assigned by the submitter of the Trade Capture Report message and is used as a pure message identifier. In the one-sided TCR in the applied model "one-party for pass through" the initiator and the counterparty assign their own TradeReportID independently of each other. TradeReportID must be unique across the trading day for each submitter separately. TradeReportID is assigned separately by Market Operator in outbound messages where the Trade Capture Report is generated e.g. as the notification sent to the counterparty or as a trade confirmation sent to the parties to the trade.

4.5.1.2. Secondary Trade Report ID

The SecondaryTradeReportID (818) is a unique identifier in a numeric format of a first leg to the trade (Trade Capture Report with tradeReportTransType: New).

GPW WATS provides newly generated SecondaryTradeReportID in TradeCaptureReportAck returned to exchange member each time a new first leg to the trade is submitted.

Gateway provides the same SecondaryTradeReportID in Trade Capture Report to notify counterparty of first leg to trade (tradeReportType: Alleged).

Counterparty uses the same SecondaryTradeReportID to accept the trade (initiated by first leg). Subsequently Gateway provides the same SecondaryTradeReportID in Trade Capture Report to notify both parties (initiator and counterparty) that trade has been finalized.

SecondaryTradeReportID can also be used as a unique reference to cancel a first leg to the trade (tradeReportTransType: Cancel).

4.5.1.3. Trade Report Reference ID

The TradeReportRefID (572) is assigned by the submitter when the Trade Capture Report is referred to the previous message. This field is filled in when e.g. Market Operator generates Trade Capture Report as the trade confirmation to the parties or when the Trade Capture Report is cancelled by the submitter.

4.5.2. BLOCK TRADES

Block trades are a form of pre-arranged trades brought onto Market Operator. Block trades must comply with the market regulation in terms of price and value. The value of the Block trade shall be not be lower than Large in Scale Value established in MiFID2/MiFIR regulations. In Trade Capture Report messages the Block trades are defined in the field "TrdType" (tag 828) with value set at "38" = Block Trade. Block trades are reported in two ways:

- Dual-sided Trade Capture Report
- One-sided Trade Capture Report

4.5.2.1. Dual-Sided Trade Capture Report

Dual-sided Trade Capture Report is applied when both beneficiary parties to the trade belong to the reporting firm (exchange member). In the Trade Capture Report message the value of the field TradeHandlingInstr (1123) is set at 1 = Two-party Report. The Trade Capture Report is accepted by GPW WATS Gateway by sending back Trade Capture Report Acknowledgment (TCR AR) with the field TrdRptStatus (939) = 0 (Accepted) followed by the Trade Capture Report which confirms the trade with the field MatchStatus = 0 (Matched). If Trade Capture Report is rejected, then the exchange member receives the Trade Capture Report Acknowledgment with TrdRptStatus (939) = 1 (Rejected) with the respective information about the reason for rejection provided in the field TradeReportRejectReason (751).

The table below provides dual-sided Trade Capture Report messages flow covering different scenarios which include:

- exchange member's TCR rejection by GPW WATS Gateway,
- exchange member's TCR accepted by GPW WATS Gateway.

Operation	IN / OUT	Message	TradeReportID (TradeReportRefID)	TradeID	TradeReport-TransType	TradeReport-Type	TrdType	TradeHandling-Instr	ExecType	TrdRptStatus	MatchStatus
			Fix Tag 571 (Fix Tag 572)	Fix Tag 1003	Fix Tag 487	Fix Tag 856	Fix Tag 828	Fix Tag 1123	Fix Tag 150	Fix Tag 939	Fix Tag 573
Exchange member submits dual-sided TCR	IN	TCR (AE)	Exchange member assigned	n/a	0 = New	0 = Submit	38 = Block Trade	1 = Two-party report	n/a	n/a	n/a
Dual-sided TCR accepted by TV	OUT	TCR Ack (AR)	Exchange member assigned (echoed back)	n/a	0 = New	0 = Submit	38 = Block Trade	1 = Two-party report	n/a	0 = Accepted	n/a
Dual-sided TCR rejected by TV	OUT	TCR Ack (AR)	Exchange member assigned (echoed back)	n/a	0 = New	0 = Submit	38 = Block Trade	1 = Two-party report	n/a	1 = Rejected	n/a
Trade	OUT	TCR (AE)	TV assigned	TV assigned	2 = Replace	0 = Submit	38 = Block Trade	1 = Two-party report	F = Trade	n/a	0 = Matched

4.5.2.2. One-Sided Trade Capture Report

Dual-sided Trade Capture Report is applied when the beneficiary parties to the trade belong to different exchange members. The applied one-sided model is the "one-party report for pass-through". The initiator submits a report (first leg to the trade) which is accepted by the counterparty to the trade. In the Trade Capture Report message the value of the field TradeHandlingInstr (1123) shall be set at 3 = one-party report for pass-through. Below is shown the basic TCR flow enabling exchange members to conclude the block trade.

The initiator submits the first leg to the trade by sending Trade Capture Report with the values:

- TradeReportTransType (487) = 0 (New)
- TradeReportType (856) = 0 (Submit)
- The first leg is accepted by WATS System by sending Trade Capture Report Ack (TCR AR) with the TrdRptStatus (939) = 0 (Accepted)
- WATS System forwards the Trade Capture Report to the counterparty by sending Trade Capture Report (notification) with the values:
 - TradeReportTransType (487) = 0 (New)
 - TradeReportType (856) = 1 (Alleged)
- The counterparty accepts the first leg by sending Trade Capture Report with the values:
 - TradeReportTransType (487) = 2 (Replace)
 - TradeReportType (856) = 2 (Accept)
- WATS System sends the Trade Capture Reports informing the initiator and the counterparty about the concluded trade with the values:
 - TradeHandlingInstr (1123) = 0 (Trade Confirmation)
 - MatchStatus (573) = 0 (Matched)

The table below provides one-sided Trade Capture Report messages flow covering different scenarios which include:

- Exchange member's first leg rejection by GPW WATS,
- First leg cancellation by the initiator before accepting by the counterparty,
- Counterparty declines the initiator's first leg,
- Counterparty's second leg rejection by GPW WATS.

Operation	IN / OUT	Message	TradeReportID (TradeReportRefID)	TradeID	TradeReport-TransType	TradeReportType	TrdType	TradeHandling-Instr	ExecType	TrdRptStatus for TCR Ack only	MatchStatus
			Fix Tag 571 (Fix Tag 572)	Fix Tag 1003	Fix Tag 487	Fix Tag 856	Fix Tag 828	Fix Tag 1123	Fix Tag 150	FIX Tag 939	Fix Tag 573
exchange member first leg accepted or rejected by TV											
exchange member submits first leg	IN	TCR (AE)	exchange member assigned	n/a	0 = New	0 = Submit	38 = Block Trade	3 = One-party	n/a	n/a	n/a
First leg rejected by TV	OUT	TCR Ack (AR)	exchange member assigned (echoed back)	n/a	0 = New	0 = Submit	38 = Block Trade	3 = One-party	n/a	1 = Rejected	n/a
First leg accepted by TV	OUT	TCR Ack (AR)	exchange member assigned (echoed back)	n/a	0 = New	0 = Submit	38 = Block Trade	3 = One-party	n/a	0 = Accepted	n/a
TV forwards TCR to C-pty (notification)	OUT	TCR (AE)	TV assigned	n/a	0 = New	1 = Alleged	38 = Block Trade	3 = One-party	n/a	n/a	n/a
exchange member cancels first leg before accepting by C-pty	IN	TCR (AE)	exchange member assigned (exchange member's previous)	n/a	1 = Cancel	6 = Trade Report Cancel	38 = Block Trade	3 = One-party	n/a	n/a	n/a
First leg cancellation rejected by TV	OUT	TCR Ack (AR)	exchange member assigned (exchange member's previous)	n/a	1 = Cancel	6 = Trade Report Cancel	38 = Block Trade	3 = One-party	n/a	1 = Rejected	n/a
First leg cancellation accepted by TV	OUT	TCR Ack (AR)	exchange member assigned (exchange member's previous)	n/a	1 = Cancel	6 = Trade Report Cancel	38 = Block Trade	3 = One-party	n/a	0 = Accepted	n/a
TV forwards cancelled first leg TCR to C-pty	OUT	TCR (AE)	TV assigned (571) TV previous (572)	n/a	1 = Cancel	6 = Trade Report Cancel	38 = Block Trade	3 = One-party	n/a	n/a	n/a
Counterparty declines exchange member's first leg											
C-pty declines exchange	IN	TCR (AE)	C-pty assigned	n/a	2 = Replace	3 = Decline	38 = Block Trade	3 = One-party	n/a	n/a	n/a

Operation	IN / OUT	Message	TradeReportID (TradeReportRefID)	TradeID	TradeReport-TransType	TradeReportType	TrdType	TradeHandling-Instr	ExecType	TrdRptStatus for TCR Ack only	MatchStatus
member first leg											
C-pty decline is rejected by TV	OUT	TCR Ack (AR)	C-pty assigned(echoed back)	n/a	2 = Replace	3 = Decline	38 = Block Trade	3 = One-party	n/a	1 = Rejected	n/a
C-pty decline accepted by TV	OUT	TCR Ack (AR)	C-pty assigned(echoed back)	n/a	2 = Replace	3 = Decline	38 = Block Trade	3 = One-party	n/a	0 = Accepted	n/a
TV submits TCR to exchange member (informing about C-pty decline)	OUT	TCR (AE)	TV assigned	n/a	2 = Replace	3 = Decline	38 = Block Trade	3 = One-party	n/a	n/a	n/a
Counterparty accepts exchange member's first leg											
C-pty accepts exchange member first leg	IN	TCR (AE)	C-pty assigned	n/a	2 = Replace	2 = Accept	38 = Block Trade	3 = One-party	n/a	n/a	n/a
C-pty accept is rejected by TV	OUT	TCR Ack (AR)	C-pty assigned(echoed back)	n/a	2 = Replace	2 = Accept	38 = Block Trade	3 = One-party	n/a	1 = Rejected	n/a
C-pty accept is accepted by TV	OUT	TCR Ack (AR)	C-pty assigned(echoed back)	n/a	2 = Replace	2 = Accept	38 = Block Trade	3 = One-party	n/a	0 = Accepted	n/a
TV match info to exchange member	OUT	TCR (AE)	TV assigned Ref to first leg	TV assigned	2 = Replace	0 = Submit	38 = Block Trade	0 = Trade Confirm	F = Trade	n/a	0 = Matched
TV match info to C-pty	OUT	TCR (AE)	TV assigned Ref to second leg	TV assigned	2 = Replace	0 = Submit	38 = Block Trade	0 = Trade Confirm	F = Trade	n/a	0 = Matched

4.5.3. CROSS TRADES

Cross Trades are privately negotiated trades brought onto Market Operator. Cross Trades must comply with Market Operator rules in terms of price and value. Cross Trades are reported by using dual-sided Trade Capture Report. Cross Trades in the Trade Capture Report messages are defined in the field TrdType (tag 828) with the value set at "22" (Privately negotiated trade).

4.6. QUOTE HANDLING

Market makers send quotes into a market using a MassQuote message. For a given instrument, it is possible to send one quotation. The MassQuote message enables the sending of up to 30 quotations.

The trading platform sends a confirmation of the accepted and rejected quotations by transmitting a MassQuoteResponse MassQuoteAck message.

Quotations rules:

- Two-sided quotes are allowed,
- Only one market maker quotation is allowed per instrument,
- MassQuoteResponse message includes all submitted quotations along with their acceptance status (e.g. Accepted, Rejected),
- Modification of submitted quotations involves placing another MassQuote message with changed values for price or quantity,
- Cancellation of submitted quotations involves placing another MassQuote message with the quantity values set to 0.

4.7. CANCEL ON DISCONNECT

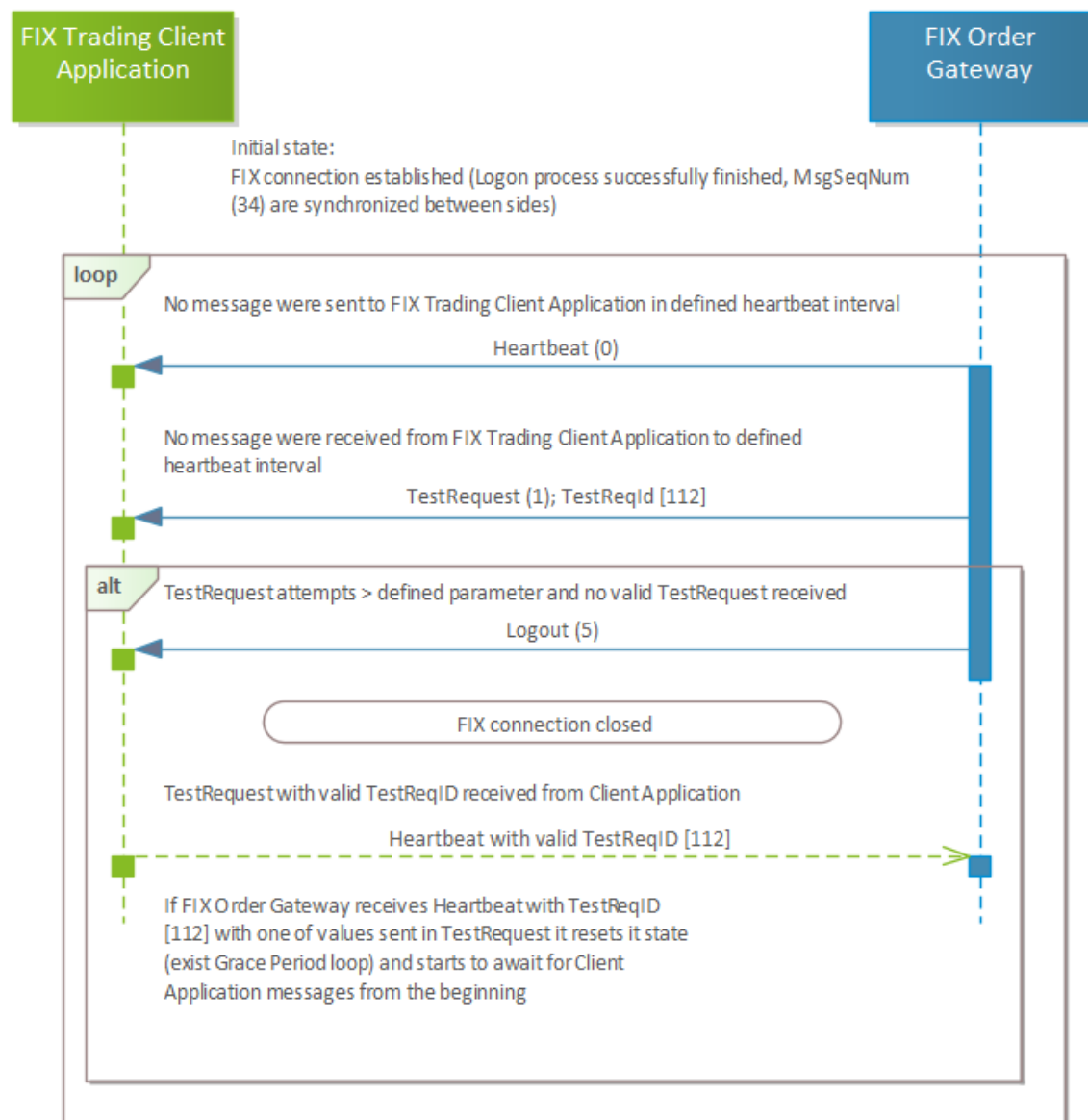
Cancel on disconnect is a mechanism designed to protect exchange member's from potential losses which can happen when they unintentionally lost control over their orders. CoD is setup on connection level not on Participant level. If the user has more than one connection, then the mechanism can be activated on each of these connections or only on chosen connections. The GPW is responsible for the configuration of the service and performs it at the exchange member request. When it is switched on and FIX Gateway discovers member's unintentional unresponsiveness it cancels all open orders which belong to this Participant. As soon as client application reestablishes FIX connection with FIX Gateway it receives Execution Reports with information of unsolicited cancellations (ExecType (150)=4 - Canceled, OrdStatus (39) = 4 - Canceled, ExecTypeReason (2431) = 106 - Cancel on Disconnect. CoD mechanism cancels orders with the following TIFs: DAY, GTT, VFA and VFC. Orders with following TIFs: IOC, FOK, GTC, GTD are excluded from cancellation.

Gateway assess that Client Application lost control in two situations:

- Connection with the Client Application was dropped without Logout procedure, but not from GPW WATS initiative. When Client Application performed Logout successfully or GPW WATS failover Cancel on Disconnect is not activated.
- Physical connection is active but Participant does not send Heartbeat messages for a period of 30 seconds (so-called grace period)

When Gateway detects no message (including heartbeat) from Client Application for a period longer than heartbeat interval it sends Test Request and starts Grace Period process (figure below). GPW WATS sends Test Requests (1) up to 3 times once every 30 seconds and if Client Application does not respond with Heartbeat (0) message Gateway sends Logout and terminates FIX connection. When responding to Test message, Client Application should include TestReqID (112) field with value received in the last received Test Request from GPW WATS.

Figure 3. – Grace period mechanism



4.8. REJECT MESSAGES

After each message is received GPW WATS performs a number of checks in a specific order. This is crucial to report the most basic issues at first, and limit processing steps for broken messages. Depends on detected issue FIX Gateway can:

- Immediately close connection without any Logout

This happens when BeginString (8), SenderCompID (49), TargetCompID (56) in Logon message is invalid or system receives Logon for SenderCompID (49) with an already open FIX connection.

- Ignore message without any response (system treats such message as it was never received). System treats message as garbled when:
 - BeginString (8), BodyLength (9), MsgType (35) are not the first three tags in received message, Checksum (10) is not the last tag in message
 - At least one of tags: BeginString (8), BodyLength (9), Checksum (10) has invalid value

In such a case (Gateway ignores incoming message) when Gateway receives next valid message with incremented MsgSeqNum (34) it detects a gap and requests exchange member to resend missing messages. by sending ResendRequest (2). To avoid resent infinity loop after 3 attempts if FIX Gateway still detects gap in MsgSeqNum (34) it sends Logout message and close the connection.

- Session level reject (35=3) message

Session level reject message consists of RefSeqNum (45) with MsgSeqNum of rejected message, SessionRejectReason (373) with code which identifies detailed reject reason.

Reject (35=3) message is returned by GPW WATS in case:

- incoming message type is invalid (tag 35) → SessionRejectReason (373) = 11
 - message does not contain mandatory, properly formatted field → SessionRejectReason (373) = 1
 - message contain invalid, undefined or undefined for this message type tag → SessionRejectReason (373) = 0,3,2
 - message contains tag without value, value is incorrect or provided value format is incorrect → SessionRejectReason (373) = 4, 5,6
 - message contains tag which appears more than once → SessionRejectReason (373) = 13
 - message contains tags in repeating group in invalid order or number of groups is invalid → SessionRejectReason (373) = 15, 16
 - SendingTime (52) in message is out of acceptable tolerance (either from the future or in the past) → SessionRejectReason (373) = 10
 - SenderCompID (49) for message different than Logon is different than expected → SessionRejectReason (373) = 9
- Reject message on Business level reject (35 = j)

If validation on business level detects an error condition then a rejection is returned.

5. CONNECTIVITY

This chapter provides information regarding: logon authentication, failover and recovery.

5.1. SESSION IDENTIFICATION

Establishing a FIX session with GPW WATS requires logging into System. To do this, it is necessary to have a token associated with a unique SenderCompID (49). The token and SenderCompID (49) are fields of the Logon(A) message. Both parameters are obtained from GPW. The token value should be provided in FIX field RawData (96) in Logon message.. SenderCompID (49) with assigned value should be included in each FIX message as it is a part of FIX standard header.

The limit of failed login attempts is set to 5. If Participant exceeds the number of login attempts, Gateway blocks the account and the Logon message in will be set to AccountLocked. In the situation of a failed login attempt, the exchange member is advised to contact the GPW service desk for further assistance.

Client can connect and disconnect with FIX Gateway in sequence many times but no more frequently than once every 3 seconds. it's not allowed to establish multiple FIX connections simultaneously on the same SenderCompID (49). After successful Logon (A) any other FIX connection attempts with Logon message with the same SenderCompID (49) causes disconnection.

5.2. FAILOVER AND RECOVERY

To reduce operational risk within GPW WATS, the following failover and recovery mechanism has been designed.

When there is temporary and short unavailability of Gateway, the exchange member should try to re-establish the connection and login to Gateway. Then the messages will be resynchronized between the exchange member and Gateway based on MsgSeqNum (34) attribute included in the Logon and Logon Acknowledge messages. Paragraph 7.1.1.3 contains a detailed description of message synchronization after successful Logon.

When there is a long-term unavailability of the primary Gateway, the procedure to switch to a secondary will be launched. In such a situation, the exchange member will receive information from GPW concerning the need to switch to a secondary Gateway. The connection will be established with the same login parameters but on a new IP address. After logging in, business messages will be synchronized in the same manner as in the first case.

5.3. MESSAGE THROTTLING

The throttling mechanism protects Gateway against overloading with an excessive number of input messages.

There are 2 limits set in Gateway:

1. business throttling - after exceeding the business limit level, next messages are rejected using Reject (j) messages with BusinessRejectReason (380) = 8 (Throttle limit exceeded).
2. technical throttling - after exceeding the technical limit level FIX Gateway sends Reject (j) message with BusinessRejectReason (380) = 9 (Throttle limit exceeded, session is disconnected) and then

sends Logout (5) message with error message Text (58) = "Anti-flooding penalty" and connection is closed.

Limit calculations take into account all messages, however, messages Logout (35=5) and OrderCancel (35=F) aren't rejected.

The MassQuote messages are treated differently. When calculating limits for MassQuote, each side of each quotation is counted as a separate message. For example, a MassQuote message containing 8 quotations is counted as 16 messages.

The above limits are calculated over a period of time configurable in Gateway (i.e., measurement period).

- The throttling parameters are configurable at Gateway service as follows:
- The business limit level is configured for each connection and depends on the level of service provision agreed upon between System Operator and the exchange member. The measurement period - last 5 seconds.
- The technical limit level is configured as an internal Gateway parameter and is set to 5000 messages per second. The measurement period - last 100 milliseconds.

In addition after exceeding the technical limit level, it is not possible to log into System for 10 seconds.

Note: the measurement period means the period of time during which messages are counted and the limits, which have possibly been exceeded, are determined.

5.4. CONNECTIVITY POLICY

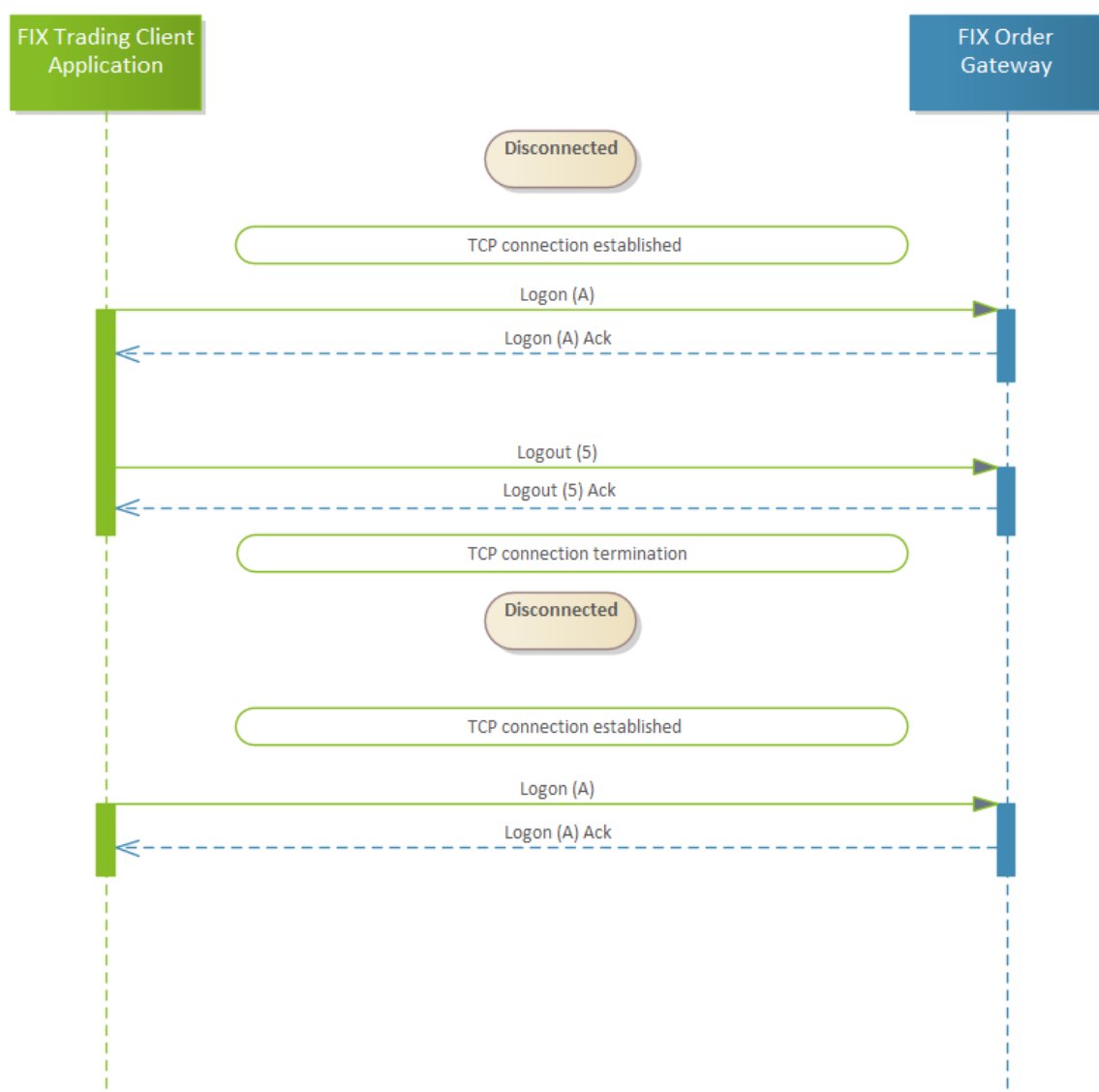
Connectivity policy is described in the document **GPW WATS 6.01 CONNECTIVITY**.

6. SESSION LAYER

This chapter aims to elaborate FIX session layer implemented in GPW WATS.

A FIX session is a bi-directional stream of sequenced messages between two parties of which one establishes the connection and initiates the session (exchange member) and another accepts the connection (GPW WATS FIX Gateway). Each party sends messages identified by a unique (across FIX session) sequence number. A single FIX session can exist across multiple sequential (not concurrent) FIX connections, thus peers may intentionally or unintentionally connect and disconnect numerous times while maintaining a single FIX session.

Figure 4. - FIX session with multiple connections



A FIX connection consists of three parts: logon process, message exchange (inclusive of resynchronization of state), and logout process over a transport layer connection. A FIX connection may be concluded by either a logout process or the unrecoverable loss of the transport layer, a system failure, or an application failure.

Each FIX session exists in boundaries of one day. It starts on first Logon and ends in the end of trading day.

6.1. ESTABLISHING A FIX CONNECTION

Establishing a FIX Connection consists of three subsequent stages:

1. Exchange member establishes a TCP connection with FIX Order Gateway. There is limited time since TCP connection is established to perform the second step where the exchange member sends Logon. If Logon is not sent in 5 seconds then FIX Gateway drops the TCP connection.
2. Exchange member sends Logon (35=A) message, which should always be the first message transmitted when new FIX connection is established. FIX Gateway authenticates and when authentication is successful it accepts connection and sends a Logon Acknowledgement message to exchange member. Exchange member should not transmit any messages until Logon Acknowledgement has been received and processed by exchange member.

Logon message should include HeartBtInt (108) set to 30 representing heartbeat interval.

If authentication fails, FIX Gateway (depending on cause) either sends Logout with information describing the reason for failure and drops connection or drops connection without any message in return. In the majority of cases FIX Gateway sends Logout (35=5) message prior to the connection being terminated and provides a descriptive reason of Logon process failure so exchange member can diagnose the issue.

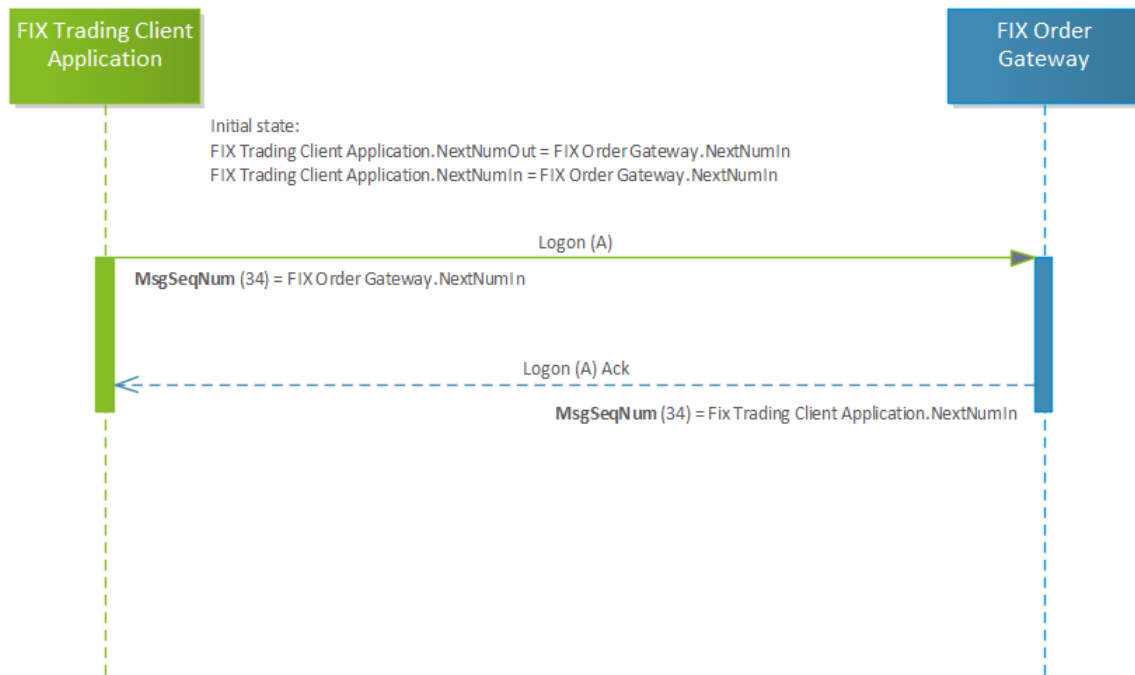
Possible GPW WATS responses to Logon message sent by exchange member:

- Logon Acknowledge when authentication is successful
- Logout message and immediate connection termination in case MsgSeqNum (34) present in Logon message is too low or when MsgSeqNum (34) is not present
- Logout message and immediate connection termination when HeartBtInt (108) value sent by Client Application is not valid. FIX Gateway provides reason for terminating the connection in Text (58) field by setting it to: "Invalid HeartBtInt (108), expected value 30 seconds"
- Terminating immediate connection without sending Logout message when BeginString (8), SenderCompID (49), TargetCompID (56), or IP address in the Logon (35=A) request is invalid
- Terminating immediate connection without sending Logout message when there is an existing opened FIX connection for SenderCompID (49).

When TCP connection is established and exchange member sends message different than valid Logon (35=A) FIX Gateway terminates connection without sending Logout (35=5) message.

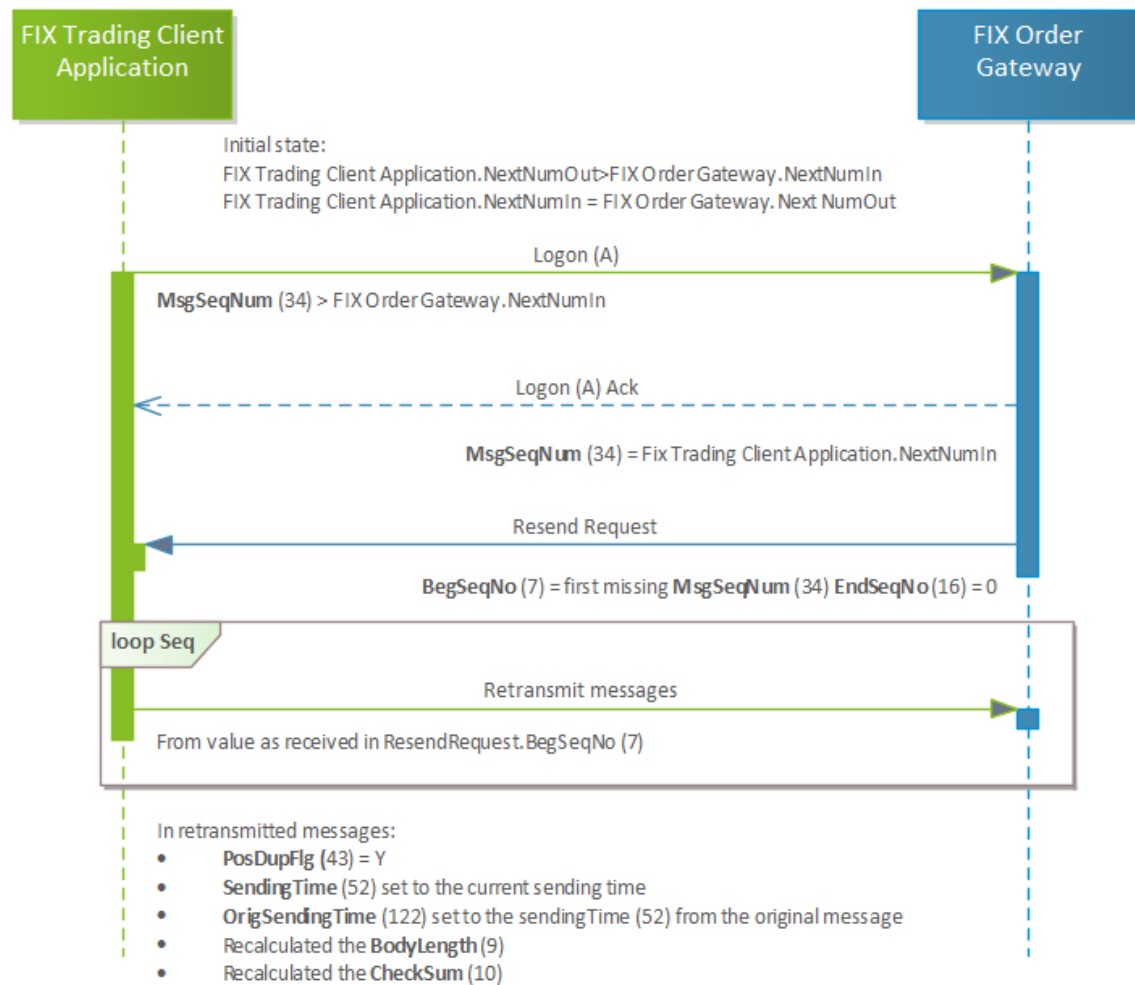
3. If authentication was successful (Logon Acknowledge was sent by FIX Gateway) both exchange member and FIX Gateway should synchronize their messages before they continue to send any messages (new or queued). FIX Gateway compares MsgSeqNum (34) received in Logon message to NextNumIn (sequence maintained on FIX Gateway side). The initiator should compare MsgSeqNum (34) received in Logon Ack to NextNumIn maintained on exchange member site.

If MsgSeqNum (34) from received messages (Logon and Logon Acknowledge) equals to NextNumIn (MsgSeqNum expected to be received) maintained on both sites, then messages are synchronized.

Figure 5. – Successful logon

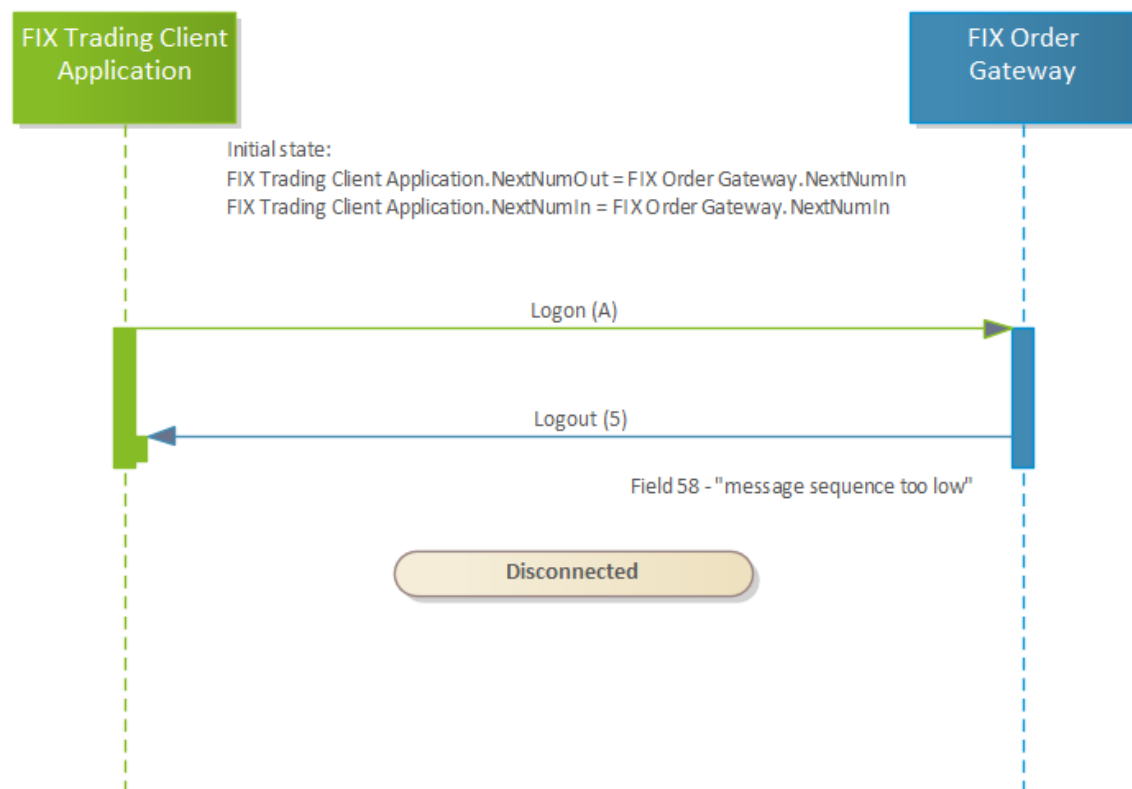
When exchange member sends Logon with MsgSeqNum (34) higher than expected by FIX Gateway, FIX Gateway requests exchange member to resend missing messages and sends ResendRequest (35=2). When FIX Gateway sends Logon Acknowledge with MsgSeqNum (34) higher than expected by exchange member, exchange member should request FIX Gateway to resend missing messages.

Figure 6 – Logon with gap fill procedure



When exchange member sends Logon with MsgSeqNum (34) lower than expected by FIX Gateway, FIX Gateway sends Logout message and terminates TCP connection. In such a situation connection GPW/WATS terminates connection assuming there is an invalid session state on one of sites (exchange member or FIX Gateway).

Figure 7. – Unsuccessful login due to message sequence number too low

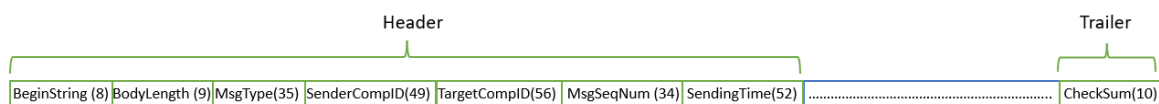


6.2. MAINTAINING A FIX SESSION

GPW FIX Gateway implements a number of mechanisms to enable successful message exchange between exchange member and FIX Gateway since FIX session is established until it is finished. FIX Gateway controls such aspects as if exchange member is active and responsive, all messages sent by exchange member are received, messages are not garbled, received message is correct. These control mechanisms are based on:

- dedicated messages such as Heartbeat, TestRequest, ResentRequests
- fields present in each message in header and trailer section:

Figure 8. – Header and trailer section



6.2.1. SEQUENCE NUMBER

Each party sends messages identified by unique (across FIX session) and independent sequence numbers. Within each session sequence number should start from 1 and should be incremented by 1 when each message is sent. FIX Gateway maintains NextNumIn (next expected MsgSeqNum (34)) and for each incoming message verifies if MsgSeqNum (34) = NextNumIn.

When MsgSeqNum (34) equals NextNumIn and message is not garbled GPW WATS continues with message processing and increments NextNumIn.

When MsgSeqNum (34) is missing or lower than expected and PossDupFlag (43) is not set to "Y" FIX Gateway sends Logout message (35=5) with Text (58) field describing error and terminates connection.

When MsgSeqNum (34) is higher than expected FIX Gateway assumes there are lost messages from exchange member and sends ResendRequest (35=2) to rebuild gap in messages.

Identical validation should be done by exchange member. Sequence number maintenance and control is crucial to prevent FIX message losses or duplicate messages.

6.2.2. BODY LENGTH AND CHECKSUM

Each FIX message received by GPW WATS contains mandatory fields BodyLength (9) and CheckSum (10).

When any of these fields are not present or their value is invalid GPW WATS treats such message as garbled, ignores and does not increment NextNumIn.

6.2.3. HEARTBEAT

Heartbeat (35=0) is single directional message (there are no acknowledgements for this message) designed to maintain a live connection during periods of inactivity (other messages are not sent).

Order Gateway sends Heartbeat messages to Client Application each time a message (from Client Application) is not sent in set heartbeat time interval. The heartbeat time interval is set to 30 seconds. It is expected from Client Application to implement the same heartbeat logic.

When Gateway detects no message (including heartbeat) from Client Application for a period longer than heartbeat interval (30 seconds) it sends TestRequest (1) and starts Grace Period process (see chapter 6.2.3). If Client Application does not respond with Heartbeat(0) message in 90 seconds period Gateway sends Logout and terminates FIX connection.

Heartbeat(0) message sent in response to TestRequest (1) should include TestReqID (112) as received in the last TestRequest (1) message.

6.3. TERMINATING A FIX CONNECTION

Both the Participant and GPW WATS can, at any time, terminate FIX connection by sending Logout request. For normal termination once Logout request is sent, requestor should wait 3 seconds until Logout acknowledgement and then disconnect TCP connection. It is important that Participant waits before terminating connection – it gives GPW WATS Server an opportunity to perform gap fill operations. When MsgSeqNum (34) in Logout message from Client is higher than expected GPW WATS sends ResendRequest (35=2) to rebuild gap in messages. When Client disconnects without Logout process GPW WATS recognizes it as an inappropriate state and starts Cancel on Disconnect mechanism (if enabled for Participant).

At the end of each trading day, before GPW WATS server shuts down, Participant should terminate fix connection by sending Logout message and wait for GPW WATS Logout Acknowledgement.

6.4. RE-ESTABLISHING A FIX SESSION

During trading day FIX connection can be established and terminated many times in scope of the same FIX session. Despite if the Fix connection termination being normal (with valid Logout process) or no new FIX connection is established with incremented MsgSeqNum (34).

6.5. RECOVERY

When exchange member or Gateway discovers MsgSeqNum (34) is higher than expected, it should start gap fill procedure to recover lost messages by sending ResendRequest (35=2). Detailed expected behavior depends on FIX connection state and message type.

When MsgSeqNum (34) is higher than expected in Logon (35=A) message then receiver (either exchange member or Gateway) should perform Logon process, send Logon Acknowledgement and then send a ResendRequest (35=2) message.

When MsgSeqNum (34) is higher than expected in Logout (35=5) message then receiver (either exchange member or Gateway) should send a ResendRequest (35=2) message, receive missing messages and then conclude FIX connection by sending Logout acknowledge message.

When MsgSeqNum (34) is higher than expected in ResendRequest (35=2) message then receiver (either exchange member or Gateway) should send a ResendRequest (35=2) message, receive missing messages and then process retransmission of requested messages.

In any other cases receiver (either exchange member or Gateway) should send ResendRequest (35=2) and process gap filling procedure.

ResendRequest (35=2) message consists of a Standard Header, BeginSeqNo (7), EndSeqNo (16) and Standard Trailer. When requestor wants one message to be retransmitted EndSeqNo should be equal to BeginSeqNo. When EndSeqNo is set to 0 then all messages since BeginSeqNo should be retransmitted.

Figure 9. – Resend Request message



In response to ResendRequest (35=2) message counterparty should resend messages from requested scope. Each retransmitted message should have:

- PossDupFlag (43) set to Y
- OrigSendingTime (122) set to the value from the original message (SendingTime (52))
- SendingTime (52) set to current time from the point of resending message
- BodyLength (9) recalculated after above changes
- CheckSum (10) recalculated after above changes

Each session layer messages (Logon, Logout, Heartbeat, TestRequest, ResendRequest, SequenceReset) should be replaced with SequenceReset (35=4) GapFillFlag (123) set to "Y".

7. MESSAGES

7.1. STANDARD HEADER

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values	Comments
8	BeginString	R		Identifies beginning of new message and protocol version. Always first field in message.	string		FIXT.1.1	Always unencrypted.
9	BodyLength	R		Message length in bytes, forward to the CheckSum field. Always second field in message.	length		integer	Always unencrypted.
35	MsgType	R		Defines message type. Always third field in message.	string		alphanumeric, 0-9, A-Z	Always unencrypted. 0 - HEARTBEAT 1 - TEST_REQUEST 2 - RESEND_REQUEST 3 - REJECT 4 - SEQUENCE_RESET 5 - LOGOUT 8 - EXECUTION_REPORT 9 - ORDER_CANCEL_REJECT A - LOGON D - NEW_ORDER_SINGLE F - ORDER_CANCEL_REQUEST G - ORDER_CANCEL_REPLACE_REQUEST j - BUSINESS_MESSAGE_REJECT AE - TRADE_CAPTURE_REPORT AR - TRADE_CAPTURE_REPORT_ACK
1128	ApplVerID				string		0 FIX27 1 FIX30	

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values	Comments
							2 FIX40 3 FIX41 4 FIX42 5 FIX43 6 FIX44 7 FIX50 8 FIX50SP1 9 FIX50SP2 10 FIXLatest	
49	SenderCompID	R		Assigned value used to identify firm sending message.	string	8 characters	alphanumeric, 0-9, A-Z	Always unencrypted.
56	TargetCompID	R		Assigned value used to identify receiving firm.	string	8 characters	alphanumeric, 0-9, A-Z	Always unencrypted.
115	OnBehalfOfCompID	O		Assigned value used to identify firm originating message if the message was delivered by a third party i.e. the third party firm identifier would be delivered in the SenderCompID field and the firm originating the message in this field.	string	8 characters	alphanumeric, 0-9, A-Z	Can be embedded within encrypted data section.
34	MsgSeqNum	R		Integer message sequence number.	seqnum		integer	Can be embedded within encrypted data section.
50	SenderSubID	O		Assigned value used to identify specific message originator (desk, trader, etc.).	string	8 characters	alphanumeric, 0-9, A-Z	Not used, can be used in future.
57	TargetSubID	O		Assigned value used to identify specific individual or unit intended to receive message. "ADMIN" reserved for	string	8 characters	alphanumeric, 0-9, A-Z	Not used, can be used in future.

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values	Comments
				administrative messages not intended for a specific user.				
43	PossDupFlag	O	Always required for retransmitted messages, whether prompted by the sending system or as a result of a resend request.	Indicates possible retransmission of message with this sequence number.	boolean	1	N = Original Transmission Y = Possible Duplicate	Can be embedded within encrypted data section.
97	PossResend	O	Required when message may be duplicate of another message sent under a different sequence number.	Indicates that message may contain information that has been sent under another sequence number.	boolean	1	N = Original Transmission Y = Possible Resend	Can be embedded within encrypted data section.
52	SendingTime	R		Time of message transmission (always expressed in UTC).	UTC timestamp		From Gateway: YYYYMMDD-HH:MM:SS.nnnnnnnnn From exchange member all listed below formats are accepted: YYYYMMDD-HH:MM:SS.nnnnnnnnn YYYYMMDD-HH:MM:SS.ffffff YYYYMMDD-HH:MM:SS.sss	Can be embedded within encrypted data section.

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values	Comments
122	OrigSendingTime	O	Required for message resent as a result of a ResendRequest. If data is not available, set to same value as SendingTime.	Original time of message transmission (always expressed in UTC) when transmitting orders as a result of a resend request.	UTC timestamp		From Gateway: YYYYMMDD- HH:MM:SS.nnnnnnnnn From exchange member all listed below formats are accepted: YYYYMMDD- HH:MM:SS.nnnnnnnnn YYYYMMDD- HH:MM:SS.ffffff YYYYMMDD- HH:MM:SS.sss	Can be embedded within encrypted data section.
369	LastMsgSeqNumProcessed	O		The last MsgSeqNum (34) value received by the FIX engine and processed by downstream application, such as trading engine or order routing system. Can be specified on each message sent. Useful for detecting a backlog with a counterparty.	seqnum			

7.2. STANDARD TRAILER

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values	Comments
10	Checksum	R		Modulo 256 checksum represented on three digits.	string	3 characters		Always unencrypted.

7.3. LOGON

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages.
98	EncryptMethod	R		Method of encryption.	int	1 character	0 = None / Other
108	HeartBtInt	R		Heartbeat interval (seconds).	int		30 (seconds)
95	RawDataLenght	R			length	1 character	Length of 96 tag with security data.
96	RawData	R			data	8 characters	The security data required for authentication. It is a key received by the exchange member in the registration process.
141	ResetSeqNumFlag	O		Indicates that both sides of the FIX session should reset sequence numbers.	boolean	1 character	N = No Y = Yes, reset sequence numbers Reserved for future use, not yet implemented.
789	NextExpectedMsgSeqNum	O		Next expected MsgSeqNum value to be received.	seqnum		
1409	SessionStatus	O	Available only in Logon response (Logon Acknowledge).	Available to provide status of a FIX session in logon acknowledge(response to a logon) message sent by FIX Gateway. Should not be used in Logon send by Client Application initiating FIX connection..	int		0 - Session active 4 - Session logout complete 6 - Account locked 7 - Logons are not allowed at this time 9 - Received MsgSeqNum (34) is too low 10 - Received NextExpectedMsgSeqNum (789) is too high 11 - Invalid user token in RawData (96)
1137	DefaultAppVerID	R		The default version of FIX messages used in this session.	string		9 = FIX50SP2
58	Text	O	Available only in Logon response (Logon Acknowledge).	Populated in logon when used as a logon acknowledgement from Gateway to Client Application.. Should not be used in Logon send by Client Application initiating FIX connection.	string	18 characters	ASCII 32-126
C	StandardTrailer	R					Trailer used in FIX TP messages.

7.4. HEARTBEAT (0)

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages.
112	TestReqID	C	Required when the heartbeat is the result of a Test Request message.	Identifier provided in Test Request message, returned in resulting heartbeat.	string		
C	StandardTrailer	R					Trailer used in FIX TP messages

7.5. TEST REQUEST (1)

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages
112	TestReqID	R		Identifier included in Test Request message to be returned in resulting heartbeat.	string		
C	StandardTrailer	R					Trailer used in FIX TP messages

7.6. RESEND REQUEST (2)

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages.
7	BeginSeqNo	R		Message sequence number of first message in range to be resent.	seqnum		
16	EndSeqNo	R		Message sequence number of last message in range to be resent. If request is for a single message BeginSeqNo (7) = EndSeqNo. If request is for all messages subsequent to a particular message, EndSeqNo = "0" (representing infinity).	seqnum		
C	StandardTrailer	R					Trailer used in FIX TP messages

7.7. SEQUENCE RESET (4)

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages.
123	GapFillFlag	O		Indicates if the Sequence Reset message is replacing administrative or application messages (which will not be resent) or reset sequence to a new value.	boolean		N = Sequence Reset, sequence number is being reset to a new value provided in NewSeqNo(36). Y = Gap Fill Message, replacing session layer or application messages which will not be resent when filling gaps in the sequence number.
36	NewSeqNo	R		New sequence number.	seqnum		
C	StandardTrailer	R					Trailer used in FIX TP messages

7.8. LOGOUT (5)

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages.
1409	SessionStatus	O	Available only in Logon response (Logon Acknowledge).	Available to provide status of a FIX session in logon acknowledge(response to a logon) message sent by FIX Gateway. Should not be used in Logon send by Client Application initiating FIX connection..	int		0 Session Active 4 Session Logout Complete 6 Account Locked 7 Logons Are Not Allowed At This Time 9 Received Msg Seq Num Too Low 10 Received Next Expected Msg Seq Num Too High 11 Invalid User Token
58	Text	O		Populated with error message when the Logout(5) is sent due to a problem with a FIX connection.	string	18 characters	ASCII 32-126
C	StandardTrailer	R					Trailer used in FIX TP messages

7.9. BUSINESS MESSAGE REJECT (J)

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages.
45	RefSeqNum	O		Reference message sequence number.	seqnum		MsgSeqNum of rejected message.
372	RefMsgType	R		The MsgType (35) of the FIX message being referenced.	string	2 chars	
380	BusinessRejectReason	R		Code to identify reason for a Business Message Reject message.	int	4	0 - Other 2 - Unknown security 3 - Unsupported message type 4 - Application not available 5 - Conditionally required field missing 6 - Not authorized 8 - Throttle limit exceeded 9 - Throttle limit exceeded, session will be disconnected 1001 - NoPartyIDs (453) must be equal to 2, 3, 4 1003 - Duplicated PartyRole (452) value
58	Text	O		Free format text string, additional description of reject reason.	string	18 characters	Where possible, message to explain reason for rejection.
C	StandardTrailer	R					Trailer used in FIX TP messages

7.10. REJECT (3)

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages
45	RefSeqNum	O		Reference message sequence number.	seqnum		MsgSeqNum of rejected message.
372	RefMsgType	R		The MsgType (35) of the FIX message being referenced.	string	2 characters	

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
373	BusinessRejectReason	R		Code to identify reason for a Business Message Reject message.	int	4	0 - INVALID_TAG_NUMBER - 1 - REQUIRED_TAG_MISSING - 10 - SENDING_TIME_ACCURACY_PROBLEM - 11 - INVALID_MSG_TYPE - 12 - XML_VALIDATION_ERROR - 13 - TAG_APPEARS_MORE_THAN_ONCE - 14 - TAG_SPECIFIED_OUT_OF_REQUIRED_ORDER - 15 - REPEATING_GROUP_FIELDS_OUT_OF_ORDER - 16 - INCORRECT_NUM_IN_GROUP_COUNT_FOR_REPEATING_GROUP 17 - NON_DATA_VALUE_INCLUDES_FIELD_DELIMITER 18 - INVALID_OR_UNSUPPORTED_APPLICATION_VERSION 2 - TAG_NOT_DEFINED_FOR_THIS_MESSAGE_TYPE 3 - UNDEFINED_TAG 4 - TAG_SPECIFIED_WITHOUT_A_VALUE 5 - VALUE_IS_INCORRECT 6 - INCORRECT_DATA_FORMAT_FOR_VALUE 7 - DECRYPTION_PROBLEM 8 - SIGNATURE_PROBLEM 9 - COMP_ID_PROBLEM 99 - OTHER
58	Text	O		Free format text string, additional description of reject reason.	string	18 characters	Where possible, message to explain reason for rejection.
C	StandardTrailer	R					Trailer used in FIX TP messages

7.11. NEW ORDER SINGLE (D)

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
9		StandardHeader	R					
11		ClOrdID	R		Unique identifier for an order assigned by the trading member. Trading Members that submit GTD / GTC orders must ensure uniqueness across	string	20 characters	alphanumeric, 0-9, A-Z, a-z

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
					days, for example by embedding a date within the COrdID value.			
C		Parties	R					
453		NoPartyIDs	R		Number of repetitions of PartyID (448), PartyIDSource (447), PartyRole (452), PartyRoleQualifier (2376) groupings.	numingroup		integer
448	→	PartyID	R		Identification of the party. The short code representing the client, execution decision maker or investment decision maker represented by this block (depending on the PartyRole value).	string	4 bytes	integer 4 - 4.294.967.295 (4 bajts) The following values are reserved for applicable use. Applicable to PartyRole = 3 : 0 = NONE (no client for this order) 1 = AGGR (an aggregation of multiple client orders) 2 = PNAL (clients are pending allocation) Applicable to PartyRole = 12 : 3 = NORE (timing and location of the execution determined by the client of the participant)
447	→	PartyIDSource	R		Used to identify classification source.	char		D = Proprietary / Custom code B = BIC N = Legal Entity Identifier (LEI) P = Short code identifier
452	→	PartyRole	R		Identifies the type of PartyID (448). Specifies the role represented by the short code provided in the PartyID tag.	int		For PartyIDSource (447) = D: 4 = Clearing Firm For PartyIDSource (447) = P: 3 = Client ID 12 = Executing Trader 122 = Investment Decision Maker
2376	→	PartyRoleQualifier	C	Conditionally required for PartyIDSource (447) = P (Short code identifier).	Provides further qualification of the PartyRole (452) value.	int		0 = None (only valid for reserved short codes) 22 = Algorithm (applicable to PartyRole values 12 or 122) 23 = Firm or legal entity (LEI) (applicable to PartyRole value 3) 24 = Natural person (applicable to PartyRole values 3, 12, 122)

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1		Account	O		Account provided by exchange member for clearing purposes.	string	16 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
581		AccountType	O		Type of account associated with the order.	int		1 = Account is carried on customer side of the books 3 = House trader
2362		SelfMatchPreventionID	O		Identifies orders that should not be matched against each other if both of them contain the same SelfMatchPreventionID (2362) for the same instrument and were submitted by the same Exchange member.	num	1 byte	integer
C		DisplayInstruction						
1138		DisplayQty	C	Conditionally required for Iceberg orders. Only allowed for OrdType (40) = 2 (Limit).	Initially displayed quantity for Iceberg order in Market Data public order book (Peak Quantity).	qty	4 bytes	integer
C		Instrument	R					
48		SecurityID	R		Security identifier value of SecurityIDSource (22). Numerical (non-ISIN) instrument identifier provided on a daily basis via Market Data reference services.	string	4 bytes	integer
22		SecurityIDSource	R		Identifies class or source of the SecurityID (48) value.	string	1 character	8 = exchange symbol
54		Side	R		Side of order.	char	1 character	1 = Buy 2 = Sell
60		TransactTime	R		Time the order submission process was initiated by exchange member. This time should be earlier than message SendingTime (52) provided in the header.	UTC timestamp		The following formats are accepted: YYYYMMDD-HH:MM:SS.nnnnnnnnn YYYYMMDD-HH:MM:SS.uuuuuu YYYYMMDD-HH:MM:SS.sss
C		OrderQtyData	R					
38		OrderQty	R		Initial total order quantity (expressed in number of Lots).	qty	4 bytes	integer

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
40		OrdType	R		Type of order.	char	1 character	1 = Market 2 = Limit 3 = Stop Loss 4 = Stop Limit K = Market With Left Over as Limit
44		Price	C	Conditionally required if OrdType = 2 (Limit) or 4 (Stop Limit).	Price per unit of quantity (e.g. 1 Lot).	price	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
C		TriggeringInstruction	C					
1102		TriggerPrice	C	Conditionally required for OrdType (40) = 3 (Stop Loss) or 4 (Stop Limit).	The price at which the order is activated and inserted into the order book.	price	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
59		TimeInForce	R		Specifies when the order should become expired (Day, Good Till Time, Good Till Date, Good Till Cancel), when it should be activated (At The Opening, At The Closing), or how it should be executed (Immediate Or Cancel, Fill Or Kill).	char	1 character	0 = Day 1 = Good Till Cancel (GTC) 2 = At The Opening (VFA) 3 = Immediate Or Cancel (IOC) 4 = Fill Or Kill (FOK) 6 = Good Till Date (GTD) 7 = At The Closing (VFC)
432		ExpireDate	C	Conditionally required if TimeInForce = 6 and ExpireTime (126) is not specified.	Date of order expiry (last day the order can trade), always expressed in terms of the local market date. ExpireDate and ExpireTime tags are mutually exclusive.	localmktdate		The following formats are accepted: YYYYMMDD (local)
126		ExpireTime	C	Conditionally required if TimeInForce = 6 and ExpireDate (432) is not specified.	Time of order expiry (always expressed in UTC (Universal Time Coordinated, also known as "GMT"). It always refers to the current trading day. ExpireDate and ExpireTime tags are mutually exclusive.	UTC timestamp		The following formats are accepted: YYYYMMDD-HH:MM:SS (UTC)

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
528		OrderCapacity	R		Designates the capacity of the Trading Firm placing the order.	char	1 character	A = Agency (mapped to AOTC) P = Principal (mapped to DEAL) R = Riskless Principal (mapped to MTCH)
529		OrderRestrictions	O		Designation required for regular Liquidity Providers (as opposite to Issuer Liquidity Providers). When an Issuer Liquidity Provider orders tag should be omitted.	char	1 character	5 = Acting as Market Maker or Specialist in the security
C		OrderAttributeGrp						
2593		NoOrderAttributes	R		Number of order attribute entries.	int	1 byte	1
2594	→	OrderAttributeType	R		The type of order attribute.	int	1 byte	2 = Liquidity provision activity order
2595	→	OrderAttributeValue	R		The value associated with the order attribute type specified in OrderAttributeType (2594). As opposed to OrderRestriction tag specifies whether the order poses any type of liquidity provision activity (either regular or issuer).	string	1 character	Y = Yes N = No
58		Text	O		Free format text string for Exchange member use.	string	18 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
1724		OrderOrigination	O		Designation required for orders submitted via Direct Electronic Access (both DMA or Sponsored Access). In case of regular trading access this tag should be omitted.	int		5 = Order received from a direct access or sponsored access customer
C		StandardTrailer	R					Trailer used in FIX TP messages

7.12. EXECUTION REPORT (8)

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
9		StandardHeader	R					Header used in FIX TP messages

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
37		OrderID	R		Unique identifier for an order assigned by Trading System. OrderID is unique across all trading days, all order books, and all exchange members, and it is persistent for the entire life of an order (i.e. it does not change after order modification). OrderID is generated for each order as a concatenation of connection ID + sequential number of a trading day (starting from 1) + sequence number of the message.	string	8 bytes	numeric
278		MDEntryID	R		Identifier for an order assigned by Trading System for Market Data purposes. Corresponds to the publicOrderID field in Market Data messages. MDEntryID is unique across all order books and all exchange members, but only within a single trading day. On the next trading day the numbering is reset and starts afresh (i.e. MDEntryID is not persistent for the entire life of an order). Moreover it also changes with each refill of an Iceberg order.	string	8 bytes	numeric
11		ClOrdID	R		Unique identifier for an order assigned by exchange member. Exchange members who submit GTD / GTC orders must ensure uniqueness across days, for example by embedding a date within the ClOrdID value.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
41		OrigClOrdID	O	Provided in response to a Cancel Request or Cancel/Replace Request (ExecType (150) = 4 or 5).	OrigClOrdID is the last ClOrdID in the modification chain for an order.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
C		Parties	R					
453		NoPartyIDs	R		Number of repetitions of PartyID (448), PartyIDSource (447), PartyRole (452), PartyRoleQualifier (2376) groupings.	numingroup		integer

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
448	→	PartyID	R		Identification of the party. The short code representing the client, execution decision maker or investment decision maker represented by this block (depending on the PartyRole value).	string	4 bytes	integer 4 - 4.294.967.295 (4 bajts) The following values are reserved for applicable use. Applicable to PartyRole = 3: 0 = NONE (no client for this order) 1 = AGGR (an aggregation of multiple client orders) 2 = PNAL (clients are pending allocation) Applicable to PartyRole = 12: 3 = NORE (timing and location of the execution determined by the client of the participant)
447	→	PartyIDSource	R		Used to identify classification source.	char		B = BIC D = Proprietary / Custom code N = Legal Entity Identifier (LEI) P = Short code identifier
452	→	PartyRole	R		Identifies the type of PartyID (448). Specifies the role represented by the short code provided in the PartyID tag.	int		For PartyIDSource (447) = D: 4 = Clearing Firm For PartyIDSource (447) = P: 3 = Client ID 12 = Executing Trader 122 = Investment Decision Maker
2376	→	PartyRoleQualifier	C	Provided for PartyIDSource (447) = P (Short code identifier).	Provides further qualification of the PartyRole (452) value.	int		0 = None (only valid for reserved short codes or in case of Clearing Firm) 22 = Algorithm (applicable to PartyRole values 12 or 122) 23 = Firm or legal entity (LEI) (applicable to PartyRole value 3) 24 = Natural person (applicable to PartyRole values 3, 12, 122)
880		TrdMatchID	C	Provided for trades (ExecType (150) = F or H).	Identifier assigned to a trade by a matching system.	string	4 bytes	numeric

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
17		ExecID	R		Unique identifier of Execution Report (8) message.	string	18 characters	alphanumeric, 0-9, A-Z, a-z
19		ExecRefID	C	Provided for ExecType (150) = H (Trade Cancel).	ExecID (17) value of Execution Report (8) message informing about a trade that is being cancelled.	string	18 characters	alphanumeric, 0-9, A-Z, a-z
150		ExecType	R		Describes the type of Execution Report (8) message, while OrderStatus (39) provides the current status of an order in Trading System.	char	1 character	0 = New 4 = Canceled 5 = Replaced 8 = Rejected C = Expired D = Restated F = Trade H = Trade Cancel L = Triggered
39		OrdStatus	R		Provides the current status of an order in Trading System.	char	1 character	0 = New 1 = Partially Filled 2 = Filled 4 = Canceled 8 = Rejected C = Expired
103		OrdRejReason	C	Provided for ExecType (150) = 8 (Rejected).	Code to identify reason for order rejection.	int	4 chars	Please see FIX Rejection Codes document for the list of available codes with their descriptions.
1328		RejectText	C	Provided for ExecType (150) = 8 (Rejected).	Text description of rejection code provided in OrdRejReason (103) tag.	string	50 characters	Please see FIX Rejection Codes document for the list of available codes with their descriptions.
2431		ExecTypeReason	C	Provided in case of ExecType (150) = 4 (Cancelled) or D (Restated) or F (Trade)	Provides detailed information on ExecType (150) in case of order restatement, Iceberg order refill, unsolicited order cancellation, trade resulting from aggressive order matching.	int	4 bytes	100 - GT order restatement 101 - Iceberg order refill 102 - Cancel IOC/FOK order 103 - Cancel by STP 104 - Cancel by Market Operations 105 - Cancel on Trading Halt 106 - Cancel on Disconnect

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
								107 - Cancel by Corporate Action 110 - First trade on aggressive order.
2667		AlgorithmicTradeIndicator	C	Provided for ExecType (150) = F (Trade) or H (Trade Cancel).	Specifies that the trade originates from a computer program or algorithm requiring little-to-no human intervention. In the context of ESMA MiFID II, a trade has to be flagged as "algorithmic" if at least one of the matched orders was submitted by a trading algorithm.	int		0 = Non-algorithmic trade 1 = Algorithmic trade
1		Account	O		Account provided by exchange member for clearing purposes.	string	16 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
581		AccountType	O		Type of account associated with the order.	int		1 = Account is carried on customer side of the books 3 = House trader
C		Instrument	R					
48		SecurityID	R		Security identifier value of SecurityIDSource (22). Numerical (non-ISIN) instrument identifier provided on a daily basis via Market Data reference services.	string	4 bytes	integer
22		SecurityIDSource	R		Identifies class or source of the SecurityID (48) value.	string	1 character	8 = exchange symbol
54		Side	R		Side of order.	char	1 character	1 = Buy 2 = Sell
C		OrderQtyData	R					
38		OrderQty	R		Total order quantity (expressed in number of Lots).	num	4 bytes	integer
40		OrdType	R		Type of order.	char	1 character	1 = Market 2 = Limit 3 = Stop Loss 4 = Stop Limit K = Market With Left Over as Limit

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
44	Price	N	Provided for OrdType (40) = 2 (Limit) or 4 (Stop Limit).	Price per unit of quantity (e.g. 1 Lot).	price	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
C	TriggeringInstruction						
1102	TriggerPrice	C	Provided for OrdType (40) = 3 (Stop Loss) or 4 (Stop Limit).	The price at which the order is activated and inserted into the order book.	price	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
1823	Triggered	C	Provided for TimeInForce (59) = 2 (VFA) or 7 (VFC), and OrdType (40) = 3 (Stop Loss) or 4 (Stop Limit).	Indicates whether an order has been already triggered or not. When triggered VFA/VFC orders value 1, for triggered Stop orders value 2.	int	1 character	0 = Not triggered (default) 1 = Triggered 2 = Stop order triggered
15	Currency	R		Identifies currency used for price (i.e. instrument trading currency).	currency	3 bytes	
59	TimeInForce	R		Specifies when the order should become expired (Day, Good Till Time, Good Till Date, Good Till Cancel), when it should be activated (At The Opening, At The Closing), or how it should be executed (Immediate Or Cancel, Fill Or Kill).	char	1 character	0 = Day 1 = Good Till Cancel (GTC) 2 = At The Opening (VFA) 3 = Immediate Or Cancel (IOC) 4 = Fill Or Kill (FOK) 6 = Good Till Date (GTD) 7 = At The Closing (VFC)
432	ExpireDate	C	Provided for TimeInForce = 6 and ExpireTime (126) is not specified.	Date of order expiry (last day the order can trade), always expressed in terms of the local market date. ExpireDate and ExpireTime tags are mutually exclusive.	localmktdate		YYYYMMDD (local)
126	ExpireTime	C	Provided for TimeInForce = 6 and ExpireDate (432) is not specified.	Time of order expiry (always expressed in UTC (Universal Time Coordinated, also known as "GMT"). It always refers to the current trading day. ExpireDate and ExpireTime tags are mutually exclusive.	UTC timestamp		YYYYMMDD-HH:MM:SS (UTC)

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
528		OrderCapacity	R		Designates the capacity of the Trading Firm placing the order.	char	1 character	A = Agency (mapped to AOTC) P = Principal (mapped to DEAL) R = Riskless Principal (mapped to MTCH)
529		OrderRestrictions	O		Designation required for regular Liquidity Providers (as opposite to Issuer Liquidity Providers). When Issuer Liquidity Provider orders tag will not be provided.	char	1 character	5 = Acting as Market Maker or Specialist in the security
C		OrderAttributeGrp						
2593		NoOrderAttributes	R		Number of order attribute entries.	int	1 byte	1
2594	→	OrderAttributeType	R		The type of order attribute.	int	1 byte	2 = Liquidity provision activity order
2595	→	OrderAttributeValue	R		The value associated with the order attribute type specified in OrderAttributeType (2594). As opposite to OrderRestriction tag specifies whether the order poses any type of liquidity provision activity (either regular or issuer).	string	1 character	Y = Yes N = No
32		LastQty	C	Provided for ExecType (150) = F (Trade) or H (Trade Cancel).	Quantity of last execution (expressed in number of Lots).	qty	4 bytes	integer
31		LastPx	C	Provided for ExecType (150) = F (Trade) or H (Trade Cancel).	Price of last execution.	price	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
151		LeavesQty	R		Quantity remaining for further execution (expressed in number of Lots). When an order that is no longer active, i.e. OrdStatus (39) is = 4 (Canceled), C (Expired), 8 (Rejected), LeavesQty (151) is set to 0, otherwise LeavesQty (151) = OrderQty (38) - CumQty (14).	qty	4 bytes	integer
14		CumQty	R		Total executed order quantity (expressed in number of Lots).	qty	4 bytes	integer
60		TransactTime	R		Timestamp when the business event represented by the message occurred in Trading System. This time should be earlier	UTC timestamp		The following format is returned from Gateway:

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
				than message SendingTime (52) provided in the header.			YYYYMMDD-HH:MM:SS.nnnnnnnnn
2362	SelfMatchPreventionID	O		Identifies orders that should not be matched against each other if both of them contain the same SelfMatchPreventionID (2362) for the same instrument and were submitted by the same Exchange member.	num	1 byte	integer
C	DisplayInstruction						
1138	DisplayQty	O		Currently displayed quantity for Iceberg order in Market Data public order book (as opposed to initially displayed quantity).	qty	4 bytes	integer
1608	InitialDisplayQty	O		Initially display quantity requested for Iceberg order in New Order Single (D) or Order Cancel/Replace Request (G) messages.	qty	4 bytes	integer
58	Text	O		Free format text string provided by Exchange member in New Order Single (D) message.	string	18 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
851	LastLiquidityInd	C	Provided for ExecType (150) = F (Trade).	Indicates whether the order added liquidity or removed liquidity from the order book (looking from the perspective of the current execution).	int		1 = Added liquidity 2 = Removed liquidity 4 = Auction execution
1724	OrderOrigination	O		Designation required for orders submitted via Direct Electronic Access (both DMA or Sponsored Access). When regular trading access this tag should be omitted.	int		5 = Order received from a direct access or sponsored access customer
C	StandardTrailer	R					Trailer used in FIX TP messages

7.13. ORDER CANCEL REPLACE (G)

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
37	OrderID	O		Unique identifier for an order assigned by the trading system. OrderID is unique across all trading days, all order books, and all Exchange Members, and it is persistent for the entire life of an order (i.e. it does not change after order modification). OrderID is generated for each order as a concatenation of connection ID + sequential number of a trading day (starting from 1) + sequence number of the message.	string	8 bytes	numeric
41	OrigCLOrdID	O		ClOrdID (11) of the previous order (not the initial order of the day) as assigned by the institution, used to identify the previous order in cancel and cancel/replace requests.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
11	ClOrdID	R		Unique identifier for an order assigned by the Exchange Member. Exchange Members that submit GTD / GTC orders must ensure uniqueness across days, for example by embedding a date within the ClOrdID value.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
C	DisplayInstruction						
1138	DisplayQty	C	Required for Iceberg order modification.	The quantity to be displayed. Required for Iceberg orders. On orders specifies the quantity to be displayed, on execution reports the currently displayed quantity.	qty	4 bytes	integer
C	Instrument	R					
48	SecurityID	R		Security identifier value of SecurityIDSource (22). Numerical (non-ISIN) instrument identifier provided on a daily basis via Market Data reference services.	string	4 bytes	integer
22	SecurityIDSource	R		Identifies class or source of the SecurityID (48) value.	string	1 character	8 = Exchange symbol
54	Side	R		Side of order.	char	1 character	1 = Buy 2 = Sell

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
60	TransactTime	R		Time the order modification process was initiated by the Trading Member. This time should be earlier than message SendingTime (52) provided in the header.	UTC timestamp		The following formats are accepted : YYYYMMDD-HH:MM:SS.nnnnnnnnn (UTC) YYYYMMDD-HH:MM:SS.uuuuuu (UTC) YYYYMMDD-HH:MM:SS.mmm (UTC)
C	OrderQtyData	R					
38	OrderQty	R		Initial total order quantity (expressed in number of Lots)..	qty	4 bytes	integer
40	OrdType	R		Type of the order.	char	1 character	1 = Market 2 = Limit 3 = Stop Loss 4 = Stop Limit K = Market With Left Over as Limit
44	Price	C	Conditionally required if OrdType = 2 (Limit) and OrdType = 4 (Stop Limit), otherwise forbidden.	Price per unit of quantity (e.g. 1 Lot).	price	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
C	TriggeringInstruction	C					
1102	TriggerPrice	C	Conditionally required for OrdType (40) = 3 (Stop Loss) or 4 (Stop Limit).	The price at which the order is activated and inserted into the order book.	price	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
C	StandardTrailer	R					Trailer used in FIX TP messages

7.14. ORDER CANCEL REJECT (9)

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
37	OrderID	R		Unique identifier for an order assigned by the trading system. OrderID is unique across all trading days, all order books, and all trading members, and it is persistent for the entire life of an order (i.e. it does not change after order modification). OrderID is generated for each order as a concatenation of connection ID + sequential number of a trading day (starting from 1) + sequence number of the message.	string	8 bytes	numeric If CxlRejReason = "Unknown order", specify "NONE".
11	CLOrdID	R		Unique identifier for an order assigned by the trading member. Exchange Members that submit GTD / GTC orders must ensure uniqueness across days, for example by embedding a date within the CLOrdID value.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
41	OrigCLOrdID	O		CLOrdID (11) of the previous order (not the initial order of the day) as assigned by the institution, used to identify the previous order in cancel and cancel/replace requests.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
39	OrdStatus	R		OrdStatus value after this cancel reject is applied.	char		0 = New 1 = Partially Filled 2 = Filled 4 = Canceled 8 = Rejected C = Expired
60	TransactTime	R		Timestamp when the business event represented by the message occurred in the trading system. This time should be earlier than message SendingTime (52) provided in the header.	UTC timestamp		The following format is returned by Gateway: YYYYMMDD-HH:MM:SS.nnnnnnnn (UTC)
434	CxlRejResponseTo	R		Identifies the type of request that a Cancel Reject is in response to.	char		1 = Order Cancel Request 2 = Order Cancel/Replace Request
102	CxlRejReason	R		Code to identify reason for Cancel or Cancel/Replace rejection.	int	4 bytes	Please see FIX Rejection Codes document for the list of available codes with their descriptions.
1328	RejectText	R		Description of the reason for Cancel or Cancel/Replace rejection.	string	50 characters	Please see FIX Rejection Codes document for the list of available codes with their descriptions.
C	StandardTrailer	R					Trailer used in FIX TP messages

7.15. ORDER CANCEL REQUEST (F)

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C	StandardHeader	R					Header used in FIX TP messages
41	OrigClOrdID	O		ClOrdID (11) of the previous order (not the initial order of the day) as assigned by the institution, used to identify the previous order in cancel and cancel/replace requests.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
37	OrderID	O		Unique identifier for an order assigned by the trading system. OrderID is unique across all trading days, all order books, and all trading members, and it is persistent for the entire life of an order (i.e. it does not change after order modification). OrderID is generated for each order as a concatenation of connection ID + sequential number of a trading day (starting from 1) + sequence number of the message.	int	8 bytes	numeric
11	ClOrdID	R		Unique identifier for an order assigned by the trading member. Exchange Members that submit GTD / GTC orders must ensure uniqueness across days, for example by embedding a date within the ClOrdID value.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
C	Instrument	R					
48	SecurityID	R		Security identifier value of SecurityIDSource (22). Numerical (non-ISIN) instrument identifier provided on a daily basis via Market Data reference services.	string	4 characters	integer
22	SecurityIDSource	R		Identifies class or source of the SecurityID (48) value.	string	1 character	8 = Exchange symbol
54	Side	R		Side of order.	char	1 character	1 = Buy 2 = Sell
60	TransactTime	R		Time the request was initiated by the Exchange Member. This time should be earlier than message SendingTime (52) provided in the header.	UTC timestamp		The following formats are accepted : YYYYMMDD-HH:MM:SS.nnnnnnnnn (UTC) YYYYMMDD-HH:MM:SS.uuuuuu (UTC) YYYYMMDD-HH:MM:SS.mmm (UTC)
C	StandardTrailer	R					Trailer used in FIX TP messages

7.16. TRADE CAPTURE REPORT (AE)

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C		StandardHeader	R					Header used in FIX TP messages
571		TradeReportID	R		Unique identifier of Trade Capture Report.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
818		SecondaryTradeReportID	C	Required for tradeReportType=Alleged or Submit	Unique system identifier of the trade.	int	8 bytes	numeric
1003		TradeID	O	Provided in case of matched trades.	Identifier assigned to a trade by a matching system.	string	4 bytes	numeric
487		TradeReportTransType	R		Identifies Trade Report message transaction type.	int		0 = New 1 = Cancel 2 = Replace
856		TradeReportType	R		Type of Trade Report.	int		0 = Submit 1 = Alleged 2 = Accept 3 = Decline 6 = Trade Report Cancel 14 = Alleged Trade Report Cancel
828		TrdType	R		Type of trade.	int		22 = Privately negotiated trade (Cross trade) 38 = Block trade
1123		TradeHandlingInstr	R		Specifies how the Trade Capture Report (35=AE) should be handled by the respondent.	char		0 = Trade confirmation 1 = Two-party report 3 = One-party report for pass-through
150		ExecType	C	Required if MatchStatus (573) = 0 (Matched) or 1 (Unmatched).	Type of execution being reported.	char		F = Trade
572		TradeReportRefID	C		Identifier used to reference leg for cancellation or in	string	20 characters	alphanumeric, 0-9, A-Z, a-z

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
					case of reporting matched trades.			
C		Instrument	R					
48		SecurityID	R		Security identifier value of SecurityIDSource (22). Numerical (non-ISIN) instrument identifier provided on a daily basis via Market Data reference services.	string	4 bytes	integer
22		SecurityIDSource	R		Identifies class or source of the SecurityID (48) value.	string	1 character	8 = exchange symbol
32		LastQty	R		Quantity (expressed in number of Lots).	qty	4 bytes	integer
31		LastPx	R		Price per unit of quantity (e.g. 1 Lot).	price	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
60		TransactTime	R		Timestamp when the business event represented by the message occurred. This time should be earlier than message SendingTime (52) provided in the header.	UTC timestamp		The following formats are accepted : YYYYMMDD-HH:MM:SS.nnnnnnnnn (UTC) YYYYMMDD-HH:MM:SS.uuuuuu (UTC) YYYYMMDD-HH:MM:SS.mmm (UTC)
64		SettlDate	C	Required if TrdType (828) = 38 (Block trade)	Specific date of trade settlement.	localmktdate		The following formats are accepted : YYYYMMDD
573		MatchStatus	C	Provided only in outgoing messages informing about new trade or trade cancellation.	The status of trade with respect to matching or comparing.	char		0 = Matched 1 = Unmatched
C		TrdCapRptSideGrp	R					
552		NoSides	R		Number of Side repeating group instances.	int		1 = One Side 2 = Both Sides

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
54	→		Side	R		Side.	char	1 character	1 = Buy 2 = Sell
	C		Parties	R					
453	→		NoPartyIDs	R		Number of repetitions of PartyID (448), PartyIDSource (447), PartyRole (452), PartyRoleQualifier (2376) groupings.	numingroup		integer
448	→		PartyID	R		Identification of the party. The short code representing the client, execution decision maker or investment decision maker represented by this block (depending on the PartyRole value).	string	4 bytes	integer 4 - 4.294.967.295 (4 bajts) The following values are reserved for applicable use. Applicable to PartyRole = 3 : 0 = NONE (no client for this order) 1 = AGGR (an aggregation of multiple client orders) 2 = PNAL (clients are pending allocation) Applicable to PartyRole = 12 : 3 = NORE (timing and location of the execution determined by the client of the participant)
447	→		PartyIDSource	R		Used to identify classification source.	char		D = Proprietary / Custom code B = BIC N = Legal Entity Identifier (LEI) P = Short code identifier
452	→		PartyRole	R		Identifies the type of PartyID (448). Specifies the role represented by the short code provided in the PartyID tag.	int		For PartyIDSource (447) = D: 1 = Executing Firm 4 = Clearing Firm 17 = Contra Firm For PartyIDSource (447) = P: 3 = Client ID 12 = Executing Trader 122 = Investment Decision Maker
2376	→		PartyRoleQualifier	C	Conditionally required for PartyIDSource (447) = P (Short code identifier).	Provides further qualification of the PartyRole (452) value.	int		0 = None (only valid for reserved short codes) 22 = Algorithm (applicable to PartyRole values 12 or 122)

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
									23 = Firm or legal entity (LEI) (applicable to PartyRole value 3) 24 = Natural person (applicable to PartyRole values 3, 12, 122)
1	→		Account	O		Account provided by the Exchange Member for clearing purposes.	string	16 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
581	→		AccountType	O		Type of account associated with the order.	int		1 = Account is carried on customer side of the books 3 = House trader
58	→		Text	O		Free format text string for Exchange Member use.	string	18 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
	C		TradeReportOrderDetail	O					
528	→		OrderCapacity	R		Designates the capacity of the firm placing the order.	char	1 character	A = Agency (mapped to AOTC) P = Principal (mapped to DEAL) R = Riskless Principal (mapped to MTCH)
529	→		OrderRestrictions	O		Designation required for regular Liquidity Providers (as opposite to Issuer Liquidity Providers). In case of Issuer Liquidity Provider orders tag should be omitted.	char	1 character	5 = Acting as Market Maker or Specialist in the security
1724	→		OrderOrigination	O		Designation required for orders submitted via Direct Electronic Access (both DMA or Sponsored Access). In case of regular trading access this tag should be omitted.	int		5 = Order received from a direct access or sponsored access customer
	C		OrderAttributeGrp	O					
2593	→		NoOrderAttributes	R		Number of order attribute entries.	Int	1 byte	1

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
2594	→	OrderAttributeType	R		The type of order attribute.	int	1 byte	2 = Liquidity provision activity order
2595	→	OrderAttributeValue	R		The value associated with the order attribute type specified in OrderAttributeType (2594). As opposite to OrderRestriction tag specifies whether the order poses any type of liquidity provision activity (either regular or issuer).	string	1 character	Y = Yes N = No
C		StandardTrailer	R					Trailer used in FIX TP messages

7.17. TRADE CAPTURE REPORT ACK (AR)

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C		StandardHeader	R					Header used in FIX TP messages
571		TradeReportID	R		Unique identifier of Trade Capture Report.	string	18 characters	alphanumeric, 0-9, A-Z, a-z
818		SecondaryTradeReportID	C	Required for tradeReportType= Alleged or Submit	Unique system identifier of the trade.	int	8 bytes	numeric
1003		TradeID	O	Provided in case of matched trades.	Identifier assigned to a trade by a matching system.	string	4 bytes	numeric
487		TradeReportTransType	R		Identifies trade report message transaction type.	int		0 = New 1 = Cancel 2 = Replace
856		TradeReportType	R		Type of trade report.	int		0 = Submit 1 = Alleged 2 = Accept 3 = Decline 6 = Trade Report Cancel 14 = Alleged Trade Report Cancel

Tag		Field Name	Required	Conditional	Description	Data Type	Max Length	Values
828		TrdType	R		Type of trade.	int		22 = Privately negotiated trade (Cross trade) 38 = Block trade
1123		TradeHandlingInstr	R		Specifies how the Trade Capture Report (35-AE) should be handled by the respondent.	char		0 = Trade confirmation 1 = Two-party report 3 = One-party report for pass-through
572		TradeReportRefID	O		Identifier used to reference leg for cancellation or in case of reporting matched trades.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
939		TrdRptStatus	R		Status of trade report.	int		0 = Accepted 1 = Rejected
751		TradeReportRejectReason	C	Provided in case of TrdRptStatus (939) = 1 (Rejected).	Code to identify reason for trade report rejection.	int		Please see FIX Rejection Codes document for the list of available codes with their descriptions.
1328		RejectText	C	Provided in case of TrdRptStatus (939) = 1 (Rejected).	Description of the reason for Trade Capture Report rejection.	string	50 characters	Please see FIX Rejection Codes document for the list of available codes with their descriptions.
32		LastQty	R		Quantity (expressed in number of Lots).	qty	4 bytes	integer
31		LastPx	R		Price per unit of quantity (e.g. 1 Lot).	price	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
C		Instrument	R					
48		SecurityID	R		Security identifier value of SecurityIDSource (22). Numerical (non-ISIN) instrument identifier provided on a daily basis via Market Data reference services.	string	4 bytes	integer
22		SecurityIDSource	R		Identifies class or source of the SecurityID (48) value.	string	1 character	8 = exchange symbol

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
15			Currency	R		Identifies currency used for price (i.e. instrument trading currency).	currency	3 bytes	
60			TransactTime	R		Timestamp when the business event represented by the message occurred. This time should be earlier than message SendingTime (52) provided in the header.	UTC timestamp		YYYYMMDD-HH:MM:SS.nnnnnnnnn (UTC)
58			Text	O		Free format text string for Exchange Member use.	string	18 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
C			TrdCapRptAckSideGrp	R					
552			NoSides	R		Number of Side repeating group instances.	int		1 = One Side 2 = Both Sides
54	→		Side	R		Side.	char	1 character	1 = Buy 2 = Sell
	C		Parties	R					
453	→		NoPartyIDs	R		Number of repetitions of PartyID (448), PartyIDSource (447), PartyRole (452), PartyRoleQualifier (2376) groupings.	numingroup		integer
448	→		PartyID	R		Identification of the party. The short code representing the client, execution decision maker or investment decision maker represented by this block (depending on the PartyRole value).	string	4 bytes	integer 4 - 4.294.967.295 (4 bajts) The following values are reserved for applicable use. Applicable to PartyRole = 3 : 0 = NONE (no client for this order) 1 = AGGR (an aggregation of multiple client orders) 2 = PNAL (clients are pending allocation) Applicable to PartyRole = 12 :

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
									3 = NORE (timing and location of the execution determined by the client of the participant)
447		→	PartyIDSource	R		Used to identify classification source.	char		D = Proprietary / Custom code B = BIC N = Legal Entity Identifier (LEI) P = Short code identifier
452		→	PartyRole	R		Identifies the type of PartyID (448). Specifies the role represented by the short code provided in the PartyID tag.	int		For PartyIDSource (447) = D: 1 = Executing Firm 4 = Clearing Firm 17 = Contra Firm For PartyIDSource (447) = P: 3 = Client ID 12 = Executing Trader 122 = Investment Decision Maker
2376		→	PartyRoleQualifier	C	Conditionally required for PartyIDSource (447) = P (Short code identifier).	Provides further qualification of the PartyRole (452) value.	int		0 = None (only valid for reserved short codes) 22 = Algorithm (applicable to PartyRole values 12 or 122) 23 = Firm or legal entity (LEI) (applicable to PartyRole value 3) 24 = Natural person (applicable to PartyRole values 3, 12, 122)
1		→	Account	O		Account provided by the Exchange Member for clearing purposes.	string	16 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
581		→	AccountType	O		Type of account associated with the order.	int		1 = Account is carried on customer side of the books 3 = House trader
C	C		TradeReportOrderDetail	R					
528		→	OrderCapacity	R		Designates the capacity of the firm placing the order.	char	1 character	A = Agency (mapped to AOTC) P = Principal (mapped to DEAL) R = Riskless Principal (mapped to MTCH)
529		→	OrderRestrictions	O		Designation required for regular Liquidity Providers (as opposite to Issuer Liquidity Providers). In case of Issuer Liquidity	char	1 character	5 = Acting as Market Maker or Specialist in the security

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
						Provider orders tag should be omitted.			
1724	→		OrderOrigination	O		Designation required for orders submitted via Direct Electronic Access (both DMA or Sponsored Access). In case of regular trading access this tag should be omitted.	int		5 = Order received from a direct access or sponsored access customer
	C		OrderAttributeGrp	O					
2593	→		NoOrderAttributes	R		Number of order attribute entries.	int	1 byte	
2594		→	OrderAttributeType	R		The type of order attribute.	int	1 byte	2 = Liquidity provision activity order
2595		→	OrderAttributeValue	R		The value associated with the order attribute type specified in OrderAttributeType (2594). As opposite to OrderRestriction tag specifies whether the order poses any type of liquidity provision activity (either regular or issuer).	string	1 character	Y = Yes N = No
64			SettlDate	C	Provided in case of TrdType (828) = 38 (Block trade)	Specific date of trade settlement.	localmktdate		YYYYMMDD
C			StandardTrailer	R					Trailer used in FIX TP messages

7.18. MASS QUOTE (I)

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C			StandardHeader	R					
117			QuoteID	R		Unique identifier for quote.	string	20 characters	alphanumeric, 0-9, A-Z, a-z

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
537			QuoteType	R		Identifies the type of quote. Default is Indicative if not specified.			1 = Tradeable
301			QuoteResponseLevel	R		Level of response requested from receiver of quote messages. A default value should be bilaterally agreed.	int		2 = Acknowledge each quote message
C			Parties	R					
453			NoPartyIDs	R		Number of PartyID (448), PartyIDSource (447), PartyRole (452) and PartyRoleQualifier (2376) entries.	num		integer
448	→		PartyID	R	Required if NoPartyIDs (453) > 0	Identification of the party.	num	4 bytes	Integer For clients, the following values are reserved for applicable use: Applicable to PartyRole value 3: 0 = NONE (no client for this order) 1 = AGGR (an aggregation of multiple client orders) 2 = PNAL (clients are pending allocation) Applicable to PartyRole value 12: 3 = NORE (timing and location of the execution determined by the client of the participant)
447	→		PartyIDSource	R	Required if NoPartyIDs (453) > 0	Used to identify classification source.			D = Proprietary / Custom code B = BIC N = Legal Entity Identifier (LEI) P = Short code identifier

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
452	→		PartyRole	R	Required if NoPartyIDs (453) > 0	Identifies the type of PartyID (448).			For PartyIDSource (447) = D: 4 = Clearing Firm For PartyIDSource (447) = P: 3 = Client ID 12 = Executing Trader 122 = Investment Decision Maker
2376	→		PartyRoleQualifier	R	Required if NoPartyIDs (453) > 0	Used to further qualify the value of PartyRole (452).			For PartyIDSource (447) = P: 0 = None (only valid for reserved short codes) 22 = Algorithm (applicable to PartyRole values 12 or 122) 23 = Firm or legal entity (LEI) (applicable to PartyRole value 3) 24 = Natural person (applicable to PartyRole values 3, 12, 122)
1			Account	O		Account mnemonic as agreed between buy and sell sides, e.g. broker and institution or investor/intermediary and fund manager.	string	16 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
581			AccountType	O		Type of account associated with the order.	int		1 = Account is carried on customer side of the books 3 = House trader
528			OrderCapacity	R		Designates the capacity of the firm placing the order.	char	1 character	A = Agency (mapped to AOTC) P = Principal (mapped to DEAL) R = Riskless Principal (mapped to MTCH)
529			OrderRestrictions	O		Restrictions associated with an order.	char	1 character	5 = Acting as Market Maker or Specialist in the security
1724			OrderOrigination	O		Identifies the origin of the order.	num		5 = Order received from a direct access or sponsored access customer
60			TransactTime	R		Timestamp when the business transaction represented by the message occurred.	UTC timestamp		

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
58			Text	O		Free format text string.	string	18 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
C			QuoteSetGrp	R					
296			NoQuoteSets	R		The number of sets of quotes in the message.	int	1 byte	
302	→		QuoteSetID	R	Required if NoQuoteSets > 0	Sequential number for the Quote Set. For a given QuoteID - assumed to start at 1. Must be the first field in the repeating group.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
304	→		TotNoQuoteEntries	R		Total number of quotes for the quote set across all messages. Should be the sum of all NoQuoteEntries in each message that has repeating quotes that are part of the same quote set.	int	1 byte	
	C		QuoteEntryGrp	R					
295			NoQuoteEntries	R		The number of quotes for this Symbol (instrument) (QuoteSet) that follow in this message.	int	1 byte	
299		→	QuoteEntryID	R		Unique identifier for a quote. The QuoteEntryID stays with the quote as a static identifier even if the quote is updated.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
	C	→	Instrument	R					
48		→	SecurityID	R		Security identifier value of SecurityIDSource (22) type (e.g. ISIN, exchange symbol, etc). Requires SecurityIDSource.	num	4 bytes	integer
22		→	SecurityIDSource	R		Identifies class or source of the SecurityID (48) value.	char	1 character	8 = exchange symbol

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
132		→	BidPx	R		Bid price.	num	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
133		→	OfferPx	R		Offer price.	num	12 digits + decimal point	Floating point number. 6 digits for integer part, decimal point, 6 digits for decimal part. Negative values and 0 allowed.
134		→	BidSize	R	0 for quote cancellation.	Quantity of bid.	num	4 bytes	integer
135		→	OfferSize	R	0 for quote cancellation.	Quantity of offer.	num	4 bytes	integer
C			StandardTrailer	R					

7.19. MASS QUOTE ACK(B)

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
C			StandardHeader	R					
117			QuoteID	O	Required when acknowledgment is in response to a Mass Quote message.	Unique identifier for quote. Maps to QuoteID (117) of a Mass Quote.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
297			QuoteStatus	R		Status of the mass quote acknowledgement.	int		0 = Accepted 5 = Rejected
300			QuoteRejectReason	O		Reason Quote was rejected.	int		2 = Exchange (security) closed 4 = Too late to enter 6 = Duplicate quote 99 = Other
301			QuoteResponseLevel	R		Level of response requested from receiver of quote messages. Is echoed back to the counterparty.	int		2 = Acknowledge each quote message

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
537			QuoteType	R		Identifies the type of quote. Default is Indicative if not specified.			1 = Tradeable
C			Parties	R					
453			NoPartyIDs	R		Number of PartyID (448), PartyIDSource (447), PartyRole (452) and PartyRoleQualifier (2376) entries.	num		integer
448	→		PartyID	R	Required if NoPartyIDs (453) > 0	Identification of the party.	num	4 bytes	integer For clients, the following values are reserved for applicable use: Applicable to PartyRole value 3: 0 = NONE (no client for this order) 1 = AGGR (an aggregation of multiple client orders) 2 = PNAL (clients are pending allocation) Applicable to PartyRole value 12: 3 = NORE (timing and location of the execution determined by the client of the participant)
447	→		PartyIDSource	R	Required if NoPartyIDs (453) > 0	Used to identify classification source.			D = Proprietary / Custom code B = BIC N = Legal Entity Identifier (LEI) P = Short code identifier
452	→		PartyRole	R	Required if NoPartyIDs (453) > 0	Identifies the type of PartyID (448).			For PartyIDSource (447) = D: 4 = Clearing Firm For PartyIDSource (447) = P: 3 = Client ID 12 = Executing Trader 122 = Investment Decision Maker
2376	→		PartyRoleQualifier	R	Required if NoPartyIDs (453) > 0	Used to further qualify the value of PartyRole (452).			For PartyIDSource (447) = P: 0 = None (only valid for reserved short codes)

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
									22 = Algorithm (applicable to PartyRole values 12 or 122) 23 = Firm or legal entity (LEI) (applicable to PartyRole value 3) 24 = Natural person (applicable to PartyRole values 3, 12, 122)
1			Account	O		Account mnemonic as agreed between buy and sell sides, e.g. broker and institution or investor/intermediary and fund manager.	string	16 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
581			AccountType	O		Type of account associated with the order.	int		1 = Account is carried on customer side of the books 3 = House trader
60			TransactTime	R		Timestamp when the business transaction represented by the message occurred.	UTC timestamp		
528			OrderCapacity	R		Designates the capacity of the firm placing the order.	char	1 character	A = Agency (mapped to AOTC) P = Principal (mapped to DEAL) R = Riskless Principal (mapped to MTCH)
529			OrderRestrictions	O		Restrictions associated with an order.	char	1 character	5 = Acting as Market Maker or Specialist in the security
1724			OrderOrigination	O		Identifies the origin of the order.	num		5 = Order received from a direct access or sponsored access customer
58			Text	O		Free format text string.	string	18 characters	alphanumeric, 0-9, A-Z, a-z or ASCII 33-126
C			QuoteSetAckGrp	R					
296			NoQuoteSets	R		The number of sets of quotes in the message.	int	1 byte	
302	→		QuoteSetID	R	Required if NoQuoteSets > 0	Sequential number for the Quote Set. For a given QuoteID - assumed to start at 1.	string	20 characters	alphanumeric, 0-9, A-Z, a-z

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
						Must be the first field in the repeating group.			
304	→		TotNoQuoteEntries	R	Required if NoQuoteEntries > 0	Total number of quotes for the quote set across all messages. Should be the sum of all NoQuoteEntries in each message that has repeating quotes that are part of the same quote set.	int	1 byte	
1168	→		TotNoClxdQuotes	R		Total number of quotes canceled for the quote set across all messages.	int	1 byte	
1169	→		TotNoAccQuotes	R		Total number of quotes accepted for the quote set across all messages.	int	1 byte	
1170	→		TotNoRejQuotes	R		Total number of quotes rejected for the quote set across all messages.	int	1 byte	
	C		QuoteEntryAckGrp	O					
295			NoQuoteEntries	R		The number of quotes for this Symbol (instrument) (QuoteSet) that follow in this message.	int	1 byte	
299	→		QuoteEntryID	R	Required if NoQuoteEntries > 0.	Uniquely identifies the quote across the complete set of all quotes for a given quote provider. First field in repeating group.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
1747	→		BidQuoteID	O		Marketplace assigned quote identifier for the	string	20 characters	alphanumeric, 0-9, A-Z, a-z

Tag			Field Name	Required	Conditional	Description	Data Type	Max Length	Values
						bid side. Can be used to indicate priority.			
1748		→	OfferQuoteID	O		Marketplace assigned quote identifier for the offer side. Can be used to indicate priority.	string	20 characters	alphanumeric, 0-9, A-Z, a-z
1167		→	QuoteEntryStatus	O		Identifies the status of an individual quote. See also QuoteStatus (297) which is used for single Quotes.	int		0 = Accepted 5 = Rejected 6 = Removed from market 7 = Expired
368		→	QuoteEntryRejectReason	O		Reason quote entry was rejected.	int		1 = Unknown symbol (security) 4 = Too late to enter 5 = Unknown quote 6 = Duplicate quote 7 = Invalid bid/ask spread 8 = Invalid price 9 = Not authorized to quote security 10 = Price exceeds current price band 99 = Other
C			StandardTrailer	R					

8. MESSAGE KINEMATICS

Scope:

The purpose of this appendix is to present the message flow within the FIX communication protocol, as used in GPW WATS. The appendix does not include all available message flows in GPW WATS, but only those selected for presentation purposes.

Remarks:

Due to ensuring transparency of message flow diagrams, the scope of presented fields is limited to basic components and does not include the complete set of all fields that a given message may contain. The scope of the fields for the same messages may also vary depending on the specific example described in the diagram. The complete range of fields in the messages is available in the description of each individual message within the protocol.

Actors:

For the purpose of the presentations, the following actors have been utilized in the context of kinematics:

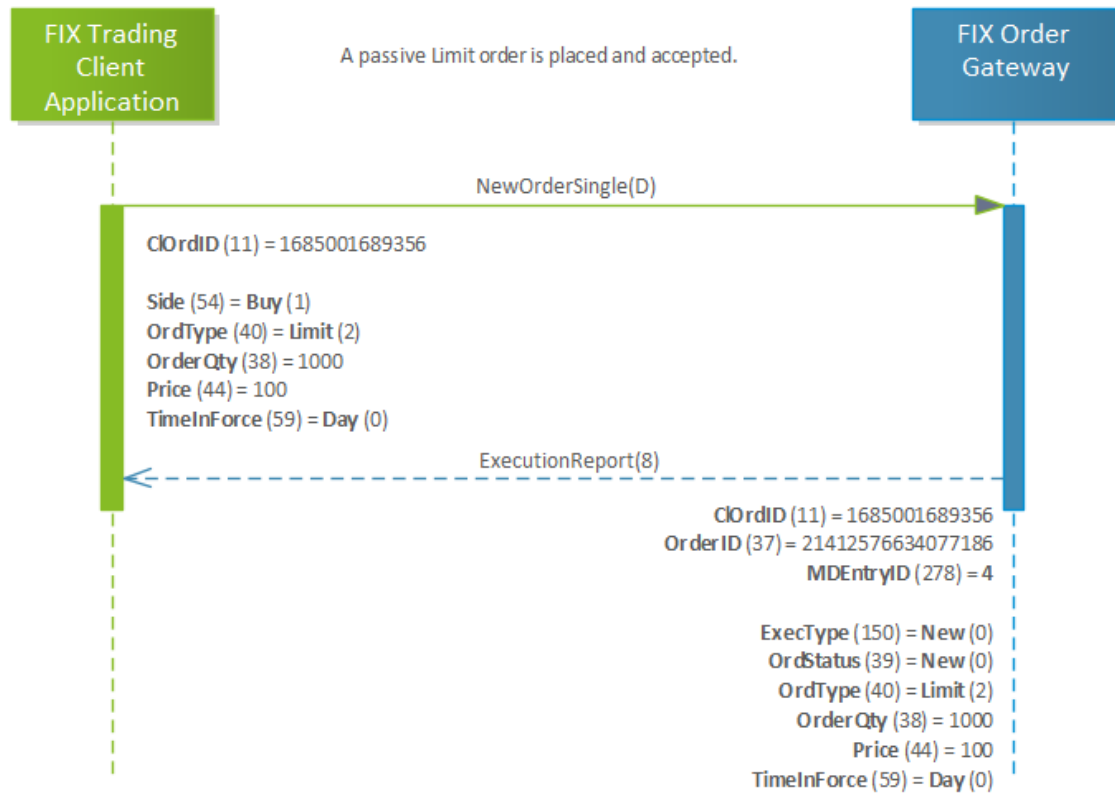
FIX Trading Client Application – an exchange member authorized to submit buy and sell orders, communicating with the system represented by the FIX Order Gateway.

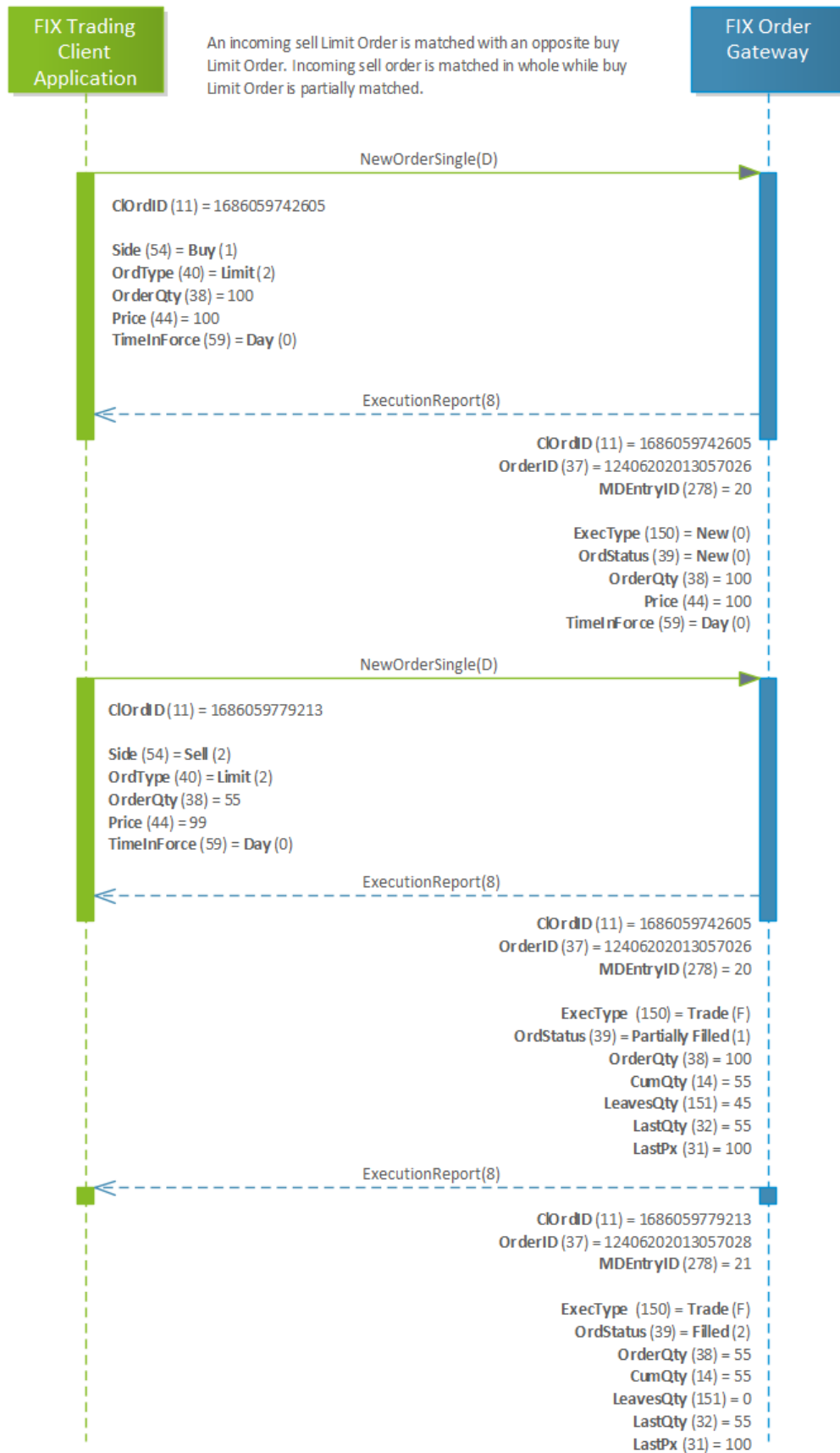
FIX Order Gateway – a system that communicates with the FIX Protocol Client Application.

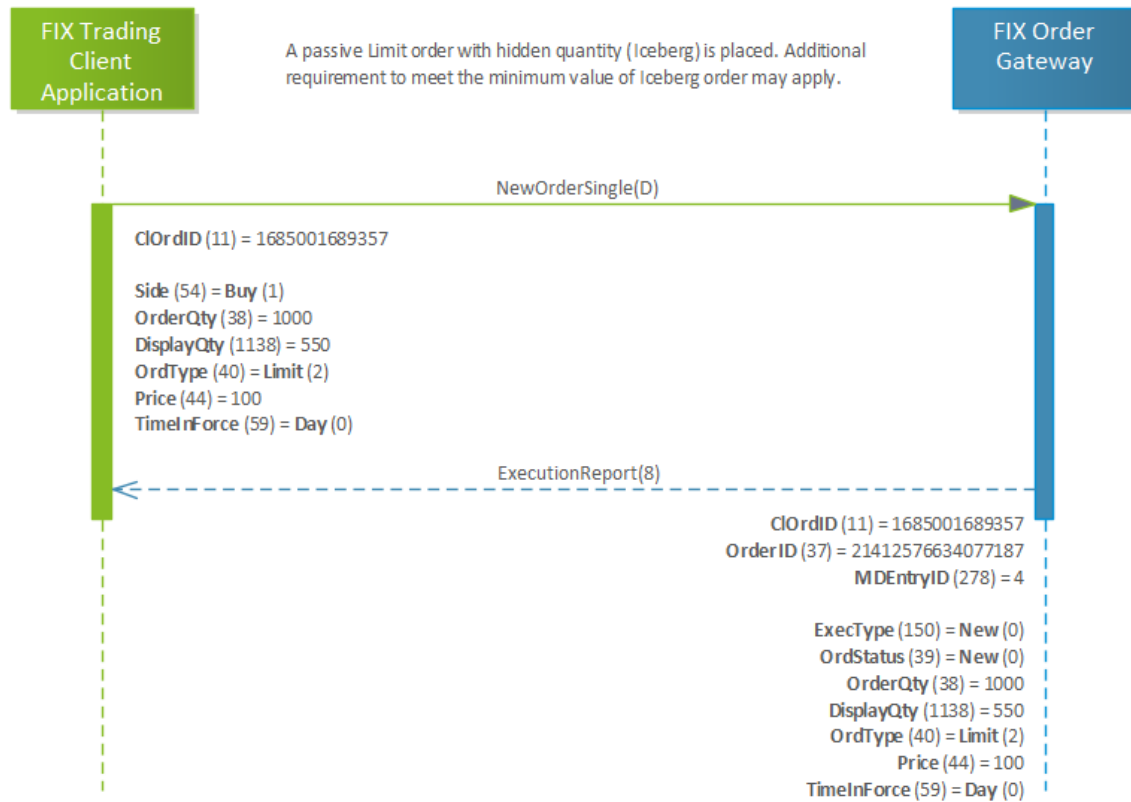
Note: presentation of simultaneous communication between multiple clients and Gateway was omitted. As a result, the FIX Trading Client Application can either represent a single exchange member or simulate the exchange of messages between two different exchange members including buying, selling and conducting transactions.

8.1. CLOB

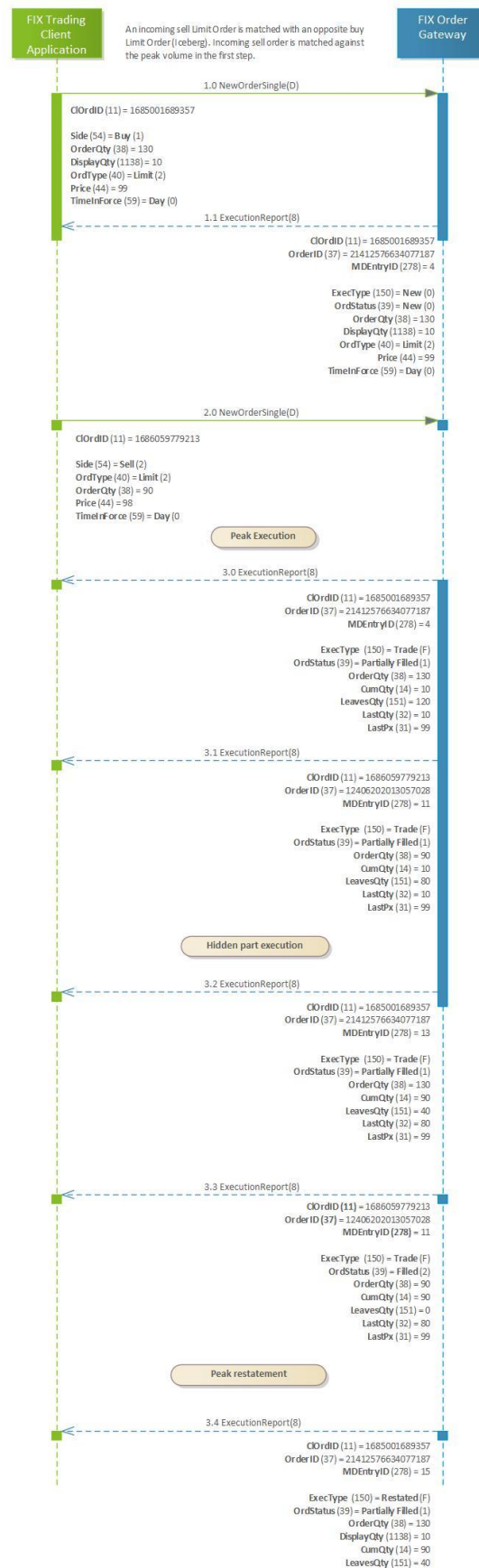
Order Accepted

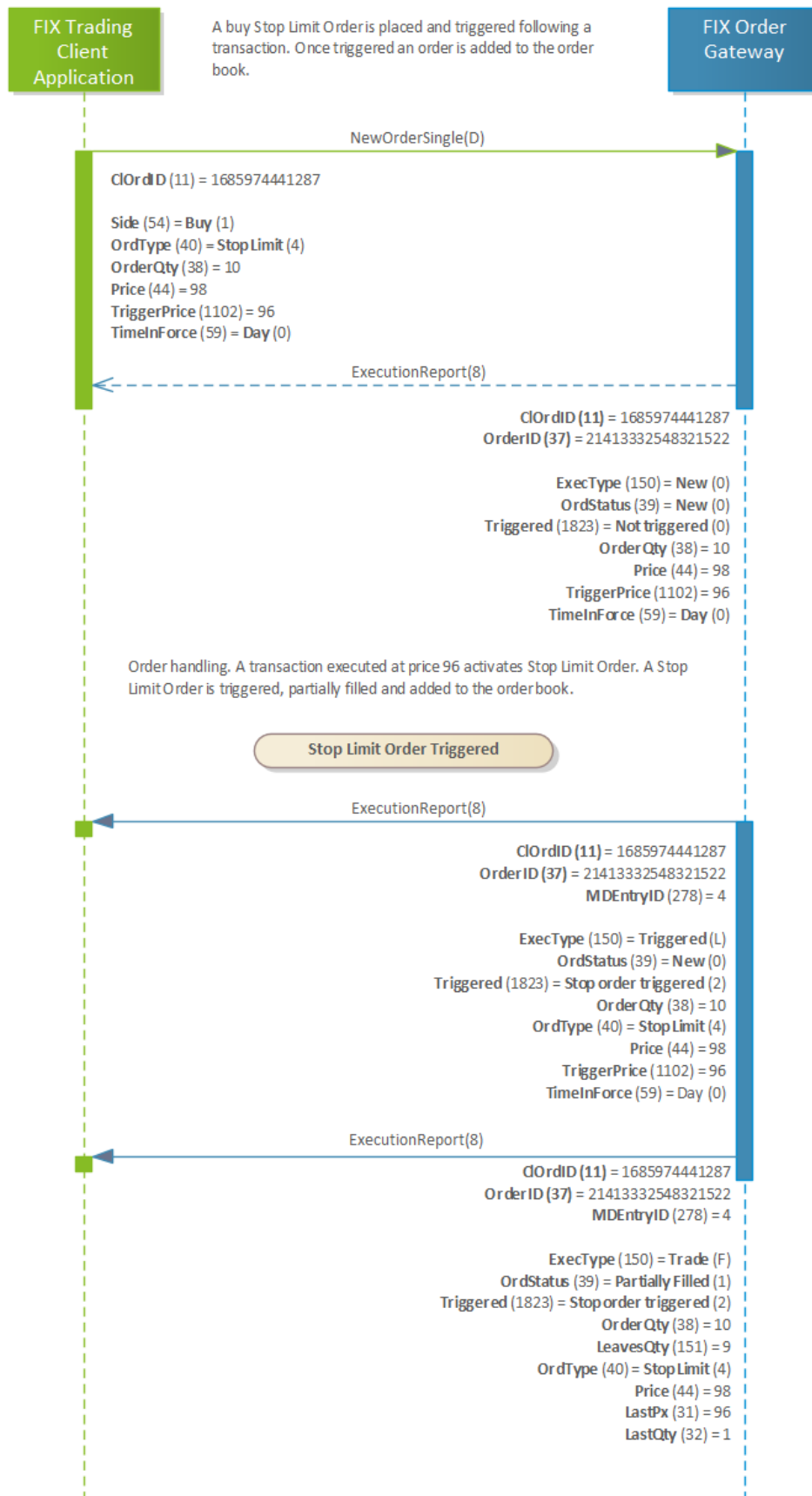


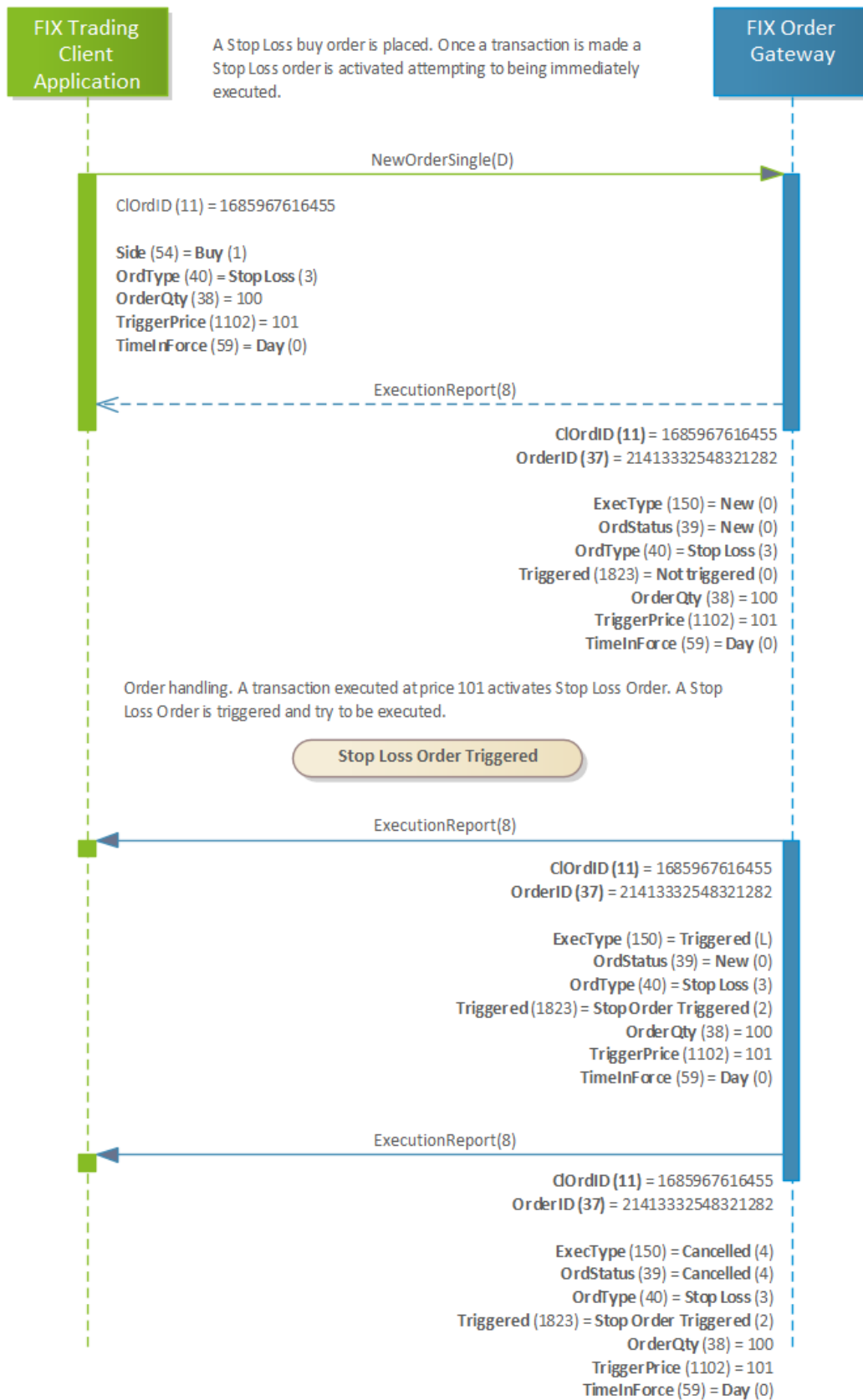
Order – Accepted and Filled – Full and Partially

Iceberg Order - Accepted

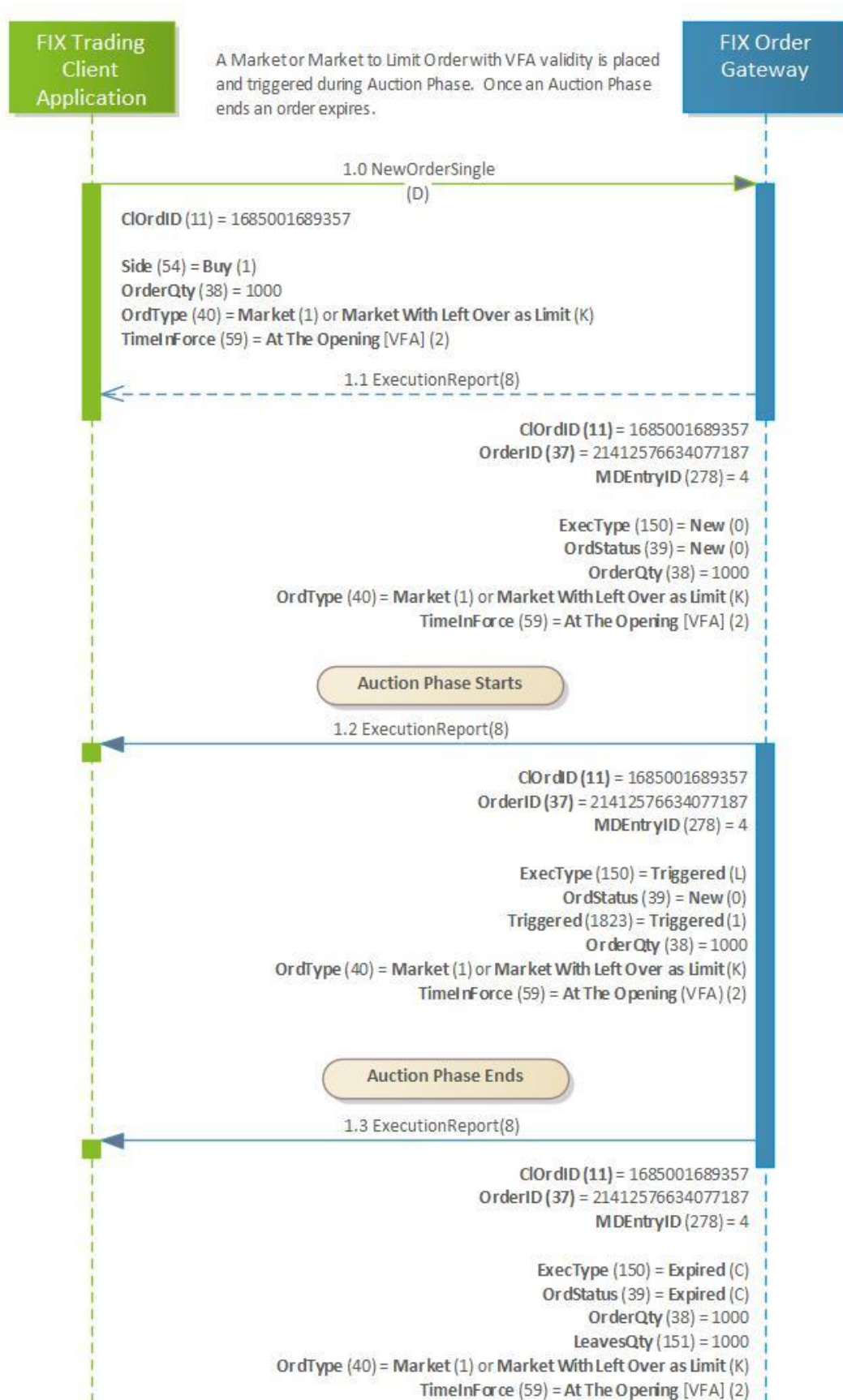
Iceberg Order – Accepted, Filled Partially and Restated

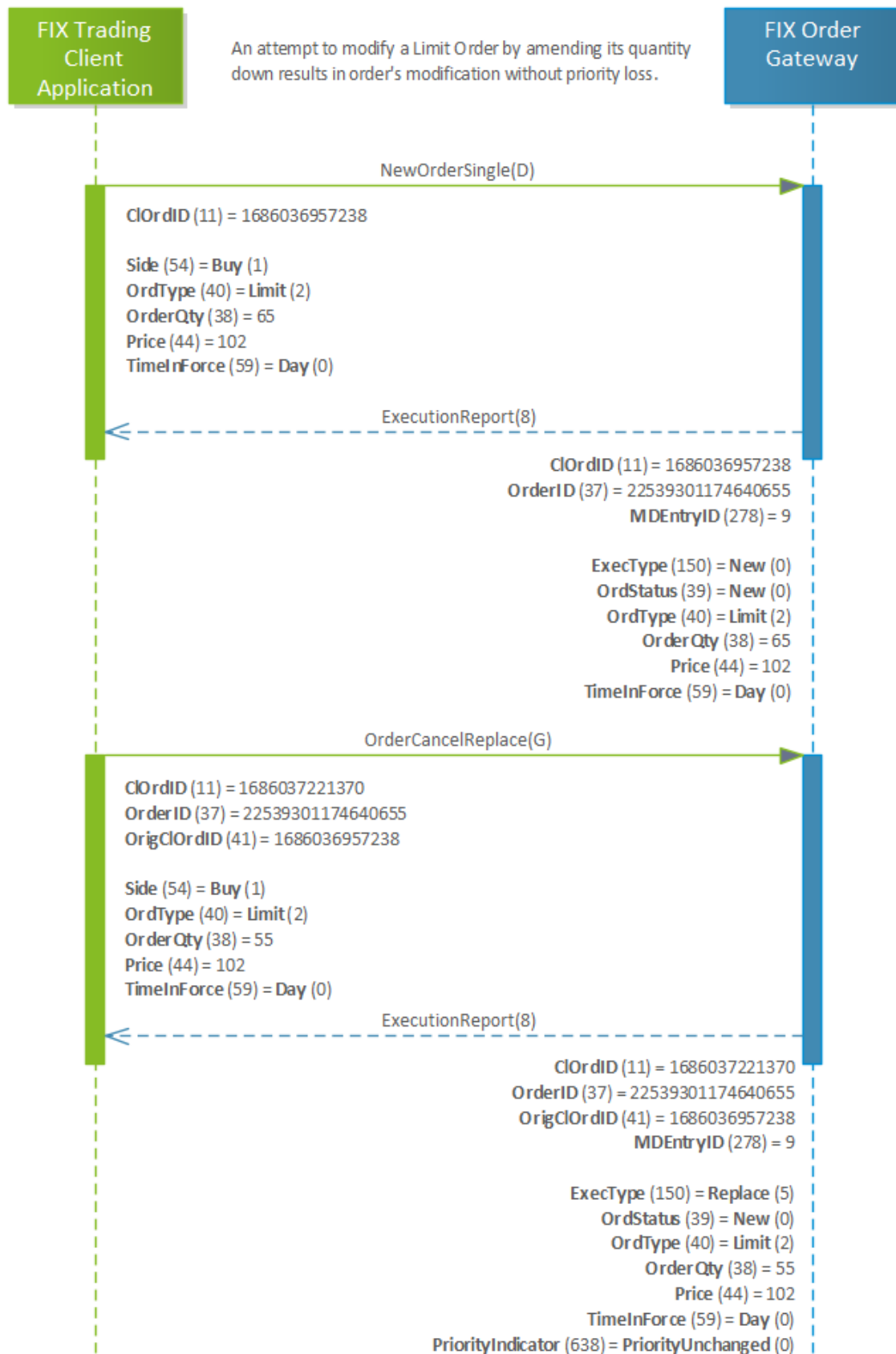


Stop Limit Order – Accepted and Triggered

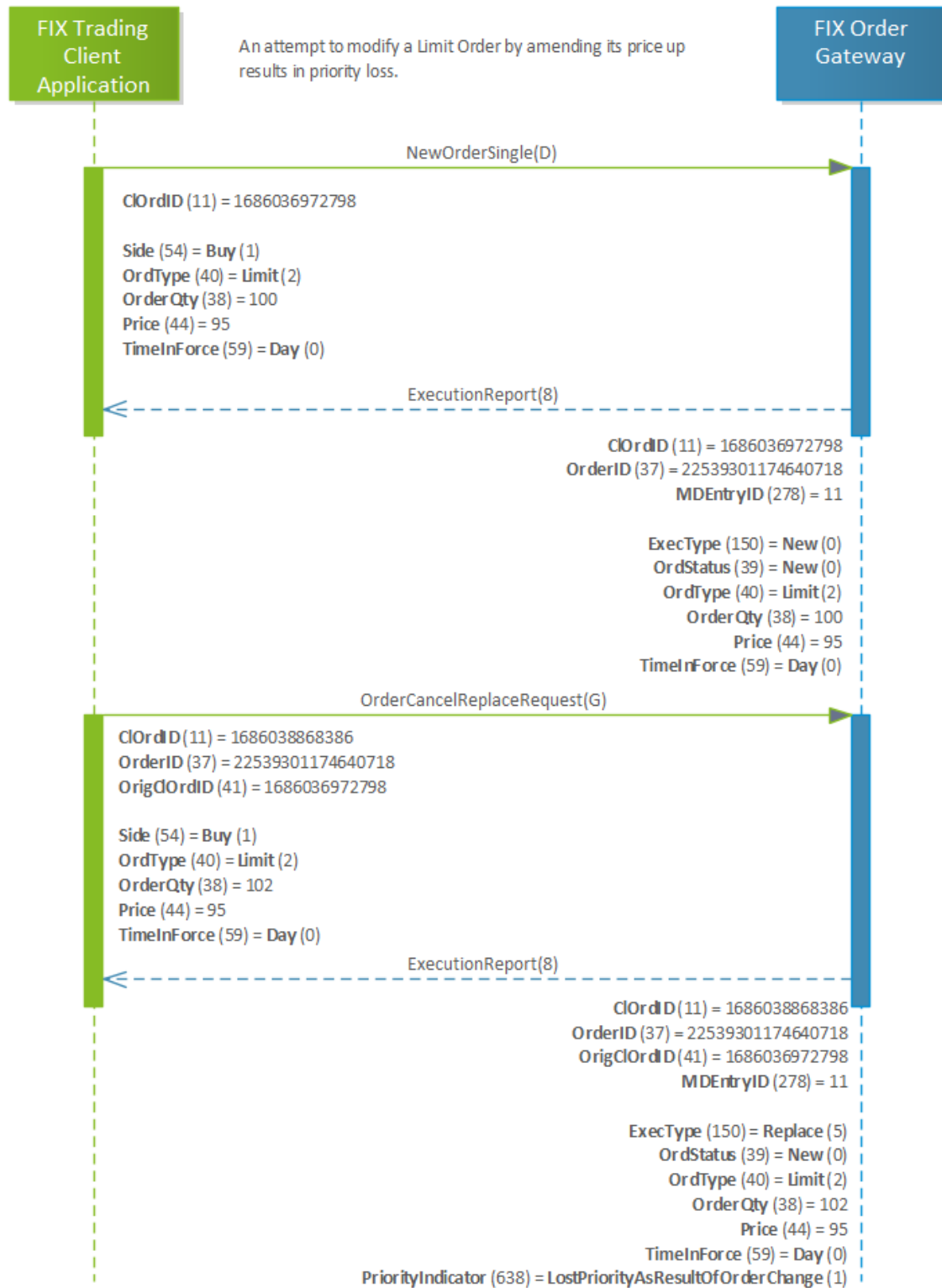
Stop Loss Order – Accepted, Triggered and Cancelled

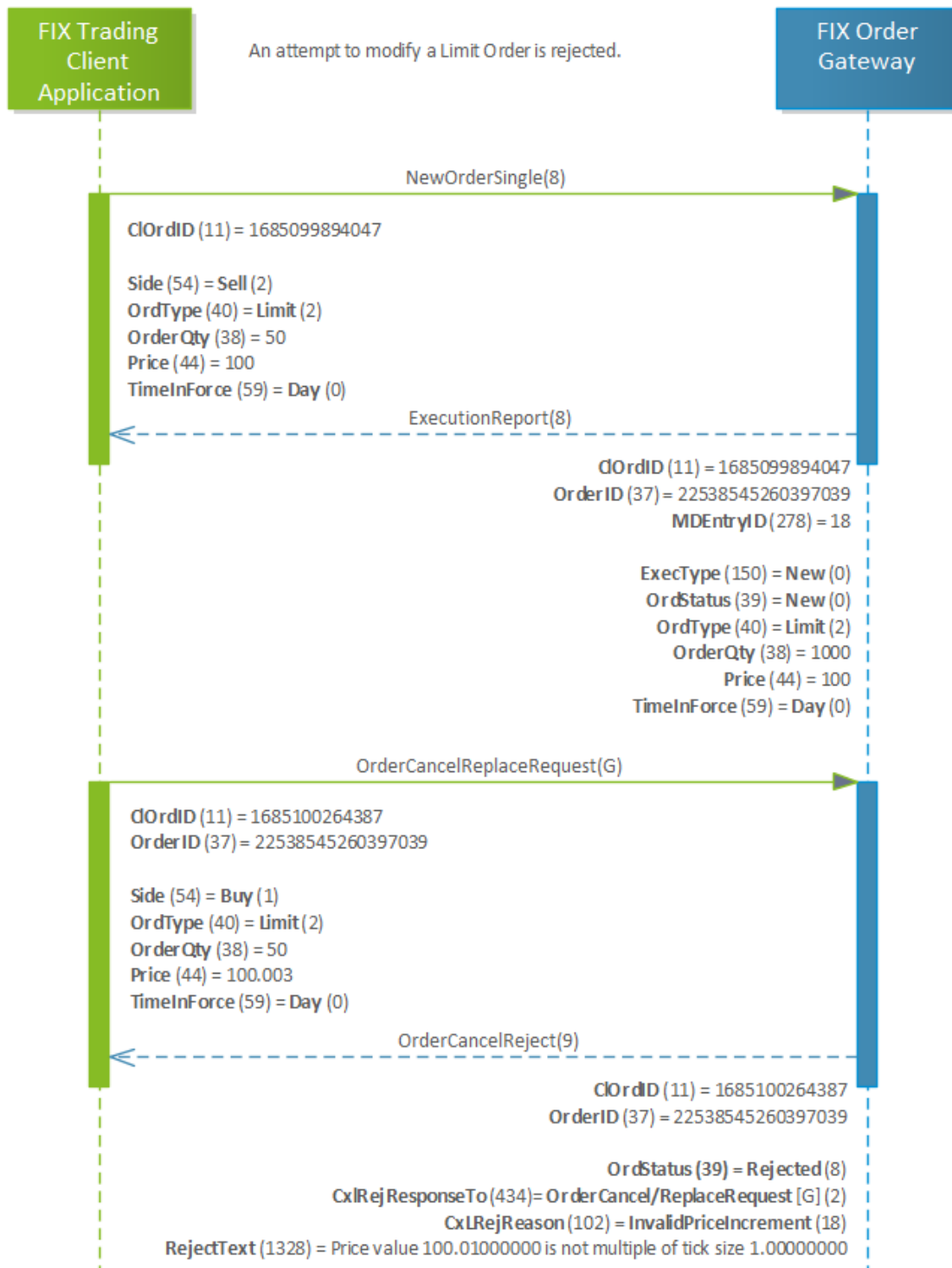
Market Order and Market to Limit Order with TIF VFA – Triggered during Auction

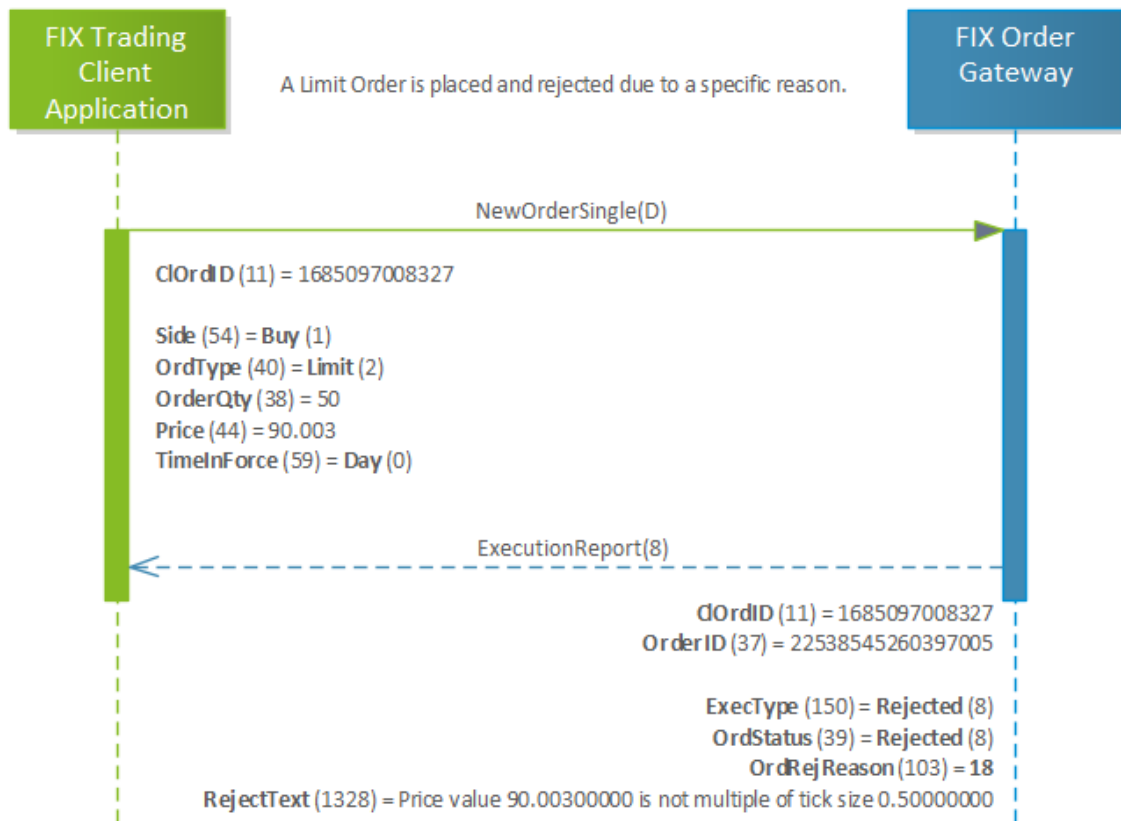


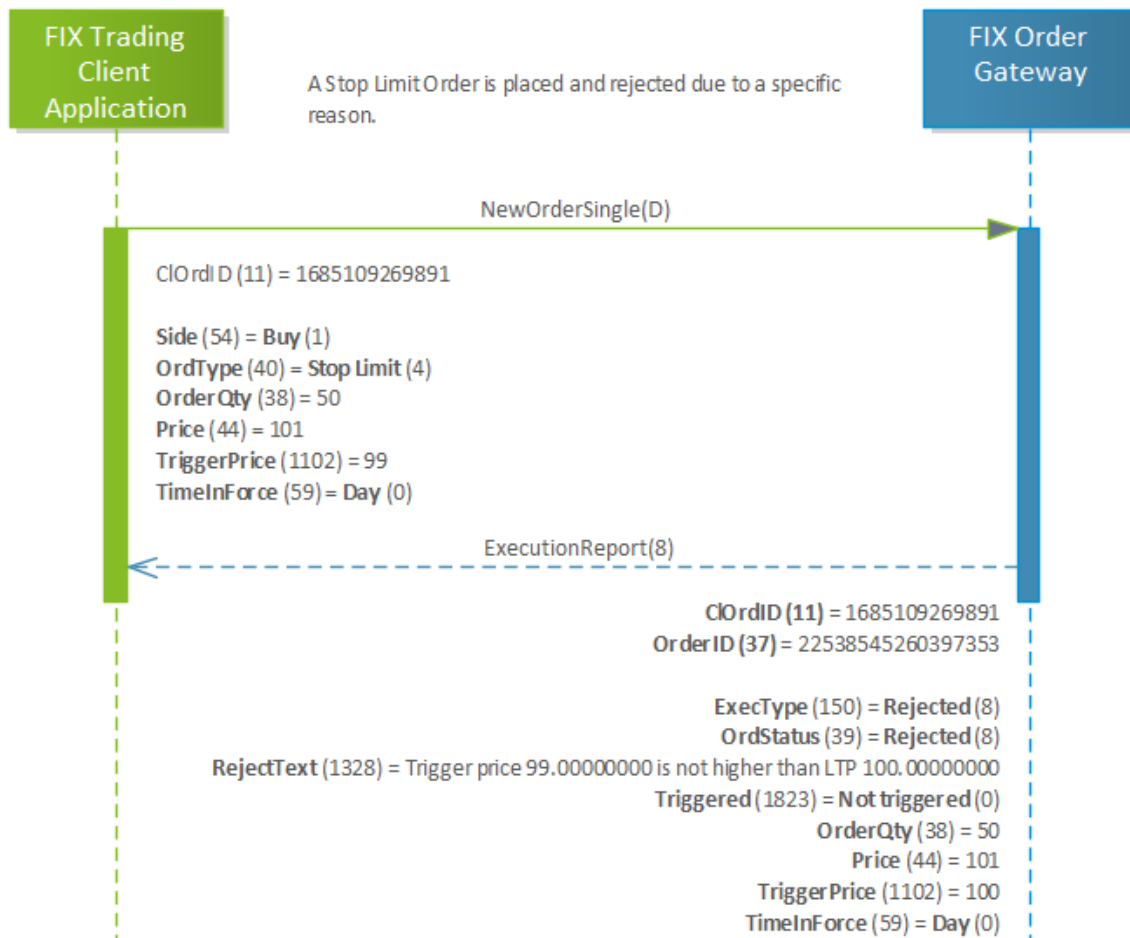
Order Modification without priority loss

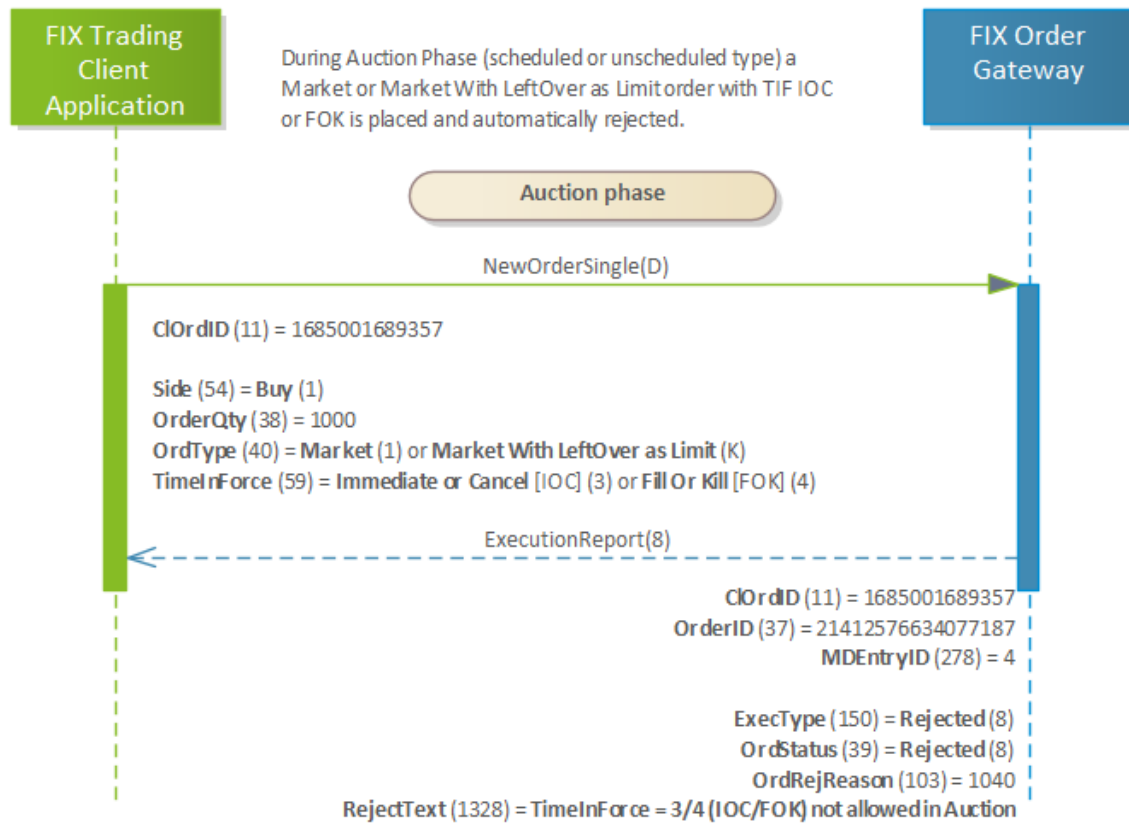
Order Modification with priority loss

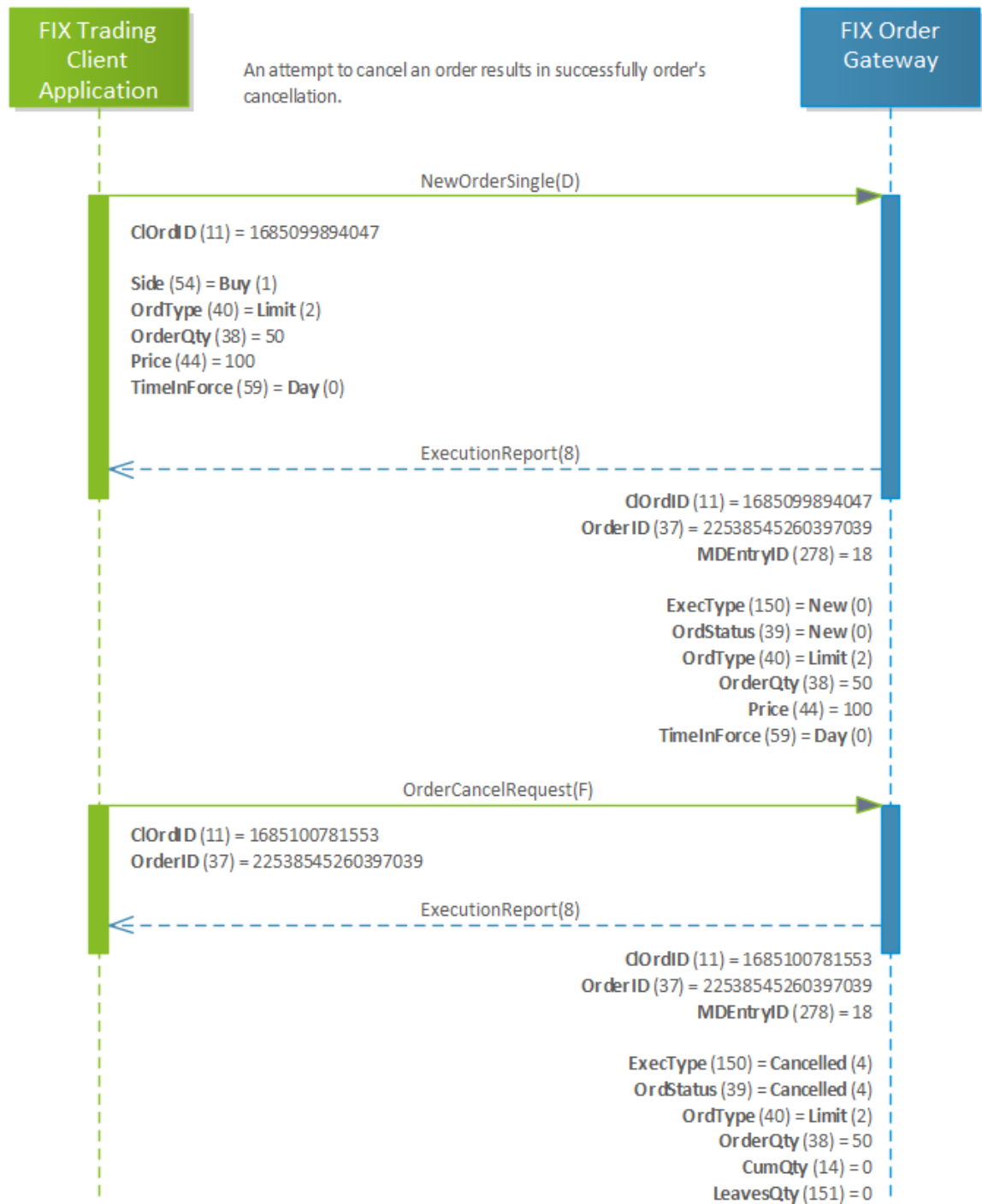


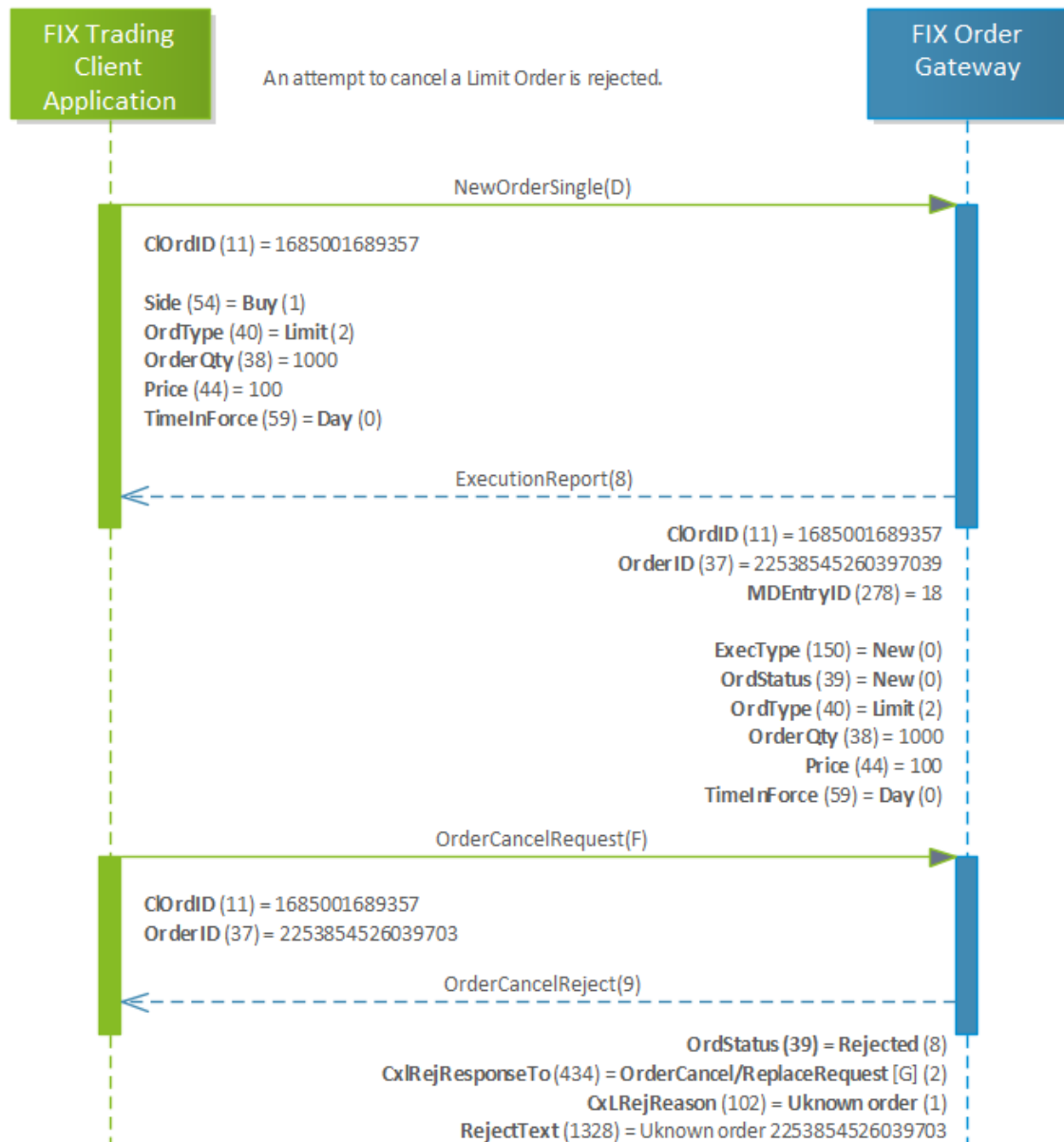
Order Modification - Rejected

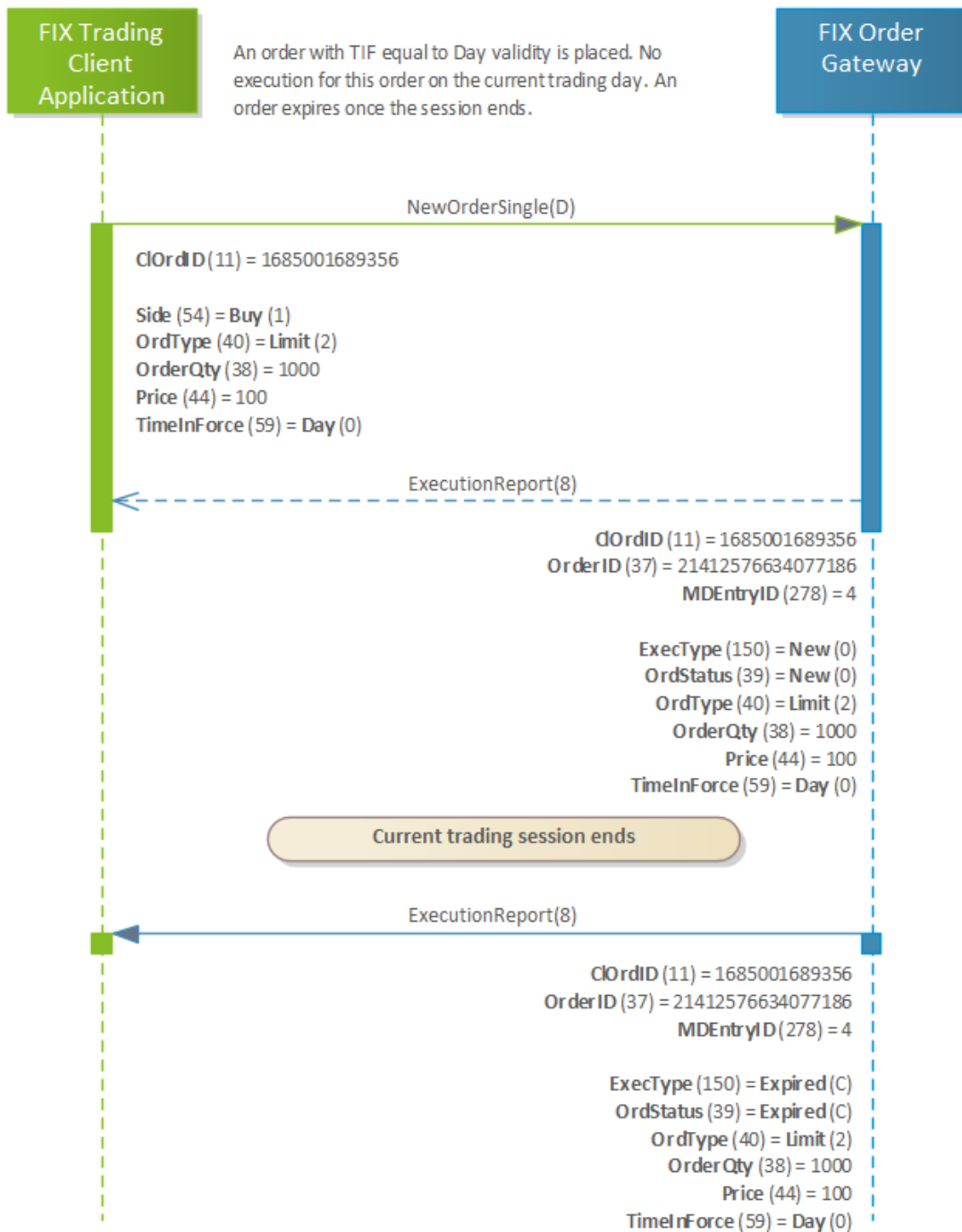
Order - Rejected

Stop Limit Order - Rejected

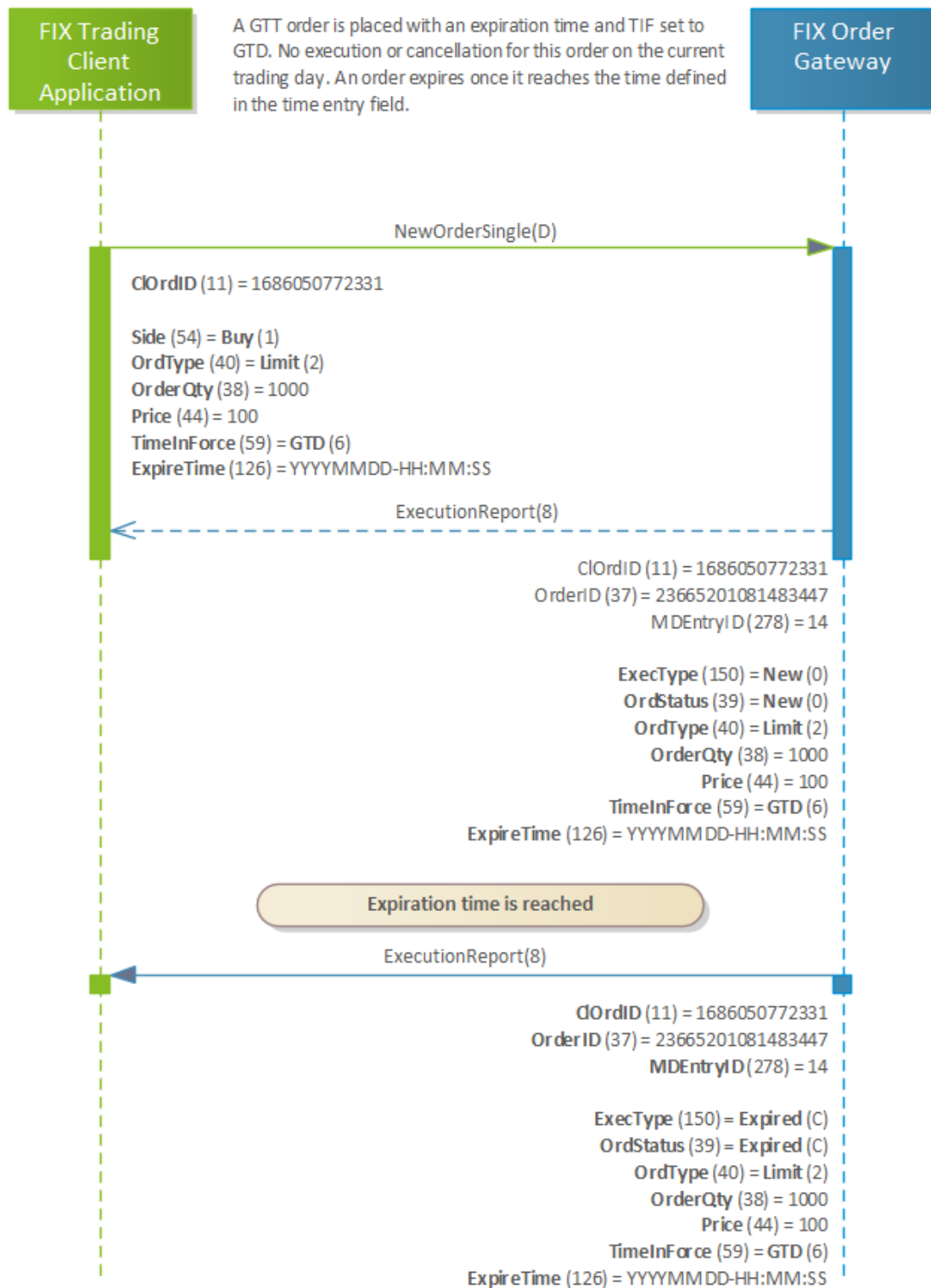
Market Order and Market to Limit Order IOC/FOK – Rejected during Auction

Order Cancellation - Accepted

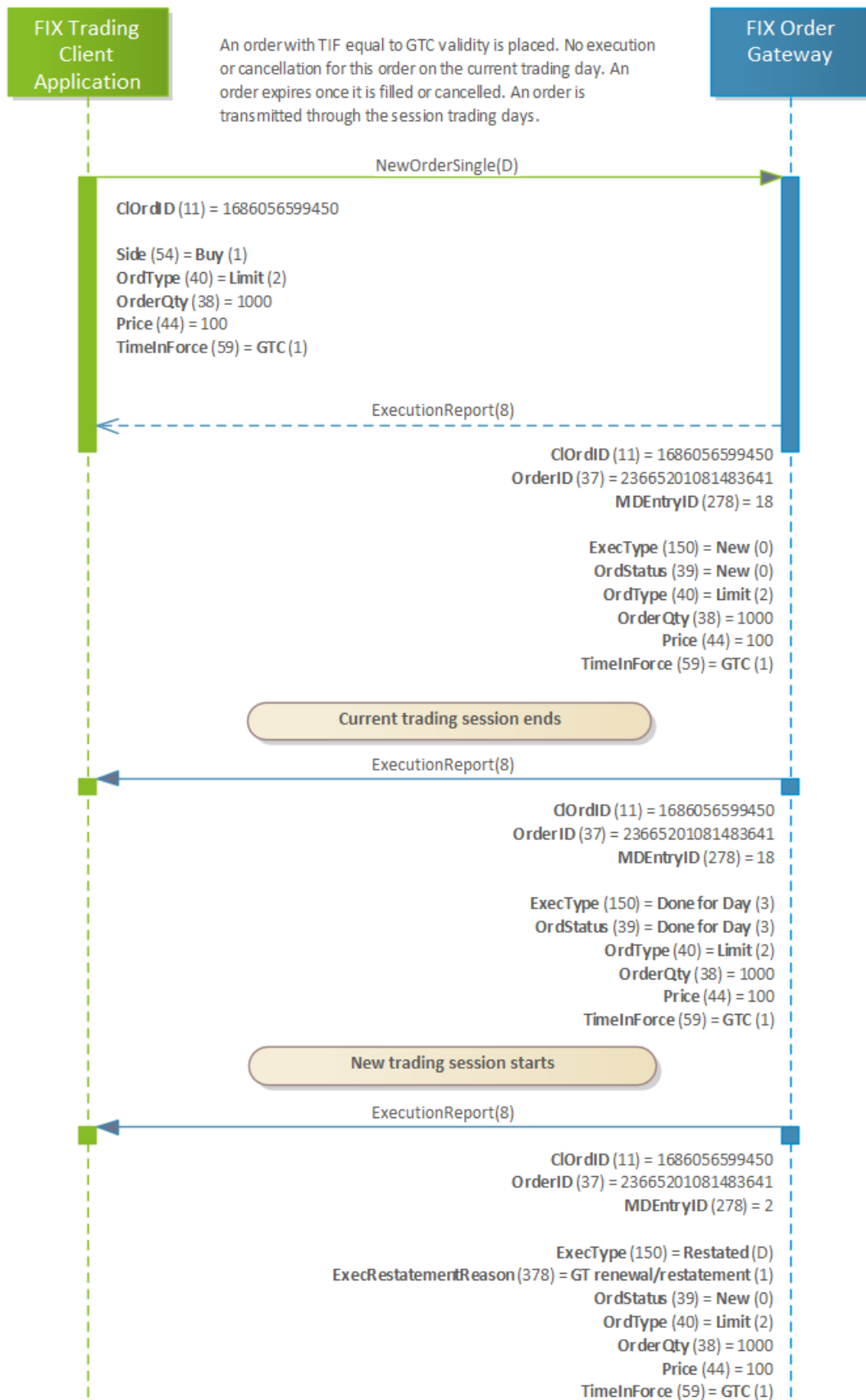
Order Cancellation - Rejected

Day Order - Expired

GTT Order - Expired

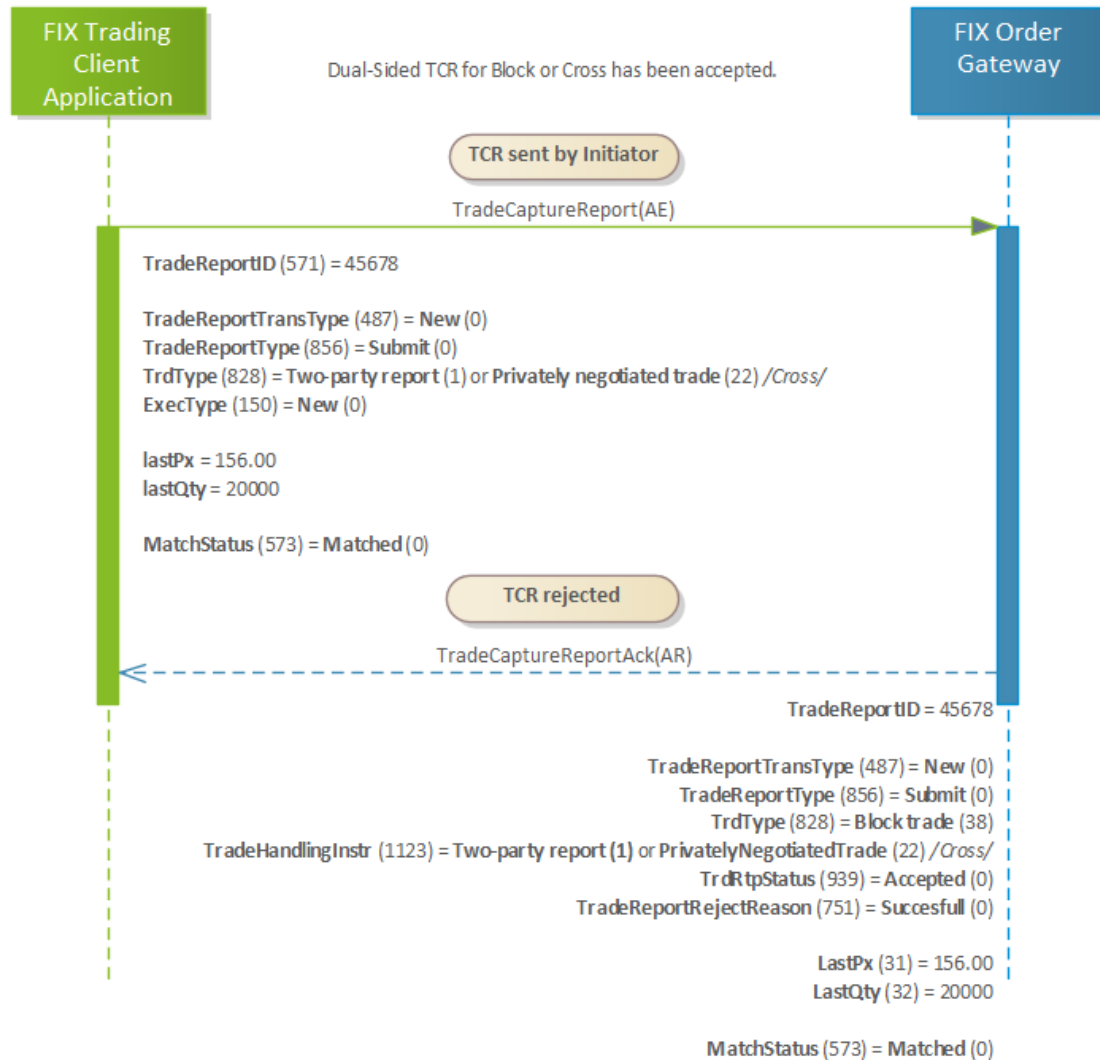


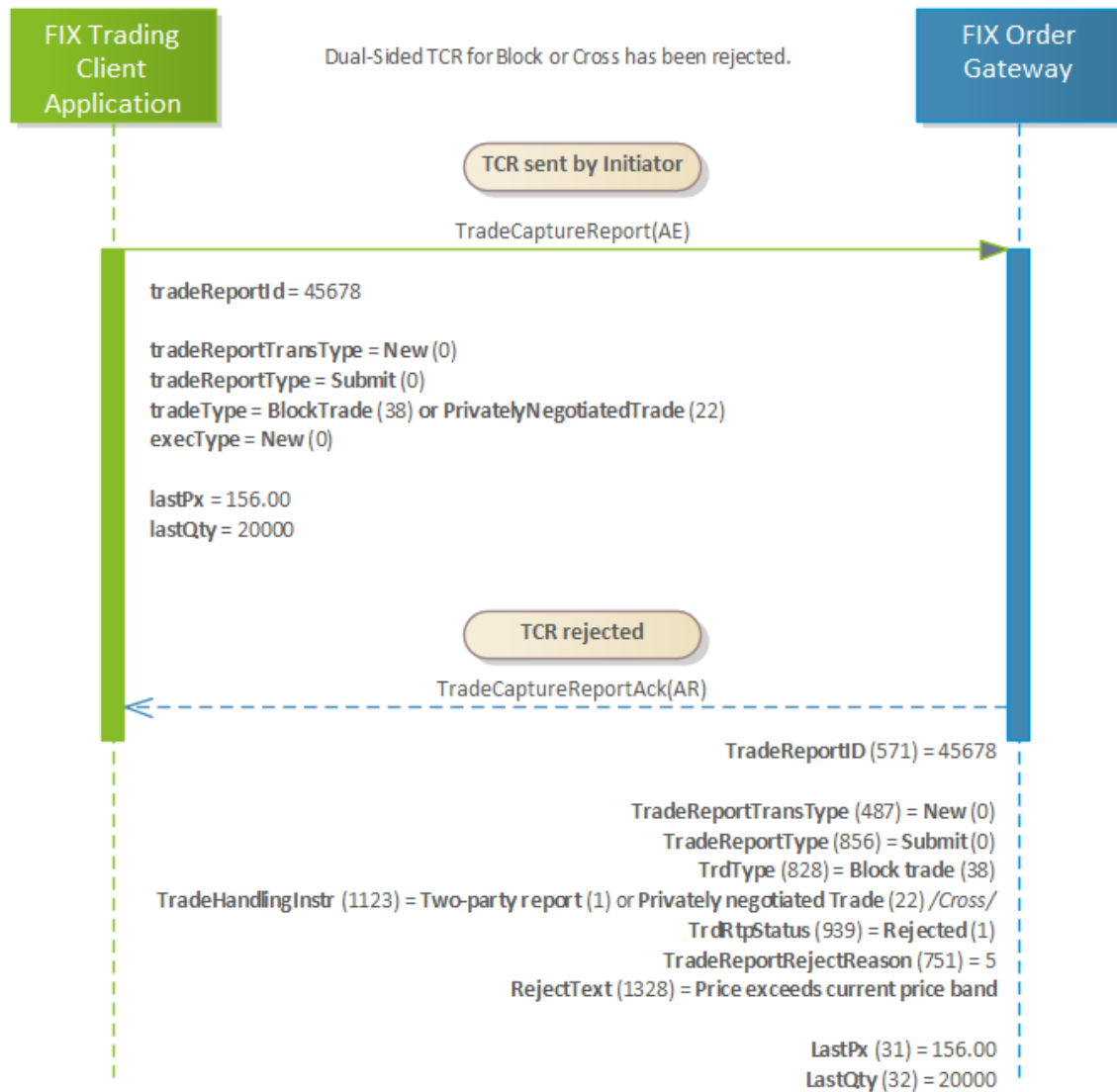
GTC Order - Reinstated

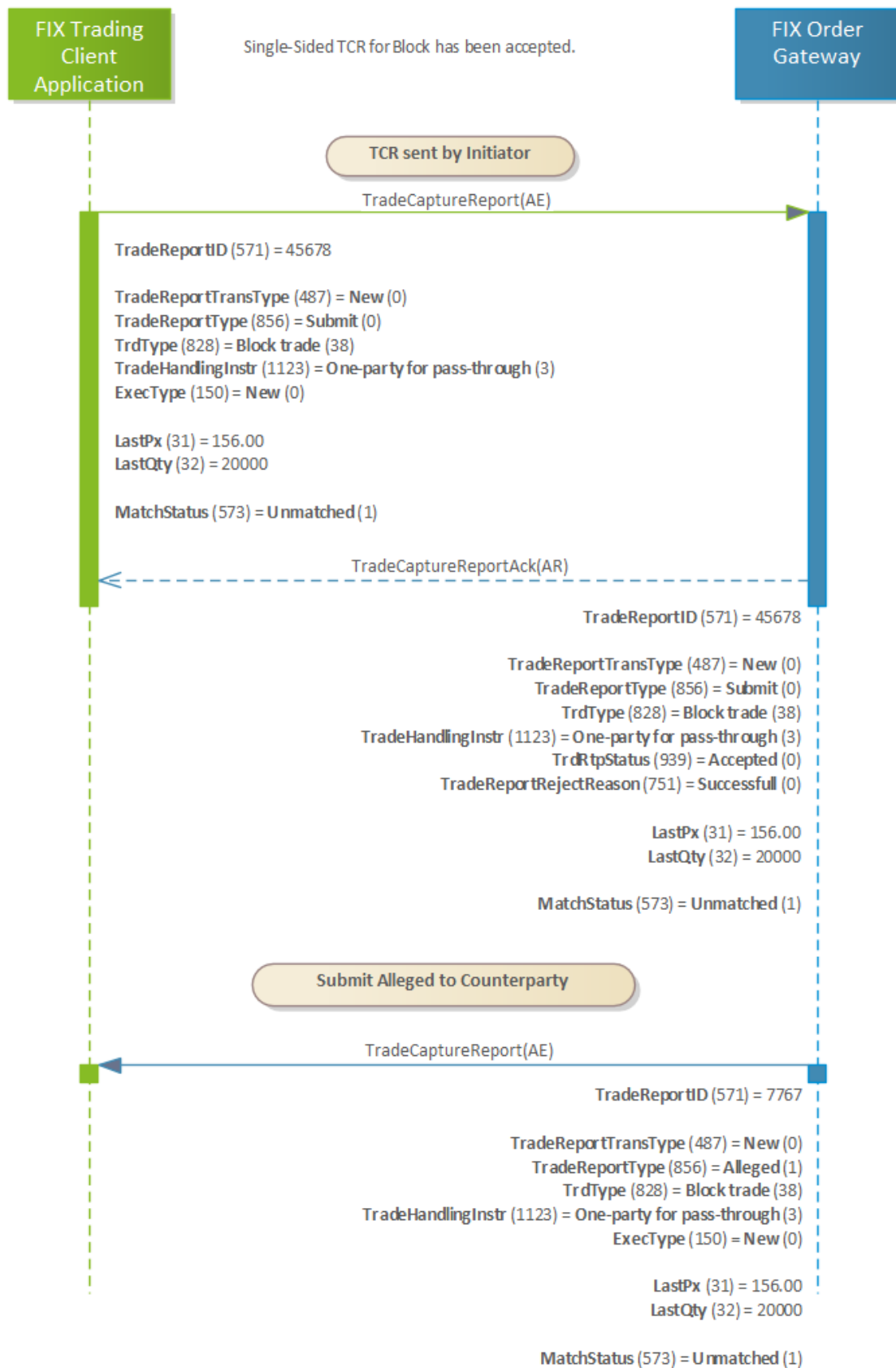


8.2. OFF-BOOK (BLOCK OR CROSS)

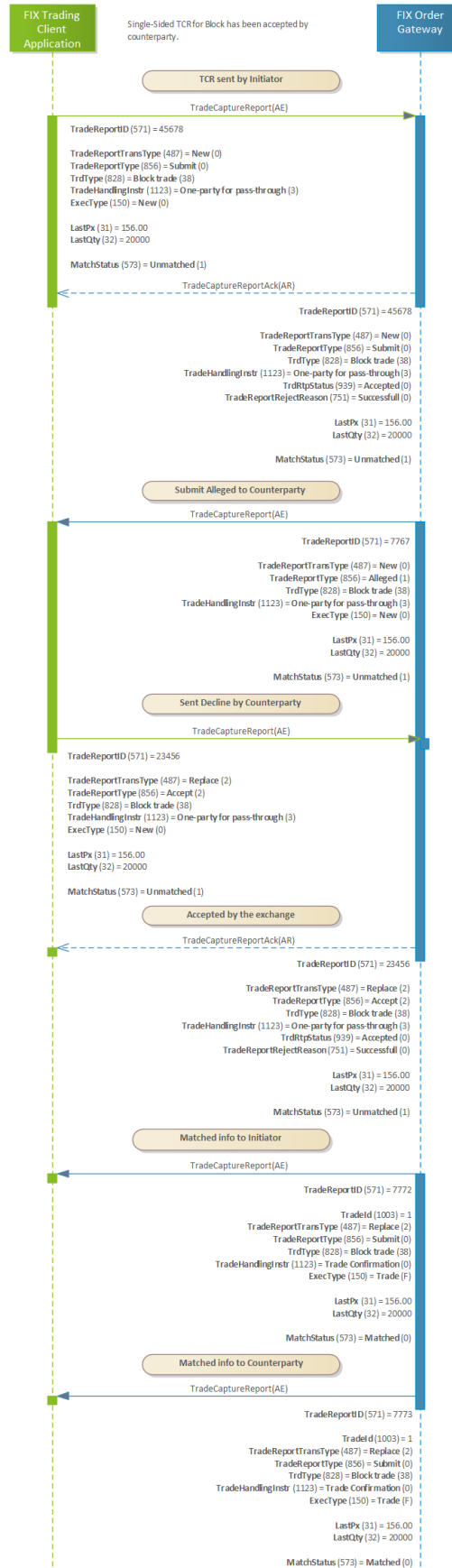
Dual Sided – Accepted



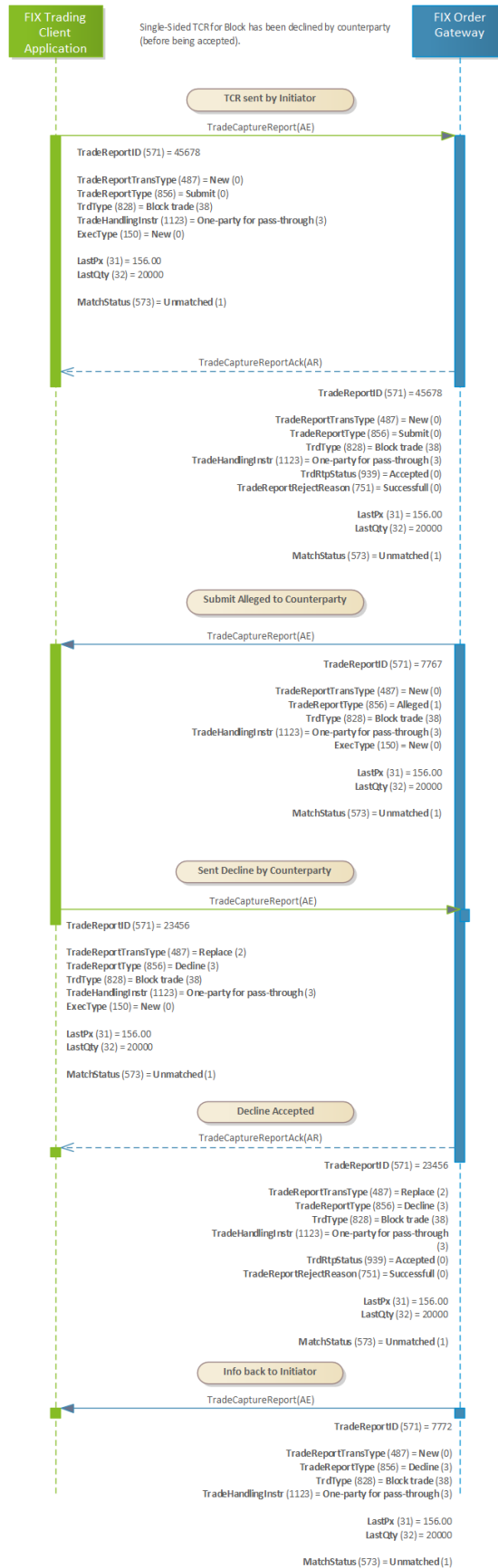
Dual Sided – Rejected

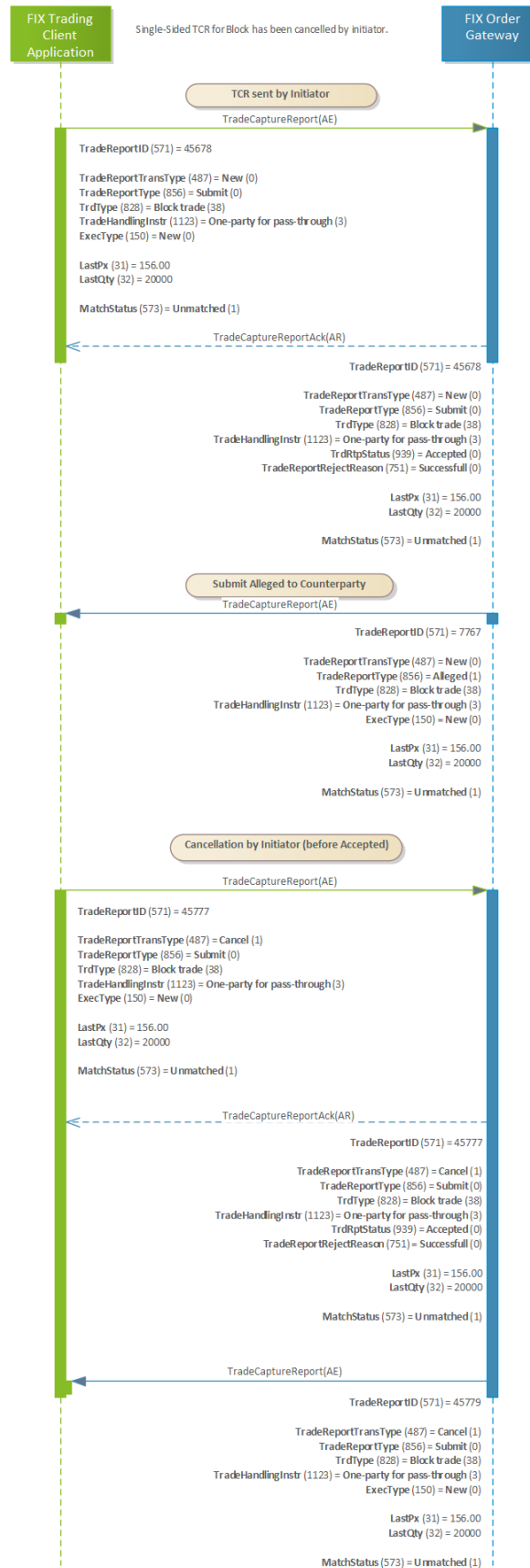
Single Sided - Accepted

Single Sided – Accepted by Counterparty



Single Sided – Denied by Counterparty



Single Sided – Cancelled by Initiator

Single Sided – Rejected